7 Structure and Synthesis of Alkenes; Elimination

Goals for Chapter 7

1 Draw and name alkenes and cycloalkenes.

2 Given a molecular formula, calculate the number of pi bonds and rings.

3 Explain how the stability of alkenes depends on their structures.

4 Write equations to show how alkenes can be synthesized by eliminations from alkyl halides and alcohols.

5 Predict the products of substitution and elimination reactions, and explain what factors favor each type of reaction.

6 Identify the differences between unimolecular and bimolecular eliminations and explain what factors determine the order of the reaction.

7 Given a set of reaction conditions, identify the possible mechanisms, and predict which mechanism(s) and product(s) are most likely.

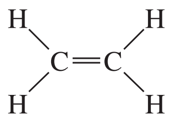
squalene

▲ **Squalene** is a 30-carbon alkene with six carbon–carbon double bonds. Squalene occurs in all plants and animals to make the steroid compounds used as hormones, vitamins, emulsifiers, and cell membrane components (Chapter 25). Squalene is used in cosmetics and as an adjuvant (makes the immune system respond more strongly) in vaccines. Squalene was originally obtained from shark livers, where sharks use it as an incompressible low-density liquid to make them neutrally buoyant in salt water. Plant sources, such as olive oil, are now used as cheaper and more sustainable sources of squalene.

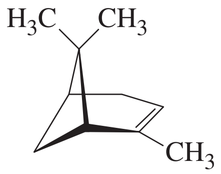
7-1 Introduction

Alkenes are hydrocarbons with carbon–carbon double bonds. Alkenes are central to organic chemistry because they are manufactured in large quantities from crude petroleum, and they are readily converted to many of the other functional groups. In this chapter, we consider the characteristics of alkenes and some of the ways they are synthesized. In studying the syntheses of alkenes, we begin with the elimination reactions of alkyl halides, which compete with the substitution reactions covered in Chapter 6. In Chapter 8, we continue the study of alkenes, emphasizing the wide variety of reactions they can undergo.

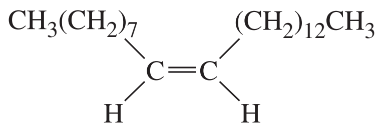
Alkenes are sometimes called **olefins,** a term derived from *olefiant gas*, meaning "oil-forming gas." This term originated with early experimentalists who noticed the oily appearance of alkene derivatives. Alkenes are among the most important industrial compounds (see Section 7-6), and many alkenes are also synthesized by plants and animals. *Ethylene* is the largest-volume industrial organic compound, used to make polyethylene and a variety of other industrial and consumer chemicals. Ethylene is also a fruit-ripening hormone found in the air released by plants. *Pinene* is a major component of *turpentine*, the paint solvent distilled from extracts of evergreen trees. *Muscalure* (*cis*-tricos-9-ene) is the sex attractant of the common housefly.



ethylene (ethene)



α-pinene



cis-tricos-9-ene, "muscalure"

A carbon–carbon double bond consists of a very stable sigma bond and a less stable pi bond. The total bond energy of a carbon–carbon double bond is about 611 kJ/mol (146 kcal/mol), compared with the single-bond energy of about 347 kJ/mol (83 kcal/mol). From these energies, we can calculate the approximate energy of a pi bond:

double-bond dissociation energy	611 kJ/mol	(146 kcal/mol)
subtract sigma bond dissociation energy	(−)347 kJ/mol	(−)(83 kcal/mol)
pi bond dissociation energy	264 kJ/mol	(63 kcal/mol)

This value of 264 kJ/mol is much less than the sigma bond energy of 347 kJ/mol, showing that pi bonds should be more reactive than sigma bonds.

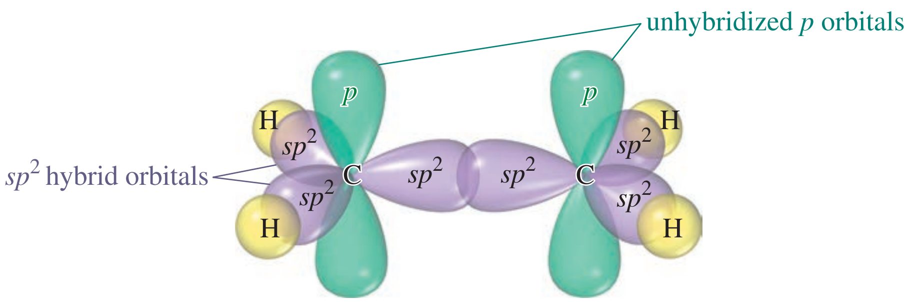
Because a carbon–carbon double bond is relatively reactive, it is considered to be a *functional group*, and alkenes are characterized by the reactions of their double bonds. In Chapter 6, we considered substitution reactions of alkyl halides in depth and briefly showed that elimination reactions can lead to alkenes. Here in Chapter 7, we study alkenes in more detail, with particular emphasis on the mechanisms of elimination reactions.

7-2 The Orbital Description of the Alkene Double Bond

In a Lewis structure, the double bond of an alkene is represented by two pairs of electrons between the carbon atoms. The Pauli exclusion principle tells us that two pairs of electrons can go into the region of space between the carbon nuclei only if each pair has its own molecular orbital. Using ethylene as an example, let's consider how the electrons are distributed in the double bond.

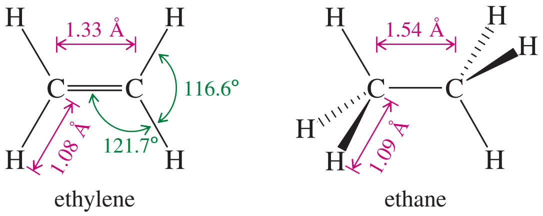
7-2A The Sigma Bond Framework

In Section 1-15, we saw how we can visualize the sigma bonds of organic molecules using hybrid atomic orbitals. In ethylene, each carbon atom is bonded to three other atoms (one carbon and two hydrogens), and there are no nonbonding electrons. Three hybrid orbitals are needed, implying sp^2 hybridization. Recall from Section 1-15 that sp^2 hybridization corresponds to bond angles of about 120°, giving optimum separation of the three atoms bonded to the carbon atom.



sigma bonding orbitals of ethylene

Each of the carbon–hydrogen sigma bonds is formed by overlap of an sp^2 hybrid orbital on carbon with the 1*s* orbital of a hydrogen atom. The C—H bond length in ethylene (1.08 Å) is slightly shorter than the C—H bond in ethane (1.09 Å) because the sp^2 orbital in ethylene has more *s* character (one-third *s*) than an sp^3 orbital (one-fourth *s*). The *s* orbital is closer to the nucleus than the *p* orbital, contributing to shorter bonds.



The remaining sp^2 orbitals overlap in the region between the carbon nuclei, providing a bonding orbital. The pair of electrons in this bonding orbital forms one of the bonds between the carbon atoms. This bond is a sigma bond because its electron density is centered along the line joining the nuclei. The C═C bond in ethylene (1.33 Å) is much shorter than the C—C bond (1.54 Å) in ethane, partly because the sigma bond of ethylene is formed from sp^2 orbitals (with more *s* character) and partly because two bonds draw the atoms closer together.

7-2B The Pi Bond

Two more electrons must go into the carbon–carbon bonding region to form the double bond in ethylene. Each carbon atom still has an unhybridized *p* orbital, and these overlap to form a pi-bonding molecular orbital. The two electrons in this orbital form the second bond between the double-bonded carbon atoms. For pi overlap to occur, these *p* orbitals must be parallel, which requires that the two carbon atoms be oriented with all their C—H bonds in a single plane (Figure 7-1). Half of the pi-bonding orbital is above the C—C sigma bond, and the other half is below the sigma bond. The pi-bonding electrons give rise to two regions of high electron density (red) in the electrostatic potential map of ethylene shown in Figure 7-1.

Figure 7-2 shows that the two ends of the ethylene molecule cannot be twisted with respect to each other without disrupting the pi bond. Unlike single bonds, a carbon–carbon double bond does not permit rotation. Six atoms, including the double-bonded carbon atoms and the four atoms bonded to them, must remain in the same plane. This is the origin of cis-trans isomerism. If two groups are on the same side of a double bond (cis), they cannot rotate to opposite sides (trans) without breaking the pi bond. Figure 7-2 shows the two distinct isomers of but-2-ene: *cis*-but-2-ene and *trans*-but-2-ene.

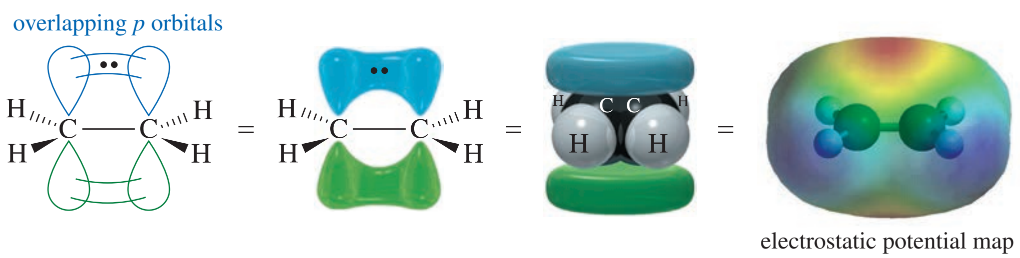


FIGURE 7-1 Parallel *p* orbitals in ethylene. The pi bond in ethylene is formed by overlap of the unhybridized *p* orbitals on the sp^2 hybrid carbon atoms. This overlap requires the two ends of the molecule to be coplanar.

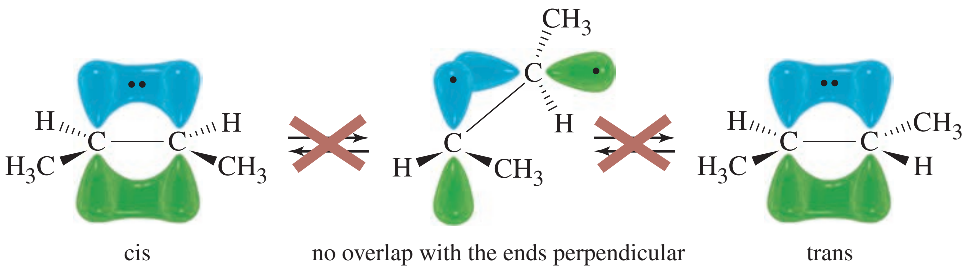


FIGURE 7-2 Distinct isomers resulting from C—C double bonds. The two isomers of but-2-ene cannot interconvert by rotation about the carbon–carbon double bond without breaking the pi bond.

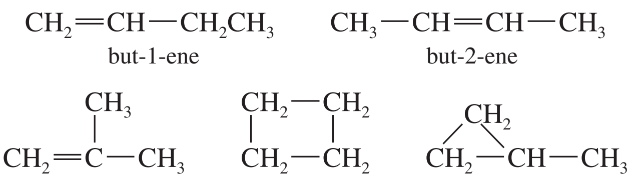
7-3 Elements of Unsaturation

7-3A Elements of Unsaturation in Hydrocarbons

Alkenes are said to be **unsaturated** because they are capable of adding hydrogen in the presence of a catalyst. The product, an alkane, is called **saturated** because it cannot react with any more hydrogen. The presence of a pi bond of an alkene (or an alkyne) or the ring of a cyclic compound decreases the number of hydrogen atoms in a molecular formula. These structural features are called **elements of unsaturation.** Each element of unsaturation corresponds to two fewer hydrogen atoms than in the "saturated" formula.

propane, C_3H_8 — saturated; propene, C_3H_6 — one element of unsaturation; cyclopropane, C_3H_6 — one element of unsaturation; propyne, C_3H_4 — two elements of unsaturation

Consider, for example, the formula C_4H_8. A saturated alkane would have a $C_nH_{(2n+2)}$ formula, or C_4H_{10}. The formula C_4H_8 is missing two hydrogen atoms, so it has one element of unsaturation, either a pi bond or a ring. There are five constitutional isomers of formula C_4H_8:



but-1-ene; but-2-ene; isobutylene; cyclobutane; methylcyclopropane

> **PROBLEM-SOLVING HINT**
> *Elements of Unsaturation*, *Degree of Unsaturation*, *Unsaturation Number*, and *Index of Hydrogen Deficiency* are all equivalent terms.

When you need a structure for a particular molecular formula, it helps to find the number of elements of unsaturation. Calculate the maximum number of hydrogen atoms from the saturated formula, $C_nH_{(2n+2)}$, and see how many are missing. The number of elements of unsaturation is simply half the number of missing hydrogens. This simple calculation allows you to consider possible structures quickly, without having to check for the correct molecular formula.

PROBLEM 7-1

(a) If a hydrocarbon has eight carbon atoms, three double bonds, and one ring, how many hydrogen atoms must it have?
(b) Calculate the number of elements of unsaturation implied by the molecular formula C_5H_{10}.
(c) Give five examples of structures with this formula (C_5H_{10}). At least one should contain a ring, and at least one should contain a double bond.

> **PROBLEM-SOLVING HINT**
> If you prefer to use a formula, elements of unsaturation
> $$= \frac{1}{2}(2C + 2 - H)$$
> C = number of carbons
> H = number of hydrogens

PROBLEM 7-2

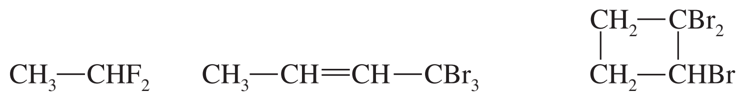
Determine the number of elements of unsaturation in the molecular formula C_3H_4. Give all three possible structures having this formula. Remember that

a double bond = one element of unsaturation
a ring = one element of unsaturation
a triple bond = two elements of unsaturation

7-3B Elements of Unsaturation with Heteroatoms

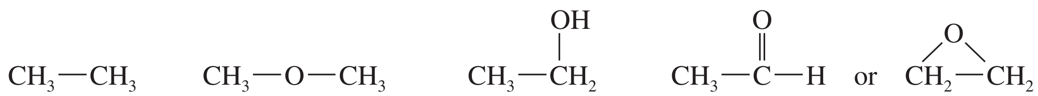
Heteroatoms (*hetero*, "different") are any atoms other than carbon and hydrogen. The rule for calculating elements of unsaturation in hydrocarbons can be extended to include heteroatoms. Let's consider how the addition of a heteroatom affects the number of hydrogen atoms in the formula.

Halogens Halogens simply substitute for hydrogen atoms in the molecular formula. The formula C_2H_6 is saturated, so the formula $C_2H_4F_2$ is also saturated. C_4H_8 has one element of unsaturation, and $C_4H_5Br_3$ also has one element of unsaturation. In calculating the number of elements of unsaturation, simply *count halogen atoms as hydrogen atoms*.



$C_2H_4F_2$ saturated | $C_4H_5Br_3$ one element of unsaturation | $C_4H_5Br_3$ one element of unsaturation

Oxygen An oxygen atom can be added to the chain (or added to a C—H bond to make a C—OH group) without changing the number of hydrogen atoms or carbon atoms. In calculating the number of elements of unsaturation, *ignore the number of oxygen atoms*.



C_2H_6, saturated | C_2H_6O, saturated | C_2H_6O, saturated | C_2H_4O, one element of unsaturation

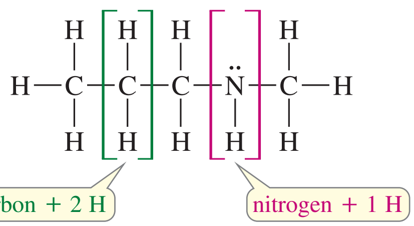
PROBLEM-SOLVING HINT

If you prefer to use a formula, the number of elements of unsaturation

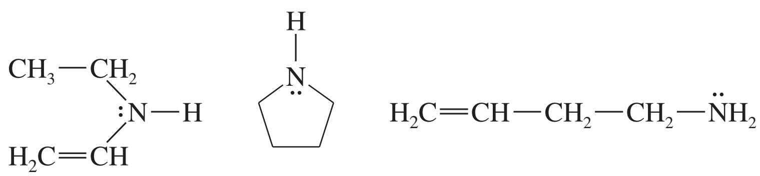
$$= \frac{1}{2}[2C + 2 + N - (H + Hal)]$$

C = number of carbons
H = number of hydrogens
N = number of nitrogens
Hal = number of halogens

Nitrogen A nitrogen atom can take the place of a carbon atom in the chain, but nitrogen is trivalent. A nitrogen atom has only one additional hydrogen atom, compared with two hydrogens for each additional carbon atom. In computing the elements of unsaturation, *count nitrogen as half a carbon atom*.



The formula C_4H_9N is like a formula with $4\frac{1}{2}$ carbon atoms, with saturated formula $C_{4.5}H_{9+2}$. The formula C_4H_9N has one element of unsaturation, because it is two hydrogen atoms short of the saturated formula.



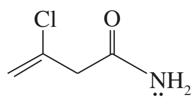
examples of formula C_4H_9N, one element of unsaturation

SOLVED PROBLEM 7-1

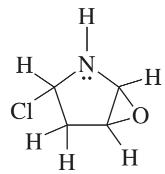
Draw at least four compounds of formula C_4H_6NOCl.

SOLUTION

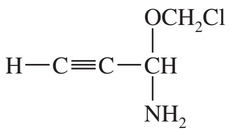
Counting the nitrogen as $\frac{1}{2}$ carbon, ignoring the oxygen, and counting chlorine as a hydrogen shows that the formula is equivalent to $C_{4.5}H_7$. The saturated formula for 4.5 carbon atoms is $C_{4.5}H_{11}$, so C_4H_6NOCl has two elements of unsaturation. These could be two double bonds, two rings, one triple bond, or a ring and a double bond. There are many possibilities, four of which are listed here.



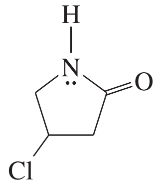
two double bonds



two rings



one triple bond



one ring,
one double bond

PROBLEM-SOLVING HINT

In figuring elements of unsaturation:

- Count halogens as hydrogens.
- Ignore oxygen.
- Count nitrogen as half a carbon.

PROBLEM 7-3

Draw five more compounds of formula C_4H_6NOCl.

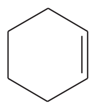
PROBLEM 7-4

For each of the following molecular formulas, determine the number of elements of unsaturation, and draw three examples.

(a) $C_4H_4Cl_2$ **(b)** C_4H_8O **(c)** $C_6H_8O_2$ **(d)** $C_5H_5NO_2$ **(e)** C_6H_3NClBr

7-4 Nomenclature of Alkenes

Simple alkenes are named much like alkanes, using the root name of the longest chain containing the double bond. The ending is changed from *-ane* to *-ene*. For example, "ethane" becomes "ethene," "propane" becomes "propene," and "cyclohexane" becomes "cyclohexene."

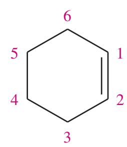
	$CH_2{=}CH_2$	$CH_2{=}CH{-}CH_3$	
IUPAC names:	ethene	propene	cyclohexene
Common names:	ethylene	propylene	

PROBLEM-SOLVING HINT

The nomenclature of alkenes is reviewed in Appendix 5, the Summary of Organic Nomenclature.

When the chain contains more than three carbon atoms, a number is used to give the location of the double bond. The chain is numbered starting from the end closest to the double bond, and the double bond is given the *lower* number of its two double-bonded carbon atoms. Cycloalkenes are assumed to have the double bond in the number 1 position.

	$\overset{1}{CH_2}{=}\overset{2}{CH}{-}\overset{3}{CH_2}{-}\overset{4}{CH_3}$	$\overset{1}{CH_2}{=}\overset{2}{CH}{-}\overset{3}{CH_2}{-}\overset{4}{CH_2}{-}\overset{5}{CH_3}$
old IUPAC names:	1-butene	1-pentene
new IUPAC names:	but-1-ene	pent-1-ene



cyclohexene

	$\overset{1}{CH_3}{-}\overset{2}{CH}{=}\overset{3}{CH}{-}\overset{4}{CH_3}$	$\overset{1}{CH_3}{-}\overset{2}{CH}{=}\overset{3}{CH}{-}\overset{4}{CH_2}{-}\overset{5}{CH_3}$
old IUPAC names:	2-butene	2-pentene
new IUPAC names:	but-2-ene	pent-2-ene

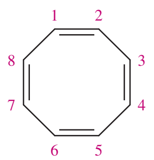
Old and New IUPAC Names

In 1993, the IUPAC recommended a logical change in the positions of the numbers used in names. Instead of placing the numbers before the root name (1-butene), it recommended placing them immediately before the part of the name they locate (but-1-ene). The new placement is helpful for clarifying the names of compounds containing multiple functional groups. You should be prepared to recognize names using either placement of the numbers, because both are widely used. In this section, names using the old number placement are printed in blue, and those using the new number placement are printed in green. Throughout this book, we will generally use the new number placement.

A compound with two double bonds is a **diene.** A **triene** has three double bonds, and a **tetraene** has four. Numbers are used to specify the locations of the double bonds.

$\overset{1}{C}H_2{=}\overset{2}{C}H{-}\overset{3}{C}H{=}\overset{4}{C}H_2$

$\overset{7}{C}H_3{-}\overset{6}{C}H{=}\overset{5}{C}H{-}\overset{4}{C}H{=}\overset{3}{C}H{-}\overset{2}{C}H{=}\overset{1}{C}H_2$



old IUPAC names:	1,3-butadiene	1,3,5-heptatriene	1,3,5,7-cyclooctatetraene
new IUPAC names:	buta-1,3-diene	hepta-1,3,5-triene	cycloocta-1,3,5,7-tetraene

Each alkyl group attached to the main chain is listed with a number to give its location. Note that the double bond is still given preference in numbering, however.

$\overset{4}{C}H_3{-}\overset{3}{C}H{=}\overset{2}{C}(CH_3){-}\overset{1}{C}H_3$

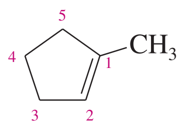
2-methyl-2-butene
2-methylbut-2-ene

$\overset{4}{C}H_3{-}\overset{3}{C}H(CH_3){-}\overset{2}{C}H{=}\overset{1}{C}H_2$

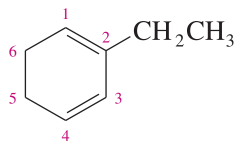
3-methyl-1-butene
3-methylbut-1-ene

$\overset{1}{C}H_3{-}\overset{2}{C}H{=}\overset{3}{C}(CH_3){-}\overset{4}{C}H_2{-}\overset{5}{C}H_2{-}\overset{6}{C}H(CH_3){-}\overset{7}{C}H_3$

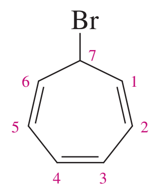
3,6-dimethyl-2-heptene
3,6-dimethylhept-2-ene



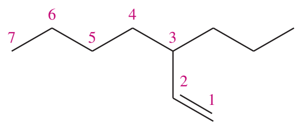
1-methylcyclopentene
1-methylcyclopentene



2-ethyl-1,3-cyclohexadiene
2-ethylcyclohexa-1,3-diene



7-bromo-1,3,5-cycloheptatriene
7-bromocyclohepta-1,3,5-triene

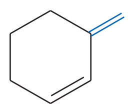


3-propyl-1-heptene
3-propylhept-1-ene

Alkenes as Substituents When a double bond is not part of the main chain, we name the group containing the double bond as a substituent called an *alkenyl group*. Alkenyl groups can be named systematically (ethenyl, propenyl, etc.) or by common names. Common alkenyl substituents are the vinyl, allyl, methylene, and phenyl groups. The phenyl group (Ph) is different from the others because it is aromatic (see Chapter 16) and does not undergo the typical reactions of alkenes.

$\wr{=}CH_2$
methylene group
(methylidene group)

$\wr{-}CH{=}CH_2$
vinyl group
(ethenyl group)

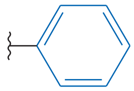


3-methylenecyclohexene

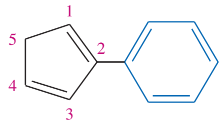
$CH_2{=}CHCH(CH{=}CH_2)CH_2CH{=}CH_2$

3-vinyl-1,5-hexadiene
3-vinylhexa-1,5-diene

$\wr\!-CH_2-CH{=}CH_2$
allyl group
2-propenyl group


phenyl group
(Ph)

$CH_2{=}CH-CH_2-Cl$
allyl chloride
3-chloropropene

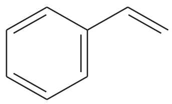

2-phenyl-1,3-cyclopentadiene
2-phenylcyclopenta-1,3-diene

Common Names Most alkenes are conveniently named by the IUPAC system, but common names are sometimes used for the simplest compounds.

	$CH_2{=}CH_2$	$CH_2{=}CH-CH_3$
IUPAC names:	ethene	propene
common name:	ethylene	propylene

$CH_2{=}C(CH_3)-CH_3$
2-methylpropene
isobutylene

$CH_2{=}C(CH_3)-CH{=}CH_2$
2-methylbuta-1,3-diene
isoprene


ethenylbenzene
styrene
(vinylbenzene)

PROBLEM 7-5

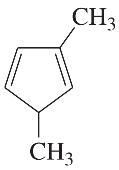
Give the systematic (IUPAC) names of the following alkenes.

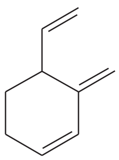
(a) $CH_2{=}CH-CH_2-CH(CH_3)_2$

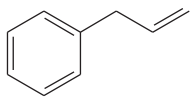
(b) $CH_3(CH_2)_3-C({=}CH_2)-CH_2CH_3$

(c) $CH_2{=}CH-CH_2-CH{=}CH_2$

(d) $CH_2{=}C{=}CH-CH{=}CH_2$

(e)


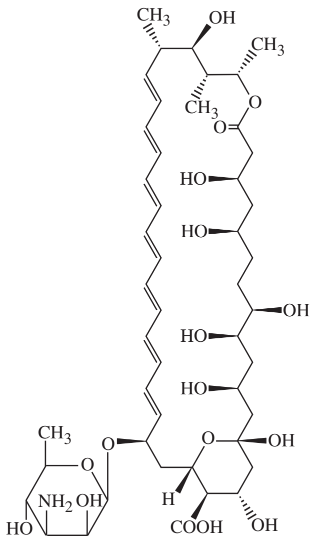
(f)


(g)


Application: Antifungal Drugs

The polyene antifungals are a group of drugs with a nonpolar region consisting of four to seven sets of alternating single and double bonds. They insert themselves in the cell membranes of fungi, causing disruption and leakiness that results in fungal cell death.

The best-known polyene antifungal drug is Amphotericin B, whose structure is shown below.


Amphotericin B

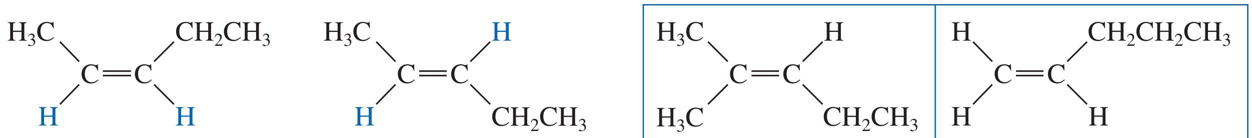
7-5 Nomenclature of Cis-Trans Isomers

Depending on their substituents, the double-bonded carbon atoms of an alkene may be stereogenic, giving rise to cis-trans stereoisomers. In that case, a complete name should also specify which of the possible stereoisomers is intended.

7-5A Cis-Trans Nomenclature

In Chapters 2 and 5, we saw how the rigidity and lack of rotation of carbon–carbon double bonds give rise to **cis-trans isomerism,** also called **geometric isomerism.** A double bond that gives rise to cis-trans stereoisomerism is called **stereogenic,** and it has two possible configurations. If two of the same groups (often hydrogen) are bonded to the same side of the carbons of the double bond, the alkene is the **cis** isomer. If the same groups are on opposite sides of the double bond, the alkene is **trans.**

Not all alkenes are capable of showing cis-trans isomerism. If either carbon of the double bond holds two identical groups, the molecule cannot have cis and trans forms.

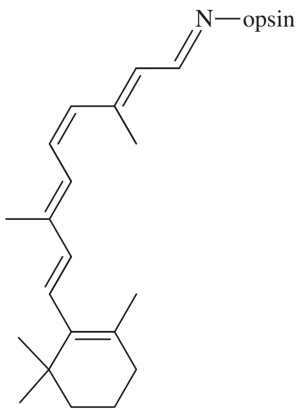


cis-2-pentene *cis*-pent-2-ene | *trans*-2-pentene *trans*-pent-2-ene | 2-methyl-2-pentene 2-methylpent-2-ene | 1-pentene pent-1-ene

(neither cis nor trans)

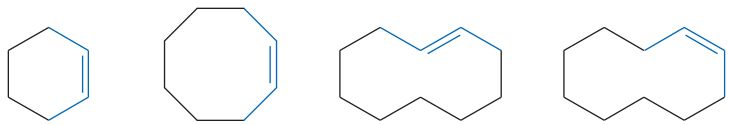
Application: Biochemistry

A double bond in rhodopsin, a visual pigment found in your eyes that enables you to see at night, is converted from the cis isomer to the trans isomer when light strikes the eye. As a result, a nerve impulse travels to the brain and you see the source of the light.



cis-rhodopsin

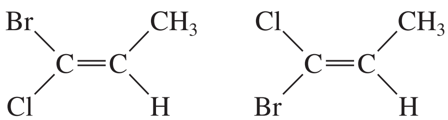
Trans cycloalkenes are unstable unless the ring is large enough (at least eight carbon atoms) to accommodate the trans double bond (Section 7-8D). Therefore, all cycloalkenes are assumed to be cis unless they are specifically named trans. The cis name is rarely used with cycloalkenes, except to distinguish a large cycloalkene from its trans isomer.



cyclohexene cyclooctene *trans*-cyclodecene *cis*-cyclodecene

7-5B *E-Z* Nomenclature

The cis-trans nomenclature for geometric isomers sometimes gives an ambiguous name. For example, the isomers of 1-bromo-1-chloropropene are not clearly cis or trans because it is not obvious which substituents are referred to as being cis or trans.



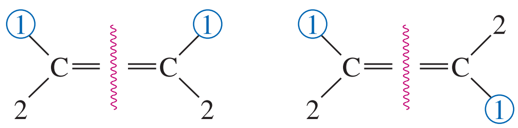
geometric isomers of 1-bromo-1-chloropropene

To deal with this problem, we use the ***E-Z* system** of nomenclature (pun intended) for stereogenic double bonds. The *E-Z* system is patterned after the Cahn–Ingold–Prelog convention for asymmetric carbon atoms (Section 5-3). It assigns a unique configuration of either *E* or *Z* to any stereogenic double bond.

To name an alkene by the *E-Z* system, mentally separate the double bond into its two ends. Recall how you used the Cahn–Ingold–Prelog rules (page 252) to assign relative priorities to groups on an asymmetric carbon atom so that you could name it (*R*) or (*S*). Consider each end of the double bond separately, and use those same rules to assign first and second priorities to the two substituent groups on that end. Do the same for the other end of the double bond. If the two first-priority atoms are *together (cis)* on the same side of the double bond, you have the ***Z*** isomer, from the German word *zusammen*, "together." If the two first-priority atoms are on *opposite (trans)* sides of the double bond, you have the ***E*** isomer, from the German word *entgegen*, "opposite."

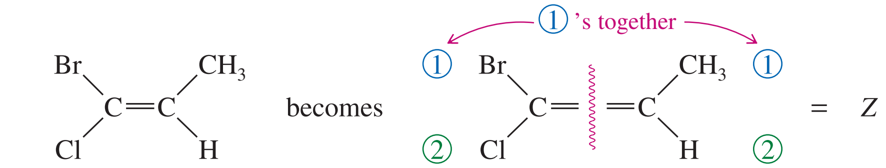
PROBLEM-SOLVING HINT

Those who are unfamiliar with German terms can remember that *Z* = "zame zide."



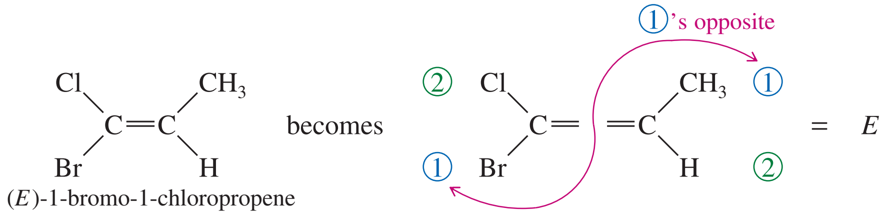
*Z*usammen (1s on same side) | *E*ntgegen (1s on opposite sides)

For example,



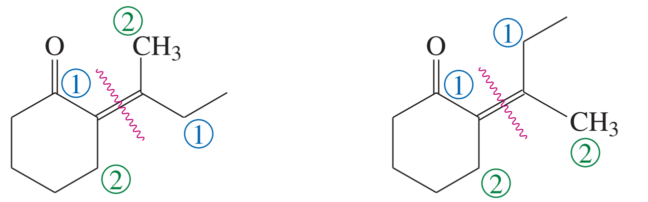
(*Z*)-1-bromo-1-chloropropene

The other isomer is named similarly:



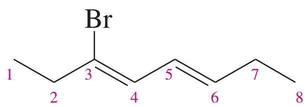
(*E*)-1-bromo-1-chloropropene

The following example shows the use of the *E-Z* nomenclature with cyclic stereoisomers that are not clearly cis or trans.



(*E*) isomer (*Z*) isomer

If the alkene has more than one double bond, the stereochemistry about each double bond should be specified. The following compound is properly named (3*Z*,5*E*)-3-bromoocta-3,5-diene:



(3*Z*,5*E*)-3-bromoocta-3,5-diene

The use of *E-Z* names (rather than cis and trans) is always an option, but it is required whenever a double bond is not clearly cis or trans. Most trisubstituted and tetrasubstituted double bonds are more clearly named *E* or *Z* rather than cis or trans.

PROBLEM-SOLVING HINT

For a compound with multiple double bonds, the maximum possible number of cis-trans stereoisomers is given by 2^n, where n = the number of stereogenic double bonds. Fewer than 2^n stereoisomers will exist if the molecule has symmetry that leads to duplications among the possible isomers. (See Problem 7-9.)

Robert Sidney Cahn (1899–1981) was a British chemist known for his contributions to stereochemistry and the nomenclature of organic molecules, in particular in the Cahn–Ingold–Prelog priority rules. He proposed these rules in 1956 in collaboration with Christopher Kelk Ingold and Vladimir Prelog, and they are still widely used today.

Sir Christopher Kelk Ingold (1893–1970) was also a British chemist. His pioneering work in reaction mechanisms and the electronic structure of organic compounds led to the introduction of key concepts such as nucleophile and electrophile and inductive and resonance effects in mainstream chemistry. For his groundbreaking work in the field of physical organic chemistry, Sir Ingold was awarded the Faraday Lectureship Prize (1961).

SUMMARY Rules for Naming Alkenes

The following rules summarize the IUPAC system for naming alkenes:

1. Select the longest chain or largest ring that contains the *largest possible number of double bonds*, and name it with the *-ene* suffix. If there are two double bonds, the suffix is *-diene*; for three, *-triene*; for four, *-tetraene*; etc.
2. Number the chain from the end closest to the double bond(s). Number a ring so that the double bond is between carbons 1 and 2. Place the numbers giving the locations of the double bonds in front of the root name (old system) or in front of the suffix *-ene, -diene*, etc. (new system).
3. Name substituent groups as in alkanes, indicating their locations by the number of the main-chain carbon to which they are attached. The ethenyl group and the propenyl group are usually called the *vinyl* group and the *allyl* group, respectively.
4. For compounds that show geometric isomerism, add the appropriate prefix: *cis-* or *trans-*, or *E-* or *Z-*. Cycloalkenes are assumed to be cis unless named otherwise.

Vladimir Prelog (1906–1998) was a Croatian-Swiss chemist who was a pioneer in the field of organic chemistry. He is renowned for the first synthesis of the molecule adamantane in 1941. For his groundbreaking work on the stereochemistry of organic molecules and reactions, he was awarded the Nobel Prize in chemistry (1975).

PROBLEM-SOLVING HINT

To see whether a compound can have cis and trans isomers, draw the structure, and then draw it again with the groups on one end of the double bond reversed. See if you can describe a difference between the two.

PROBLEM 7-6

1. Determine which of the following compounds show cis-trans isomerism.
2. Draw and name the cis and trans (or *Z* and *E*) isomers of those that do.

(a) hex-3-ene **(b)** buta-1,3-diene **(c)** hexa-2,4-diene
(d) 3-methylpent-2-ene **(e)** 2,3-dimethylpent-2-ene **(f)** 3,4-dibromocyclopentene

PROBLEM 7-7

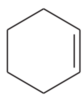
The following names are all incorrect. Draw the structure represented by the incorrect name (or a consistent structure if the name is ambiguous), and give your drawing the correct name.

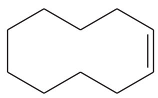
(a) *cis*-dimethylpent-2-ene **(b)** 3-vinylhex-4-ene
(c) 2-methylcyclopentene **(d)** 6-chlorocyclohexadiene
(e) 2,5-dimethylcyclohexene **(f)** *cis*-2,5-dibromo-3-ethylpent-2-ene

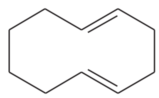
PROBLEM 7-8

Some of the following examples can show geometric isomerism, and some cannot. For the ones that can, draw all the geometric isomers, and assign complete names using the *E-Z* system.

(a) 3-bromo-2-chloropent-2-ene **(b)** 3-ethylhexa-2,4-diene
(c) 3-bromo-2-methylhex-3-ene **(d)** penta-1,3-diene
(e) 3-ethyl-5-methyloct-3-ene **(f)** 3,7-dichloroocta-2,5-diene

(g) 
cyclohexene

(h) 
cyclodecene

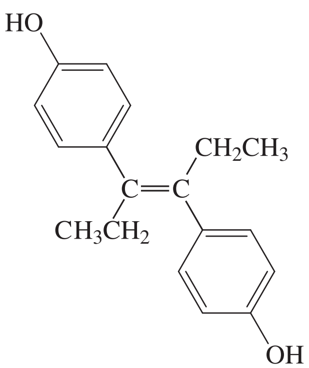
(i) 
cyclodeca-1,5-diene

PROBLEM 7-9*

(a) How many stereogenic double bonds are in octa-1,3,5-triene? How many stereocenters are there? Draw and name the four stereoisomers of octa-1,3,5-triene.
(b) Draw the six stereoisomers of octa-2,4,6-triene. Explain why there are only six stereoisomers, rather than the eight we might expect for a compound with three stereogenic double bonds.

Application: Drug Side Effects

A derivative of *trans*-stilbene known as diethylstilbestrol, or DES, was once taken by women during pregnancy to prevent miscarriages. The use of DES was discontinued because studies showed that DES increases the risk of cervical cancer in the children of women who take it.



diethylstilbestrol (DES)

7-6 Commercial Importance of Alkenes

Because the carbon–carbon double bond is readily converted to other functional groups, alkenes are important intermediates in the synthesis of plastics, drugs, pesticides, and other valuable chemicals.

Ethylene is the organic compound produced in the largest volume, at over 300 billion pounds per year worldwide. Most of this ethylene is polymerized to form about 160 billion pounds of polyethylene per year. The remainder is used to synthesize a wide variety of organic chemicals including ethanol, acetic acid, ethylene glycol, and vinyl chloride (Figure 7-3). Ethylene also serves as a plant hormone, accelerating the ripening of fruit. For example, tomatoes are harvested and shipped while green, then treated with ethylene to make them ripen and turn red just before they are placed on display.

Propylene is produced at the rate of about 160 billion pounds per year worldwide, with most of that going to make about 100 billion pounds of polypropylene. The rest is used to make propylene glycol, acetone, isopropyl alcohol, and a variety of other useful organic chemicals (Figure 7-4).

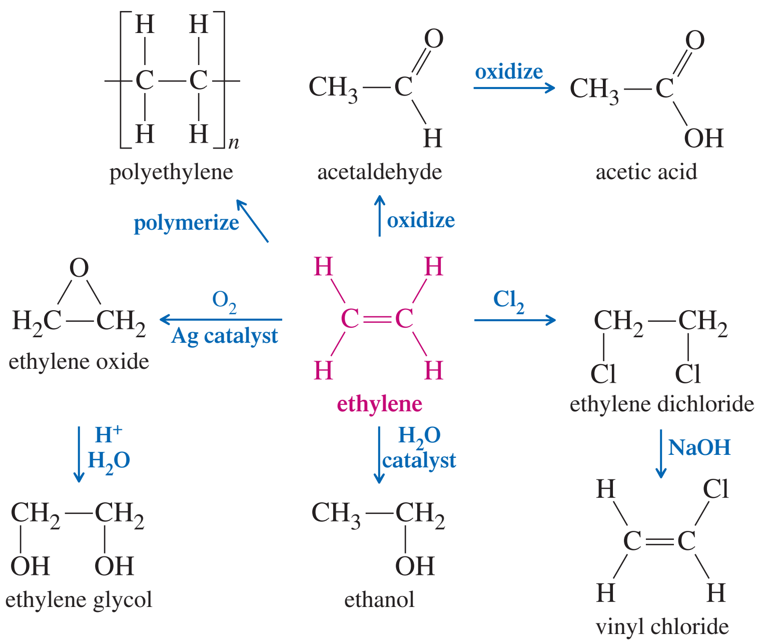


FIGURE 7-3 Uses of ethylene. Ethylene is the largest-volume industrial organic chemical. Most of the ethylene produced each year is polymerized to make polyethylene. Much of the rest is used to produce a variety of useful two-carbon compounds.

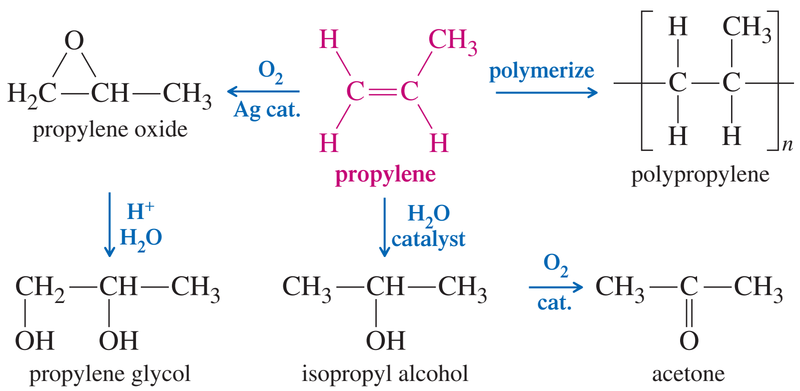


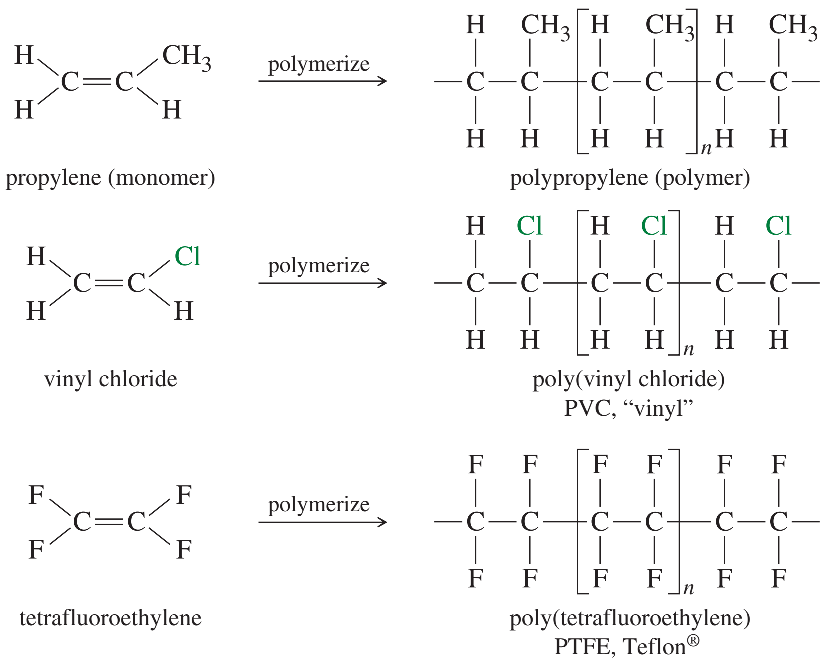
FIGURE 7-4 Uses of propylene. Most propylene is polymerized to make polypropylene. It is also used to make several important three-carbon compounds.

Many common polymers are made by polymerizing alkenes. These polymers are used in consumer products from shoes to plastic bags to car bumpers. A **polymer** (Greek, *poly*, "many," and *meros*, "parts") is a large molecule made up of many **monomer** (Greek, *mono*, "one") molecules. An alkene monomer can **polymerize** by a chain reaction where additional alkene molecules add to the end of the growing polymer chain. Because these polymers result from addition of many individual alkene units, they are called **addition polymers. Polyolefins** are polymers made from monofunctional (single functional group) alkenes such as ethylene and propylene. Figure 7-5 shows some addition polymers made from simple alkenes and haloalkenes. We discuss polymerization reactions in more detail in Chapters 8 and 26.

PROBLEM 7-10

Teflon-coated frying pans routinely endure temperatures that would cause polyethylene or polypropylene to oxidize and decompose. Decomposition of polyethylene is initiated by free-radical abstraction of a hydrogen atom by O_2. Bond-dissociation energies of C—H bonds are about 400 kJ/mol, and C—F bonds are about 460 kJ/mol. The BDE of the H—OO bond is about 192 kJ/mol, and the F—OO bond is about 63 kJ/mol. Show why Teflon (Figure 7-5) is much more resistant to oxidation than polyethylene is.

FIGURE 7-5 Addition polymers. Alkenes polymerize in this way to form many common plastics. Uses of these materials are listed in Table 26-1.



7-7 Physical Properties of Alkenes

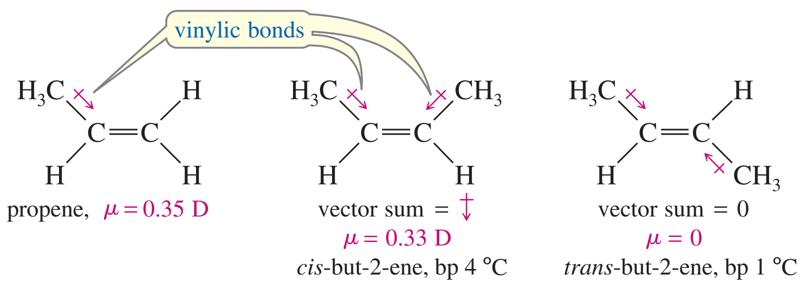
7-7A Boiling Points and Densities

Most physical properties of alkenes are similar to those of the corresponding alkanes. For example, the boiling points of but-1-ene, *cis*-but-2-ene, *trans*-but-2-ene, and *n*-butane are all close to 0 °C. Also like the alkanes, alkenes have densities around 0.6 or 0.7 g/cm^3. The boiling points and densities of some representative alkenes are listed in Table 7-1. The table shows that boiling points of alkenes increase smoothly with molecular weight. As with alkanes, increased branching leads to greater volatility and lower boiling points. For example, 2-methylpropene (isobutylene) has a boiling point of −7 °C, which is lower than the boiling point of any of the unbranched butenes.

7-7B Polarity

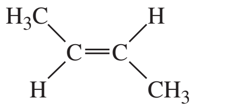
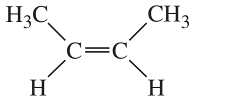
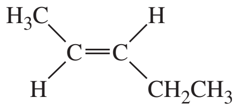
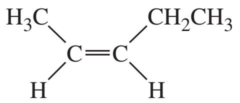
Like alkanes, alkenes are relatively nonpolar. They are insoluble in water but soluble in nonpolar solvents such as hexane, gasoline, halogenated solvents, and ethers. Alkenes tend to be slightly more polar than alkanes, however, for two reasons: The more weakly held electrons in the pi bond are more polarizable (contributing to instantaneous dipole moments), and the vinylic bonds tend to be slightly polar (contributing to a permanent dipole moment).

Alkyl groups are slightly electron-donating toward a double bond, helping to stabilize it. This donation slightly polarizes the vinylic bond, with a small partial positive charge on the alkyl group and a small negative charge on the double-bond carbon atom. For example, propene has a small dipole moment of 0.35 D.



In a cis-disubstituted alkene, the vector sum of the two dipole moments is directed perpendicular to the double bond. In a trans-disubstituted alkene, the two dipole

TABLE 7-1 Physical Properties of Some Representative Alkenes

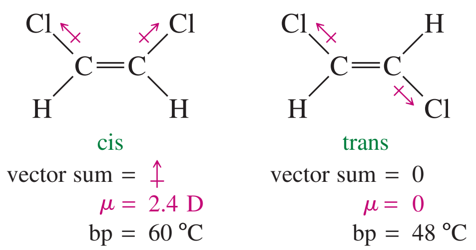
Name	Structure	Carbons	Boiling Point (°C)	Density (g/cm3)
ethene (ethylene)	$CH_2{=}CH_2$	2	−104	
propene (propylene)	$CH_3CH{=}CH_2$	3	−47	0.52
2-methylpropene (isobutylene)	$(CH_3)_2C{=}CH_2$	4	−7	0.59
but-1-ene	$CH_3CH_2CH{=}CH_2$	4	−6	0.59
trans-but-2-ene		4	1	0.60
cis-but-2-ene		4	4	0.62
3-methylbut-1-ene	$(CH_3)_2CH{-}CH{=}CH_2$	5	25	0.65
pent-1-ene	$CH_3CH_2CH_2{-}CH{=}CH_2$	5	30	0.64
trans-pent-2-ene		5	36	0.65
cis-pent-2-ene		5	37	0.66
2-methylbut-2-ene	$(CH_3)_2C{=}CH{-}CH_3$	5	39	0.66
hex-1-ene	$CH_3(CH_2)_3{-}CH{=}CH_2$	6	64	0.68
2,3-dimethylbut-2-ene	$(CH_3)_2C{=}C(CH_3)_2$	6	73	0.71
hept-1-ene	$CH_3(CH_2)_4{-}CH{=}CH_2$	7	93	0.70
oct-1-ene	$CH_3(CH_2)_5{-}CH{=}CH_2$	8	122	0.72
non-1-ene	$CH_3(CH_2)_6{-}CH{=}CH_2$	9	146	0.73
dec-1-ene	$CH_3(CH_2)_7{-}CH{=}CH_2$	10	171	0.74

moments tend to cancel out. If an alkene is symmetrically trans disubstituted, the dipole moment is zero. For example, *cis*-but-2-ene has a nonzero dipole moment, but *trans*-but-2-ene has no measurable dipole moment.

Compounds with permanent dipole moments engage in dipole–dipole attractions, while those without permanent dipole moments engage only in van der Waals attractions. *cis*-But-2-ene and *trans*-but-2-ene have similar van der Waals attractions, but only the cis isomer has dipole–dipole attractions. Because of its increased intermolecular attractions, *cis*-but-2-ene must be heated to a slightly higher temperature (4 °C versus 1 °C) before it begins to boil.

> **PROBLEM-SOLVING HINT**
>
> Boiling points are rough indicators of the strength of intermolecular forces.

The effect of bond polarity is even more apparent in the 1,2-dichloroethenes, with their strongly polar carbon–chlorine bonds. The cis isomer has a large dipole moment (2.4 D), giving it a boiling point 12 degrees higher than the trans isomer, with zero dipole moment.



PROBLEM 7-11

For each pair of compounds, predict the one with a higher boiling point. Which compounds have zero dipole moments?

(a) *cis*-1,2-dichloroethene or *cis*-1,2-dibromoethene
(b) *cis*- or *trans*-2,3-dichlorobut-2-ene
(c) cyclohexene or 1,2-dichlorocyclohexene

7-8 Stability of Alkenes

In making alkenes, we often find that the major product is the most stable alkene. Many reactions also provide opportunities for alkenes to rearrange to more stable isomers by movement of the double bonds. Therefore, we need to know how the stability of an alkene depends on its structure.

We can compare stabilities by converting different compounds to a common product and comparing the amounts of heat given off. One possibility would be to measure heats of combustion from converting alkenes to CO_2 and H_2O. However, heats of combustion are large numbers (thousands of kJ per mole), and measuring small differences in these large numbers is difficult. Instead, alkene energies are often compared by measuring the **heat of hydrogenation:** the heat given off ($\Delta H°$) during catalytic hydrogenation. Heats of hydrogenation can be measured about as easily as heats of combustion, yet they are smaller numbers and provide more accurate energy differences.

7-8A Heats of Hydrogenation

When an alkene is treated with hydrogen in the presence of a platinum catalyst, hydrogen adds to the double bond, reducing the alkene to an alkane. Hydrogenation is mildly exothermic, evolving about 110 to 136 kJ (26 to 33 kcal) of heat per mole of hydrogen consumed. Consider the hydrogenation of but-1-ene and *trans*-but-2-ene:

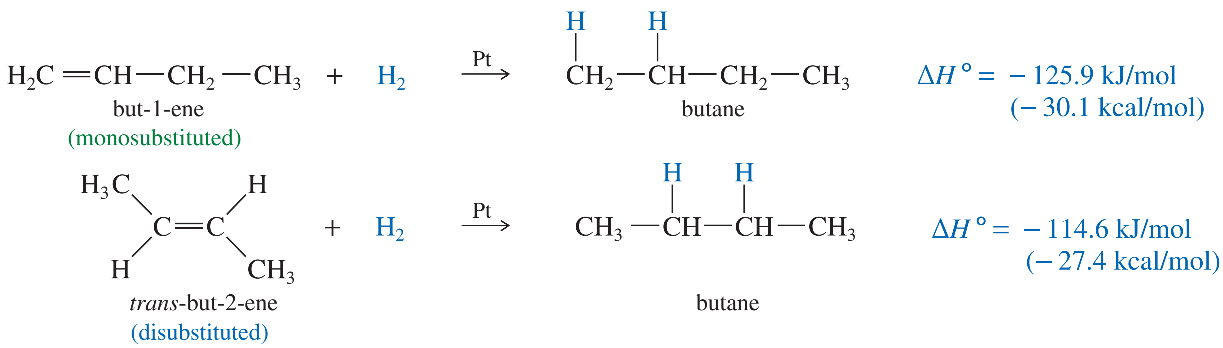


Figure 7-6 shows these heats of hydrogenation on a reaction-energy diagram. The difference in the stabilities of but-1-ene and *trans*-but-2-ene is the difference in their heats of hydrogenation. *trans*-But-2-ene is more stable by

$$125.9\ \text{kJ/mol} - 114.6\ \text{kJ/mol} = 11.3\ \text{kJ/mol}\ (2.7\ \text{kcal/mol})$$

7-8B Substitution Effects

An 11.3 kJ/mol (2.7 kcal/mol) stability difference is typical between a monosubstituted alkene (but-1-ene) and a trans-disubstituted alkene (*trans*-but-2-ene). In the following equations, we compare the monosubstituted double bond of 3-methylbut-1-ene with the trisubstituted double bond of 2-methylbut-2-ene. The trisubstituted alkene is more stable by 14.7 kJ/mol (3.5 kcal/mol).

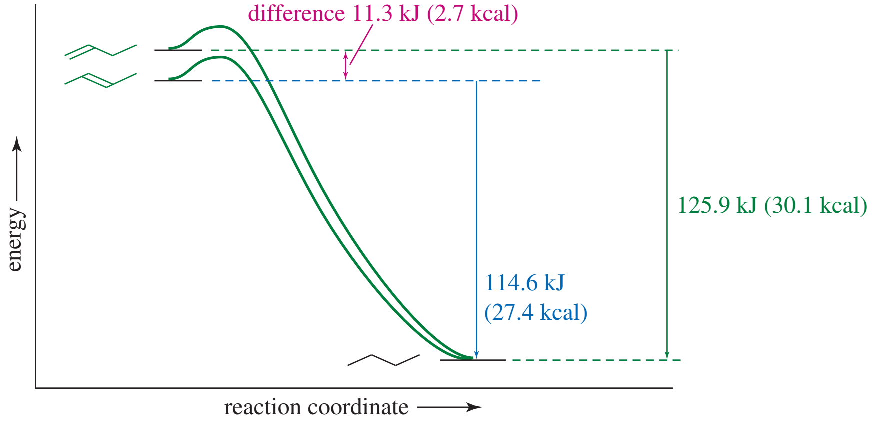


FIGURE 7-6 Relative heats of hydrogenation per mole of alkene. *trans*-But-2-ene is more stable than but-1-ene by 11.3 kJ/mol (2.7 kcal/mol).

$$\underset{\substack{\text{3-methylbut-1-ene} \\ \text{(monosubstituted)}}}{CH_2{=}CH{-}\overset{\overset{\large CH_3}{|}}{C}H{-}CH_3} \xrightarrow{H_2,\ Pt} \underset{\text{2-methylbutane}}{CH_3{-}CH_2{-}\overset{\overset{\large CH_3}{|}}{C}H{-}CH_3} \qquad \Delta H° = -126.3\text{ kJ } (-30.2\text{ kcal})$$

$$\underset{\substack{\text{2-methylbut-2-ene} \\ \text{(trisubstituted)}}}{CH_3{-}CH{=}\overset{\overset{\large CH_3}{|}}{C}{-}CH_3} \xrightarrow{H_2,\ Pt} \underset{\text{2-methylbutane}}{CH_3{-}CH_2{-}\overset{\overset{\large CH_3}{|}}{C}H{-}CH_3} \qquad \Delta H° = -111.6\text{ kJ } (-26.7\text{ kcal})$$

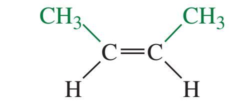
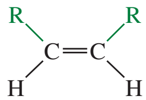
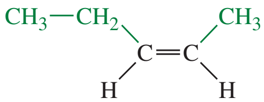
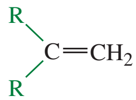
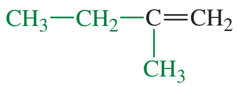
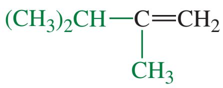
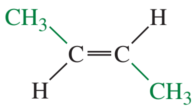
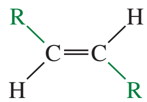
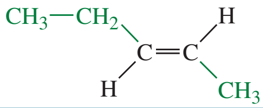
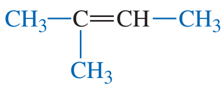
To be completely correct, we should compare heats of hydrogenation only for compounds that give the same alkane, as 3-methylbut-1-ene and 2-methylbut-2-ene do. However, most alkenes with similar substitution patterns give similar heats of hydrogenation. For example, 3,3-dimethylbut-1-ene (below) hydrogenates to give a different alkane than does 3-methylbut-1-ene or but-1-ene (above); yet these three monosubstituted alkenes have similar heats of hydrogenation because the alkanes formed have similar energies. In effect, the heat of hydrogenation is a measure of the energy content of the pi bond.

$$\underset{\substack{\text{3,3-dimethylbut-1-ene} \\ \text{(monosubstituted)}}}{CH_2{=}CH{-}\overset{\overset{\large CH_3}{|}}{\underset{\underset{\large CH_3}{|}}{C}}{-}CH_3} \xrightarrow{H_2,\ Pt} \underset{\text{2,2-dimethylbutane}}{CH_3{-}CH_2{-}\overset{\overset{\large CH_3}{|}}{\underset{\underset{\large CH_3}{|}}{C}}{-}CH_3} \qquad \Delta H° = -125.8\text{ kJ } (-30.1\text{ kcal})$$

In practice, we can use heats of hydrogenation to compare the stabilities of different alkenes as long as they hydrogenate to give alkanes of similar energies. Most acyclic alkanes and unstrained cycloalkanes have similar energies, and we can use this approximation. Table 7-2 shows the heats of hydrogenation of a variety of alkenes with different substitution. The compounds are ranked in decreasing order of their heats of hydrogenation, that is, from the least stable double bonds to the most stable. Note that the values are similar for alkenes with similar substitution patterns.

The most stable double bonds are those with the most alkyl groups attached. For example, hydrogenation of ethylene (no alkyl groups attached) evolves 136.0 kJ/mol, while but-1-ene and pent-1-ene (one alkyl group for each) give off about 126 kJ/mol. Double bonds with two alkyl groups hydrogenate to produce about 114–119 kJ/mol. Three or four alkyl substituents further stabilize the double bond, as with 2-methylbut-2-ene (trisubstituted, 111.6 kJ/mol) and 2,3-dimethylbut-2-ene (tetrasubstituted, 110.4 kJ/mol).

TABLE 7-2 Molar Heats of Hydrogenation of Alkenes

Name	Structure	Molar Heat of Hydrogenation ($-\Delta H°$) kJ	kcal	General Structure
ethene (ethylene)	$H_2C{=}CH_2$	136.0	32.5	unsubstituted
propene (propylene)	$CH_3{-}CH{=}CH_2$	123.4	29.5	monosubstituted $R{-}CH{=}CH_2$
but-1-ene	$CH_3{-}CH_2{-}CH{=}CH_2$	125.9	30.1	
pent-1-ene	$CH_3{-}CH_2{-}CH_2{-}CH{=}CH_2$	126.0	30.1	
hex-1-ene	$CH_3{-}(CH_2)_3{-}CH{=}CH_2$	125.0	29.9	
3-methylbut-1-ene	$(CH_3)_2CH{-}CH{=}CH_2$	126.3	30.2	
3,3-dimethylbut-1-ene	$(CH_3)_3C{-}CH{=}CH_2$	125.8	30.1	
cis-but-2-ene		118.5	28.3	disubstituted (cis) 
cis-pent-2-ene		117.7	28.1	
2-methylpropene (isobutylene)	$(CH_3)_2C{=}CH_2$	117.8	28.2	disubstituted (geminal) 
2-methylbut-1-ene		118.2	28.3	
2,3-dimethylbut-1-ene		116.3	27.8	
trans-but-2-ene		114.6	27.4	disubstituted (trans) 
trans-pent-2-ene		113.8	27.2	
2-methylbut-2-ene		111.6	26.7	trisubstituted $R_2C{=}CHR$
2,3-dimethylbut-2-ene	$(CH_3)_2C{=}C(CH_3)_2$	110.4	26.4	tetrasubstituted $R_2C{=}CR_2$

Note: A lower heat of hydrogenation corresponds to lower energy and greater stability of the alkene.

The values in Table 7-2 confirm **Zaitsev's rule:**

Alkenes with more highly substituted double bonds are usually more stable.

In other words, the alkyl groups attached to the double-bonded carbons stabilize the alkene.

Two factors are probably responsible for the stabilizing effect of alkyl groups on a double bond. Alkyl groups are mildly electron-donating, and they contribute electron density to the pi bond. In addition, bulky substituents such as alkyl groups are best situated as far apart as possible. In an alkane, they are separated by the tetrahedral bond angle, about 109.5°. A double bond increases this separation to about 120°. In general, alkyl groups are separated best by the most highly substituted double bond. This steric effect is illustrated in Figure 7-7 for two **double-bond isomers** (isomers that differ

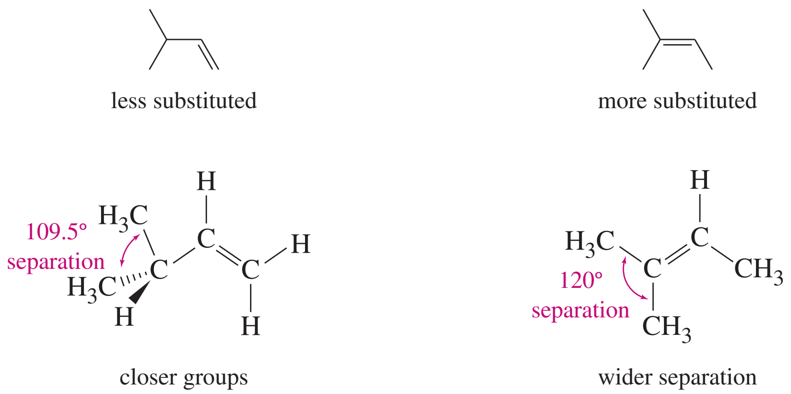


FIGURE 7-7 Bond angles in double-bond isomers. The isomer with the more substituted double bond has a larger angular separation between the bulky alkyl groups.

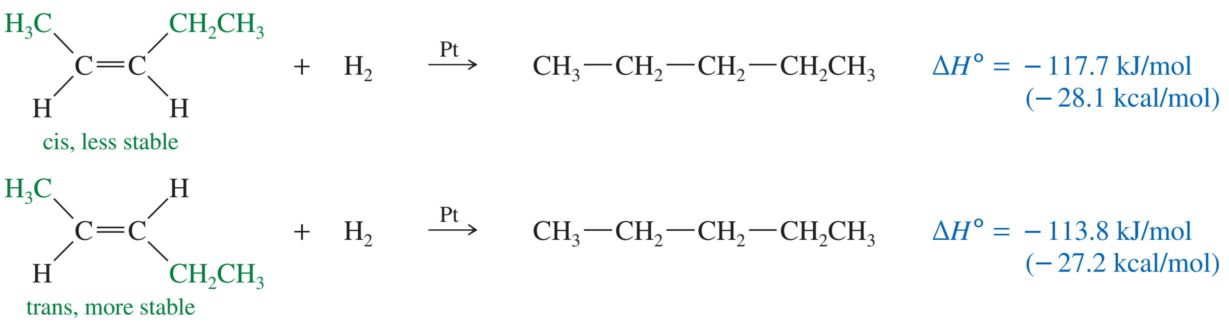
only in the position of the double bond). The isomer with the monosubstituted double bond separates the alkyl groups by only 109.5°, while the trisubstituted double bond separates them by about 120°.

> **PROBLEM 7-12**
>
> Use the data in Table 7-2 to predict the energy difference between 2,3-dimethylbut-1-ene and 2,3-dimethylbut-2-ene. Which of these double-bond isomers is more stable?

> **PROBLEM-SOLVING HINT**
>
> Heats of hydrogenation are usually exothermic. A larger amount of heat given off implies a less stable alkene, because the less stable alkene starts from a higher potential energy.

7-8C Energy Differences in cis-trans Isomers

The heats of hydrogenation in Table 7-2 show that trans isomers are generally more stable than the corresponding cis isomers. This trend seems reasonable because the alkyl substituents are separated farther in trans isomers than they are in cis isomers. The greater stability of the trans isomer is evident in the pent-2-enes, which show nearly a 4 kJ/mol (1 kcal/mol) difference between the cis and trans isomers.



A 4 kJ/mol difference between cis and trans isomers is typical for disubstituted alkenes. Figure 7-8 summarizes the relative stabilities of alkenes, comparing them with ethylene, the least stable of the simple alkenes. Geminal isomers, $CR_2{=}CH_2$, tend to fall between the cis and trans isomers in energy, but the differences are not as predictable as the cis/trans differences.

> **PROBLEM 7-13**
>
> Using Table 7-2 as a guide, predict which member of each pair is more stable, as well as by about how many kJ/mol or kcal/mol.
>
> **(a)** *cis,cis*-hexa-2,4-diene or *trans,trans*-hexa-2,4-diene
> **(b)** 2-methylbut-1-ene or 3-methylbut-1-ene
> **(c)** 2-methylbut-1-ene or 2-methylbut-2-ene
> **(d)** *cis*-4-methylpent-2-ene or 2-methylpent-2-ene

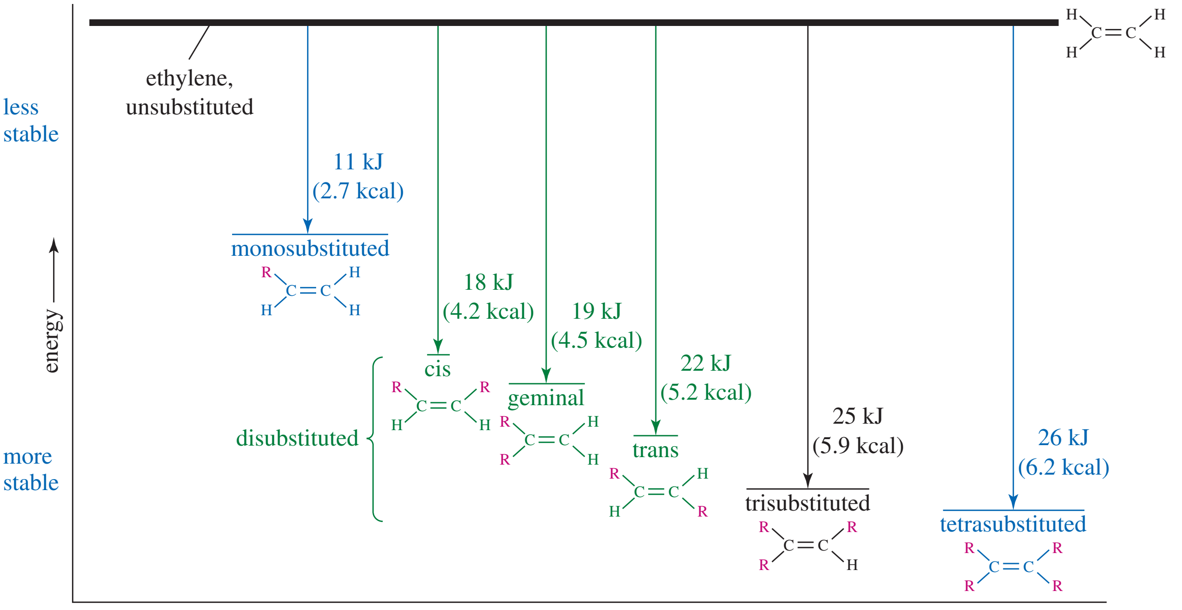
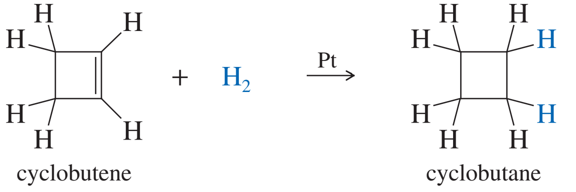


FIGURE 7-8 Relative energies of typical π bonds compared with ethylene. (The numbers are approximate.)

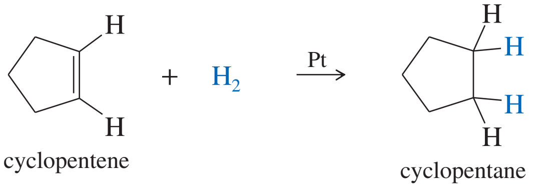
7-8D Stability of Cycloalkenes

Most cycloalkenes react like acyclic (noncyclic) alkenes. The presence of a ring makes a major difference only if there is ring strain, either because of a small ring or because of a trans double bond. Rings that are five-membered or larger can easily accommodate double bonds, and these cycloalkenes react much like straight-chain alkenes. Three- and four-membered rings show evidence of ring strain, however.

Cyclobutene Cyclobutene has a heat of hydrogenation of −129.7 kJ/mol (−31.0 kcal/mol), compared with −112.7 kJ/mol (−26.9 kcal/mol) for cyclopentene.



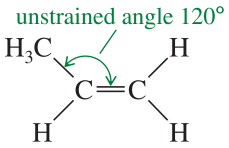
$\Delta H° = -129.7$ kJ/mol
(-31.0 kcal/mol)



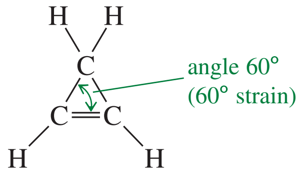
$\Delta H° = -112.7$ kJ/mol
(-26.9 kcal/mol)

The double bond in cyclobutene has about 17 kJ/mol of *extra* ring strain (in addition to the ring strain in cyclobutane) by virtue of the small ring. The 90° bond angles in cyclobutene compress the angles of the sp^2 hybrid carbons (normally 120°) more than they compress the sp^3 hybrid angles (normally 109.5°) in cyclobutane. The extra ring strain in cyclobutene makes its double bond more reactive than a typical double bond.

Cyclopropene Cyclopropene has bond angles of about 60°, compressing the bond angles of the carbon–carbon double bond to half their usual value of 120°. The double bond in cyclopropene is highly strained.

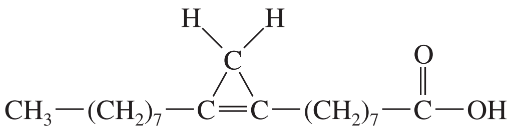


propene



cyclopropene

Many chemists once believed that a cyclopropene could never be made because it would snap open (or polymerize) immediately from the large ring strain. Cyclopropene was eventually synthesized, however, and it can be stored in the cold. Cyclopropenes were still considered to be strange, highly unusual compounds. Natural-product chemists were surprised when they found that the kernel oil of *Sterculia foelida*, a tropical tree, contains *sterculic acid*, a carboxylic acid with a cyclopropene ring.


sterculic acid

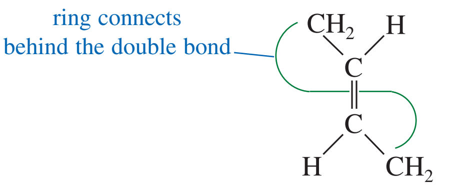
Application: Toxicology

Sterculic acid is a potent inhibitor of several enzymes that are responsible for the formation of double bonds in long-chain fatty acids. These unsaturated fatty acids are needed in the cells for energy sources and membrane components and to make other critical biological molecules.

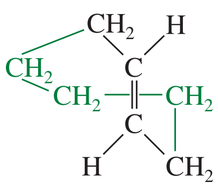
Consequently, vegetable oils containing sterculic acid must be hydrogenated or processed at high temperatures to reduce or destroy the cyclopropene ring.

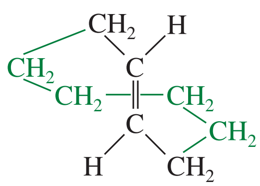
Trans Cycloalkenes Another difference between cyclic and acyclic alkenes is the relationship between cis and trans isomers. In acyclic alkenes, the trans isomers are usually more stable; but the trans isomers of small cycloalkenes are rare, and those with fewer than eight carbon atoms are unstable at room temperature. The problem with making a trans cycloalkene lies in the geometry of the trans double bond. The two alkyl groups on a trans double bond are so far apart that several carbon atoms are needed to complete the ring.

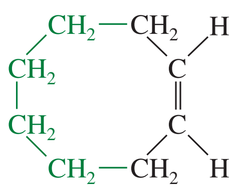
Try to make a model of *trans*-cyclohexene, being careful that the large amount of ring strain does not break your models. *trans*-Cyclohexene is too strained to be isolated, but *trans*-cycloheptene can be isolated at low temperatures. *trans*-Cyclooctene is stable at room temperature, although its cis isomer is still more stable.



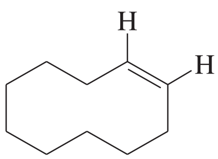
trans cyclic system

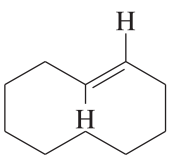

trans-cycloheptene
marginally stable


trans-cyclooctene
stable


cis-cyclooctene
more stable

Once a cycloalkene contains at least ten or more carbon atoms, it can easily accommodate a trans double bond. For cyclodecene and larger cycloalkenes, the trans isomer is nearly as stable as the cis isomer.


cis-cyclodecene


trans-cyclodecene

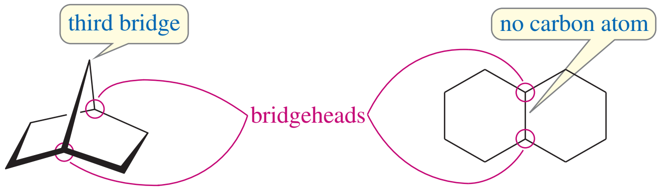
Julius Bredt (1855–1937) was a German alicyclic stereochemist who did pioneering work in the research on camphor, a natural product that sublimes. He also proposed the rule on the strain relationships in polycyclic systems with bridges, a theory now known as Bredt's rule.

7-8E Bredt's Rule

We have seen that a trans cycloalkene is not stable unless there are at least eight carbon atoms in the ring. An interesting extension of this principle is called **Bredt's rule.**

> BREDT'S RULE: A bridged bicyclic compound cannot have a double bond at a bridgehead position unless one of the rings contains at least eight carbon atoms.

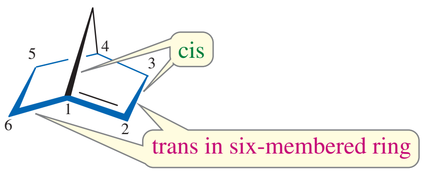
Let's review exactly what Bredt's rule means. A **bicyclic** compound contains two rings. The **bridgehead carbon atoms** are part of both rings, with three links connecting them. A **bridged bicyclic** compound has at least one carbon atom in each of the three links between the bridgehead carbons. In the following examples, the bridgehead carbon atoms are circled in red.



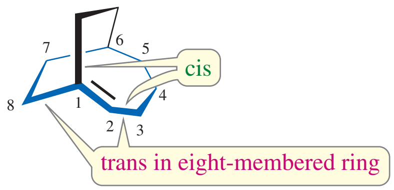
norbornane
a bridged bicyclic compound

decalin
a fused bicyclic compound

If there is a double bond at the bridgehead carbon of a bridged bicyclic system, one of the two rings contains a cis double bond and the other must contain a trans double bond. For example, the following structures show that the norbornane ring system contains a five-membered ring and a six-membered ring. If there is a double bond at the bridgehead carbon atom, the five-membered ring contains a cis double bond and the six-membered ring contains a trans double bond. This unstable arrangement is called a "Bredt's rule violation." If the larger ring contains at least eight carbon atoms, then it can contain a trans double bond and the bridgehead double bond is stable.



Bredt's rule violation



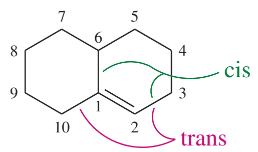
Stable: trans in an eight-membered ring

In general, compounds that violate Bredt's rule are not stable at room temperature. In a few cases, such compounds (usually with seven carbon atoms in the largest ring) have been synthesized at low temperatures.

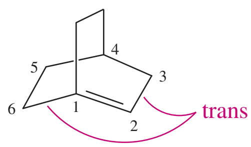
SOLVED PROBLEM 7-2

Which of the following alkenes are stable?

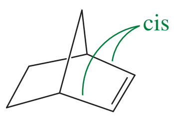
(a)



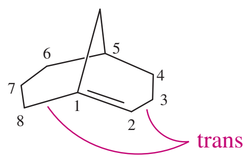
(b)



(c)



(d)



SOLUTION

Compound (a) is stable. Although the double bond is at a bridgehead, it is not a bridged bicyclic system. The trans double bond is in a ten-membered ring. Compound (b) is a Bredt's rule violation and is not stable. The largest ring contains six carbon atoms, and the trans double bond cannot be stable in this bridgehead position.

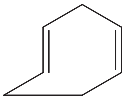
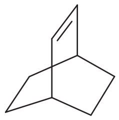
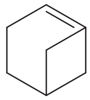
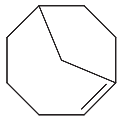
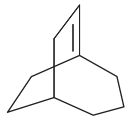
Compound (c) (norbornene) is stable. The (cis) double bond is not at a bridgehead carbon.

Compound (d) is stable. Although the double bond is at the bridgehead of a bridged bicyclic system, there is an eight-membered ring to accommodate the trans double bond.

PROBLEM 7-14

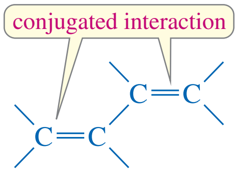
Explain why each of the following alkenes is stable or unstable.

(a) 1,2-dimethylcyclopentene **(b)** *trans*-1,2-dimethylcyclopentene
(c) *trans*-3,4-dimethylcyclopentene **(d)** *trans*-1,2-dimethylcyclodecene

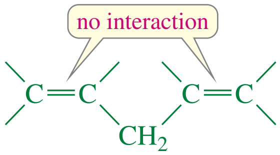
(e)  **(f)**  **(g)**  **(h)**  **(i)** 

7-8F Enhanced Stability of Conjugated Double Bonds

Double bonds can interact with each other if they are separated by just one single bond. Such interacting double bonds are said to be **conjugated.** Double bonds with two or more single bonds separating them have little interaction and are called **isolated double bonds.** For example, *trans*-hexa-1,3-diene has conjugated double bonds, while *trans*-hexa-1,4-diene has isolated double bonds.



conjugated double bonds
(more stable than isolated double bonds)



isolated double bonds


trans-hexa-1,3-diene
conjugated, more stable

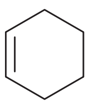

trans-hexa-1,4-diene
isolated

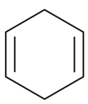
The pi-bonding electrons of conjugated double bonds are delocalized over both bonds, enhancing the stability of the conjugated system over a similar compound with isolated double bonds. In the examples above, *trans*-hexa-1,3-diene is conjugated, and it is more stable than *trans*-hexa-1,4-diene by about 17 kJ/mol (4.0 kcal/mol). This extra stability of the conjugated molecule is called its **resonance energy.** We will discuss the reasons for the stability of conjugated systems and the origin of the resonance energy in Chapter 15.

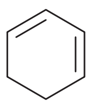
Conjugation effects are particularly important in aromatic systems, with three double bonds conjugated in a six-membered ring (in the simplest case). Benzene (below) has a huge resonance energy, 151 kJ/mol (36 kcal/mol) of enhanced stability over three isolated bonds. We will discuss the reasons for the remarkable stability of aromatic systems in Chapter 16. For now, you should recognize that aromatic systems are particularly stable, and they are not likely to react as easily as typical alkenes.

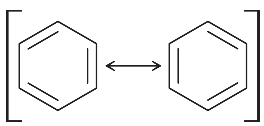
PROBLEM-SOLVING HINT

Benzene can be represented by either of two resonance forms, which are equivalent.


cyclohexene
isolated


cyclohexa-1,4-diene
isolated


cyclohexa-1,3-diene
conjugated

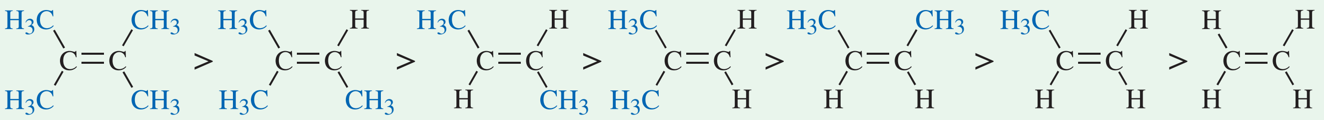

benzene
aromatic

resonance energy:	0	0	8 kJ (1.9 kcal)	151 kJ (36 kcal)

FOCUS Factors Affecting Stability of Alkenes

Several factors influence the stability of the carbon-carbon double bond.

1. In general, **more substituted double bonds are more stable** (Zaitsev's Rule). "Substituted" refers to an H replaced with an alkyl group.



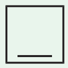
tetrasubstituted | trisubstituted | trans | geminal | cis | monosubstituted | unsubstituted

disubstituted

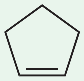
Alkenes can be disubstituted in three different patterns. Of these, the trans isomer is typically the most stable, and the cis isomer is usually the least stable.

2. Cycloalkene stability depends on **ring size and geometry**.

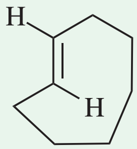
highly strained



strained

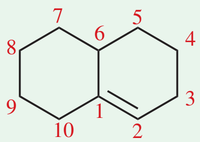


Cis double bonds have little to no ring strain in rings of five atoms or larger.

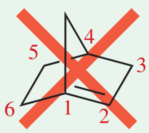


Trans double bonds require a minimum of eight atoms in the ring to be stable at room temperature.

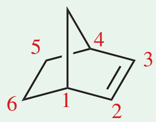
3. **Bond angle strain** can limit alkene stability. **Bredt's Rule**: In a bridged bicyclic ring system, a double bond cannot exist at a bridgehead unless the larger ring has at least 8 atoms. A *bridgehead* atom is one present in two different rings.



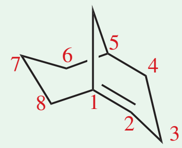
This is a fused ring, not a bridged ring. Bredt's rule does not apply.
The larger ring has 10 carbons—this alkene is stable.



The larger ring has 6 carbons—violates Bredt's rule. This alkene is not stable. (C1 and C4 are bridgeheads.)



The double bond is not at bridgehead. Bredt's rule does not apply. This alkene is stable. (C1 and C4 are bridgeheads.)

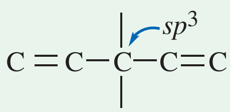


The larger ring has 8 carbons—does not violate Bredt's rule. This alkene is stable. (C1 and C5 are bridgeheads.)

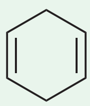
4. **Conjugated double bonds are more stable than isolated double bonds**. A conjugated system has alternating single and double bonds.

C=C—C=C—C

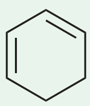
conjugated double bonds



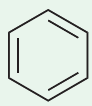
isolated double bonds



isolated double bonds



conjugated double bonds more stable than isolated



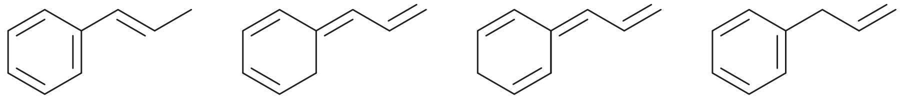
aromatic—large stabilization

The special case where three double bonds are conjugated in a six-membered ring is called an aromatic system. This arrangement has enhanced stability and is the topic of **Chapter 16**.

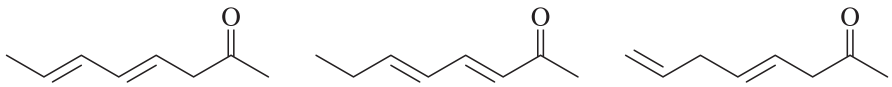
PROBLEM 7-15

For each set of isomers, choose the isomer that you expect to be most stable and the isomer you expect to be least stable.

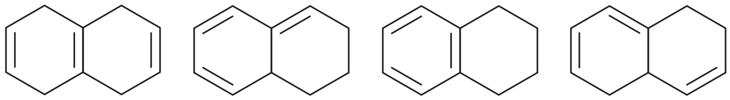
(a)



(b)



(c)

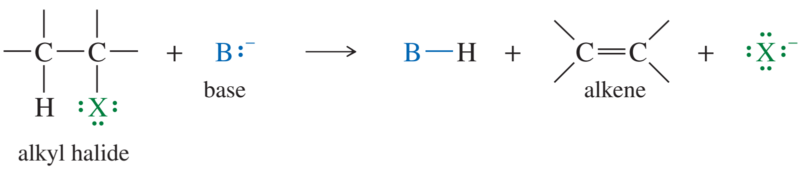


7-9 Formation of Alkenes by Dehydrohalogenation of Alkyl Halides

Elimination reactions are some of the most common methods of synthesizing alkenes. Alkyl halides undergo eliminations to alkenes under conditions that are similar to those covered in Chapter 6 for their nucleophilic substitutions. In fact, nucleophilic substitutions and eliminations often take place under the same conditions in direct competition with each other.

An **elimination** involves the loss of two atoms or groups from the substrate, usually with formation of a new π bond. Elimination of a proton and a halide ion is called **dehydrohalogenation,** and the product is an alkene.

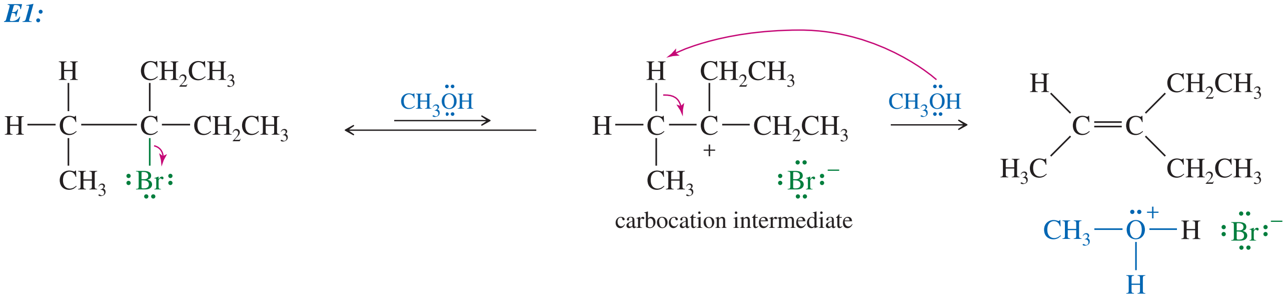
Elimination



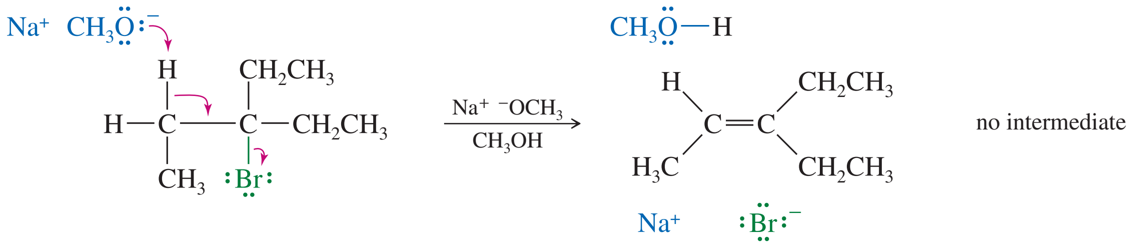
Elimination reactions often accompany and compete with substitutions. Most nucleophiles are also basic and can engage in either substitution or elimination, depending on the substrate (the alkyl halide) and the reaction conditions. By varying the reagents and conditions, we can often modify a reaction to favor substitution or to favor elimination. First, we will discuss eliminations by themselves. Then we will consider substitutions and eliminations together, trying to predict what products and which mechanisms are likely with a given set of reactants and conditions.

Depending on the reagents and conditions involved, an elimination might be a unimolecular (E1, first-order) or bimolecular (E2, second-order) process. The following examples illustrate these types of eliminations.

E1:



E2:



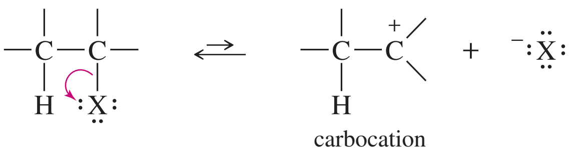
7-10 Unimolecular Elimination: The E1 Reaction

The abbreviation **E1** stands for *Elimination, unimolecular*. The mechanism is called *unimolecular* because the rate-limiting transition state involves a single molecule rather than a collision between two molecules. The slow step of an E1 reaction is the same as in the S_N1 reaction: unimolecular ionization to form a carbocation. In a fast second step, a base abstracts a proton from the carbon atom adjacent to the C^+. The electrons that once formed the carbon–hydrogen bond now form a pi bond between two carbon atoms. The general mechanism for the E1 reaction is shown in Key Mechanism 7-1.

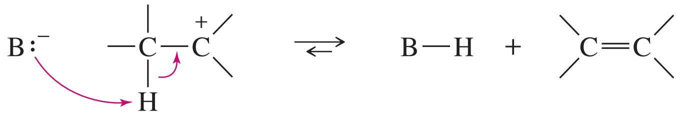
KEY MECHANISM 7-1 The E1 Reaction

The E1 reaction requires ionization to a carbocation intermediate like the S_N1. The E1 and S_N1 reactions follow the same order of reactivity based on carbocation stability: 3° >> 2° >> 1°. Tertiary alkyl halides react readily by the E1 mechanism, secondary halides react much slower, and simple primary halides do not undergo E1 or S_N1 reactions.

Step 1: Unimolecular ionization to give a carbocation (rate-limiting).

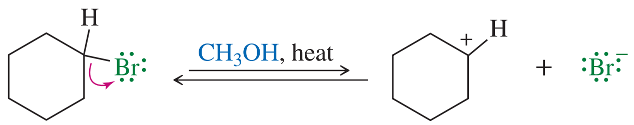


Step 2: Deprotonation by a weak base (often the solvent) gives the alkene (fast).

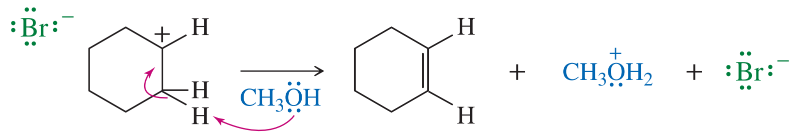


EXAMPLE: E1 elimination of bromocyclohexane in methanol.

Step 1: Ionization gives a carbocation and bromide ion in a slow step.



Step 2: Methanol abstracts a proton to give cyclohexene in a fast step.



PROBLEM 7-16

Show what happens in step 2 of the example if the solvent acts as a *nucleophile* (forming a bond to carbon) rather than as a *base* (removing a proton).

The rate-limiting step involves ionization of the alkyl halide. Because the step is unimolecular (involves only one molecule), the rate equation is first-order. The rate depends only on the concentration of the alkyl halide, and not on the strength or concentration of the base.

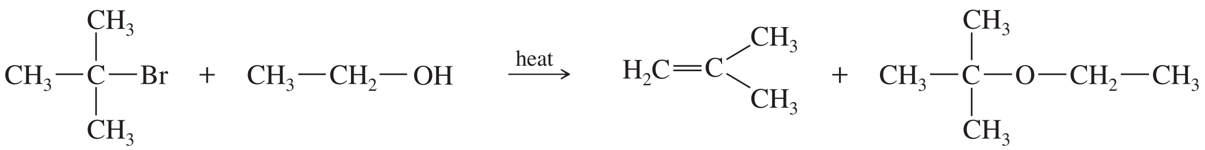
$$\text{E1 rate} = k_r[\text{RX}]$$

The weak base (often the solvent) takes part in the fast second step of the reaction.

Unimolecular (E1) dehydrohalogenations usually take place in a good ionizing solvent (such as an alcohol or water), without a strong base present to force a bimolecular (E2) reaction. The substrate for E1 dehydrohalogenation is usually a tertiary alkyl halide. If secondary halides undergo E1 reactions, they require stronger conditions and react much more slowly.

7-10A Competition between the E1 and S_N1 Reactions

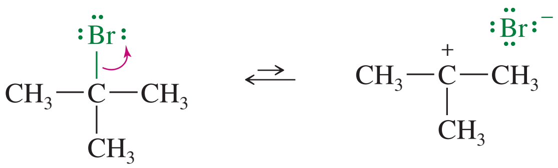
The E1 elimination almost always competes with the S_N1 substitution. Whenever a carbocation is formed, it can undergo either substitution or elimination, and mixtures of products often result. The following reaction shows the formation of both elimination and substitution products in the reaction of *tert*-butyl bromide with boiling ethanol.



tert-butyl bromide ethanol 2-methylpropene (E1 product) ethyl *tert*-butyl ether (S_N1 product)

The 2-methylpropene product results from dehydrohalogenation, an elimination of hydrogen and a halogen atom. Under these first-order conditions (the absence of a strong base), dehydrohalogenation takes place by the E1 mechanism: Ionization of the alkyl halide gives a carbocation intermediate, which loses a proton to give the alkene. Substitution results from an S_N1 nucleophilic attack on the carbocation. This kind of reaction is called a **solvolysis** (*solvo* for "solvent," *lysis* for "cleavage") because the solvent reacts with the substrate. In this example, ethanol (the solvent) serves as a base (removing a proton) in the elimination and as a nucleophile (forming a bond to carbon) in the substitution.

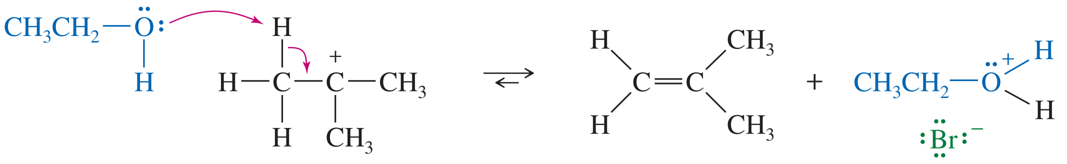
Step 1: Ionization to form a carbocation.



tert-butyl bromide carbocation intermediate

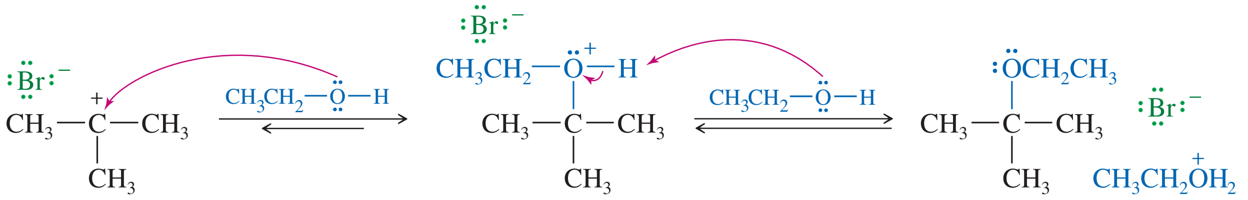
Step 2 (by the E1 mechanism): Basic attack by the solvent abstracts a proton to give an alkene.

Elimination:



or step 2 (by the S_N1 mechanism): Nucleophilic attack by the solvent on the carbocation.

Substitution:



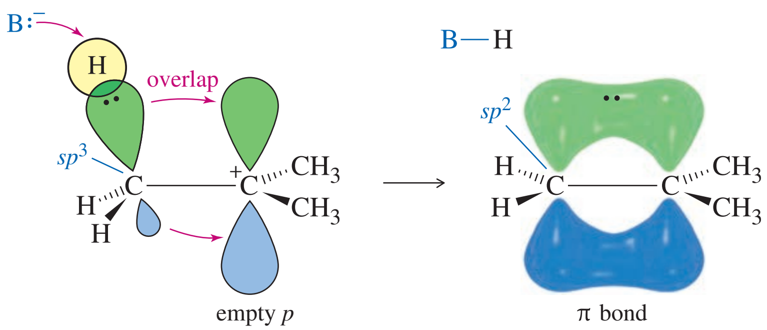
Under ideal conditions, one of these first-order reactions provides a good yield of one product or the other. Often, however, carbocation intermediates react in two or more ways to give mixtures of products. For this reason, S_N1 and E1 reactions of alkyl halides are not frequently used for organic synthesis. They have been studied in great detail to learn about the properties of carbocations, however.

PROBLEM 7-17

S_N1 substitution and E1 elimination frequently compete in the same reaction.
(a) Propose a mechanism and predict the products for the solvolysis of 2-bromo-2,3,3-trimethylbutane in methanol.
(b) Compare the function of the solvent (methanol) in the E1 and S_N1 reactions.

7-10B Orbitals and Energetics of the E1 Reaction

In the second step of the E1 mechanism, the carbon atom next to the C^+ must rehybridize to sp^2 as the base attacks the proton and electrons flow into the new pi bond.



The potential-energy diagram for the E1 reaction (Figure 7-9) is similar to that for the S_N1 reaction. The ionization step is strongly endothermic, with a rate-limiting transition state. The second step is a fast exothermic deprotonation by a base. The base is not involved in the reaction until *after* the rate-limiting step, so the rate depends only on the concentration of the alkyl halide. Weak bases are common in E1 reactions.

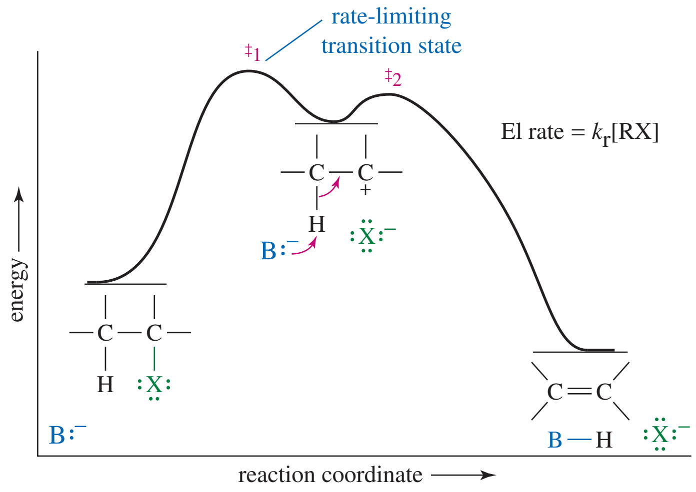


FIGURE 7-9 Reaction-energy diagram of the E1 reaction. The first step is a rate-limiting ionization. Compare this energy profile with that of the S_N1 reaction in Figure 6-8.

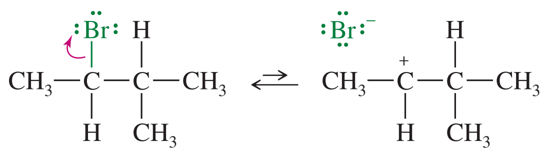
7-10C Rearrangements in E1 Reactions

Like other carbocation reactions, the E1 may be accompanied by rearrangement. Compare the following E1 reaction (with rearrangement) with the S_N1 reaction of the same substrate, shown in Mechanism 6-6. Note that the solvent acts as a *base* in the E1 reaction and a *nucleophile* in the S_N1 reaction.

MECHANISM 7-2 Rearrangement in an E1 Reaction

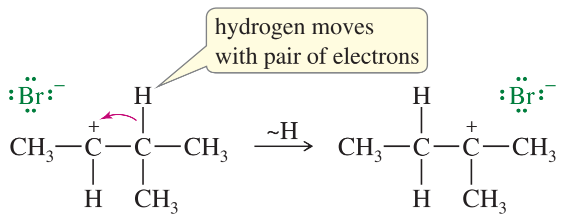
Like other reactions involving carbocations, the E1 may be accompanied by rearrangements such as hydride shifts and alkyl shifts.

Step 1: Ionization to form a carbocation. (slow)



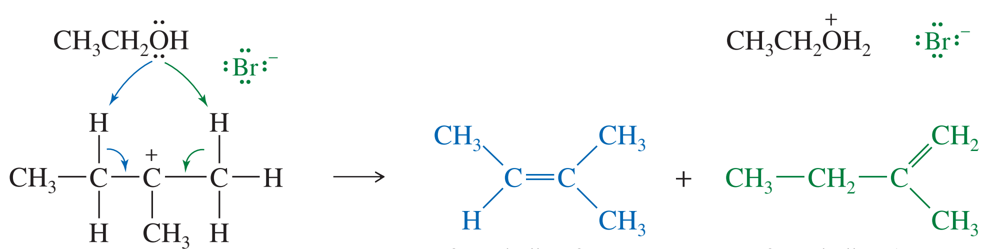
2-bromo-3-methylbutane 2° carbocation

Step 2: A hydride shift forms a more stable carbocation. (fast)



2° carbocation 3° carbocation

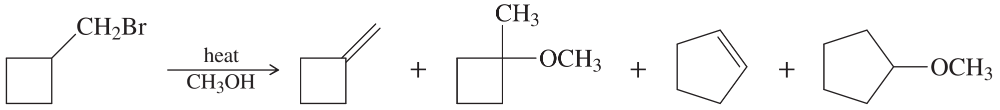
Step 3: The weakly basic solvent removes either adjacent proton. (fast)



2-methylbut-2-ene 2-methylbut-1-ene

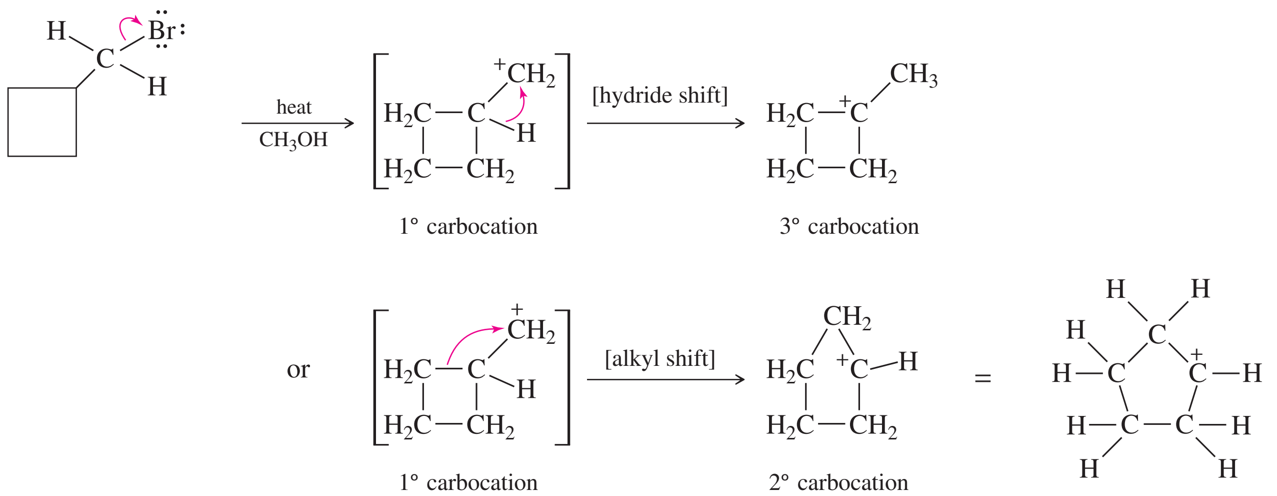
SOLVED PROBLEM 7-3

When the following compound is heated in methanol, several different products are formed. Propose mechanisms to account for the four products shown.



PARTIAL SOLUTION

With no strong base and a good ionizing solvent, we would expect a first-order reaction. But this is a primary alkyl halide, so ionization is difficult unless it rearranges. It might rearrange as it forms, but we'll imagine the cation forming then rearranging.



From these rearranged intermediates, either loss of a proton (E1) or attack by the solvent (S_N1) gives the observed products. Note that the actual reaction may give more than just these products, but the other products are not required for the problem.

PROBLEM 7-18

Finish Solved Problem 7-3 by showing how the rearranged carbocations give the four products shown in the problem. Be careful when using curved arrows to show deprotonation and/or nucleophilic attack by the solvent. The curved arrows always show movement of electrons, not movement of protons or other species.

PROBLEM 7-19

The solvolysis of 2-bromo-3-methylbutane potentially can give several products, including both E1 and S_N1 products from both the unrearranged carbocation and the rearranged carbocation. Mechanisms 6-6 (page 328) and 7-2 (previous page) show the products from the rearranged carbocation. Summarize all the possible products, showing which carbocation they come from and whether they are the products of E1 or S_N1 reactions.

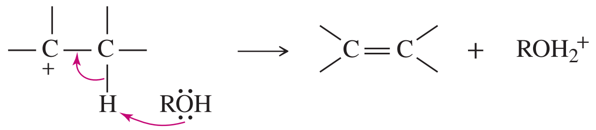
We can now summarize four ways that a carbocation can react to become more stable.

SUMMARY Carbocation Reactions

A carbocation can

1. React with its own leaving group to return to the reactant: $R^+ + :X^- \longrightarrow R{-}X$
2. React with a nucleophile to form a substitution product (S_N1): $R^+ + \text{Nuc:}^- \longrightarrow R{-}\text{Nuc}$

3. Lose a proton to form an elimination product (an alkene) (E1):



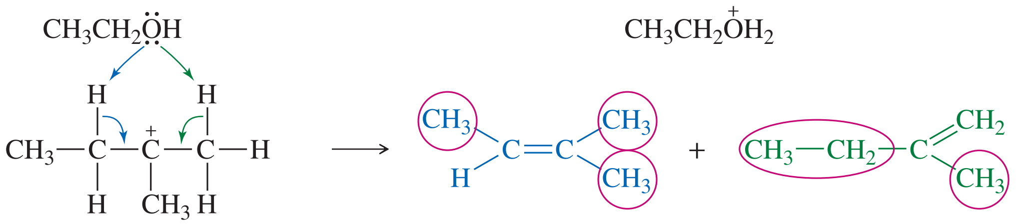
4. Rearrange to a more stable carbocation and then react further.
The order of stability of carbocations is: resonance-stabilized, 3° > 2° > 1°.

PROBLEM 7-20

Give the substitution and elimination products you would expect from the following reactions.
(a) 3-bromo-3-ethylpentane heated in methanol
(b) 1-iodo-1-phenylcyclopentane heated in ethanol
(c) 1-bromo-2-methylcyclohexane + silver nitrate in water ($AgNO_3$ forces ionization)

7-11 Positional Orientation of Elimination: Zaitsev's Rule

We have seen that many compounds can eliminate in more than one way, to give mixtures of alkenes. In many cases, we can predict which elimination product will predominate. In the example shown in Mechanism 7-2, the carbocation can lose a proton on either of two adjacent carbon atoms.



2-methylbut-2-ene
trisubstituted, major (90%)

2-methylbut-1-ene
disubstituted, minor (10%)

The first product has a *trisubstituted* double bond, with three substituents (circled) on the doubly bonded carbons. It has the general formula $R_2C{=}CHR$. The second product has a *disubstituted* double bond, with general formula $R_2C{=}CH_2$ (or $R—CH{=}CH—R$). In most E1 and E2 eliminations where there are two or more possible elimination products, *the product with the most substituted double bond will predominate*. This general principle is the applied form of **Zaitsev's rule,** which we first encountered in Section 7-8B. Reactions that give the most substituted alkene are said to follow **Zaitsev orientation.**

PROBLEM-SOLVING HINT

Zaitsev is transliterated from the Russian name, and may also be spelled ***Saytzeff.***

ZAITSEV'S RULE: In elimination reactions, the most substituted alkene usually predominates.

$R_2C{=}CR_2$	>	$R_2C{=}CHR$	>	$RHC{=}CHR$ and $R_2C{=}CH_2$	>	$RHC{=}CH_2$
tetrasubstituted		trisubstituted		disubstituted		monosubstituted

This order of preference is the same as the order of stability of alkenes.

PROBLEM-SOLVING HINT

Whenever a carbocation has the positive charge on a carbon atom next to a more highly substituted carbon, consider whether a rearrangement might occur.

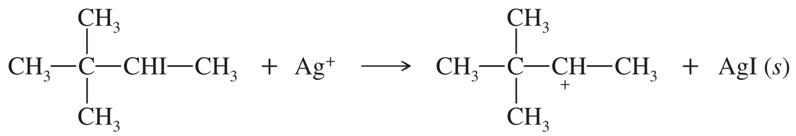
Aleksander Mikhaylovich Zaytsev (1841–1910) was a Russian chemist best known for proposing Zaitsev's rule. This rule predicts the major products in an elimination reaction, governed by the stability of the alkenes formed. As a rule of thumb, the more substituted alkenes will form predominantly.

SOLVED PROBLEM 7-4

When 3-iodo-2,2-dimethylbutane is treated with silver nitrate in ethanol, three elimination products are formed. Give their structures, and predict which ones are formed in larger amounts.

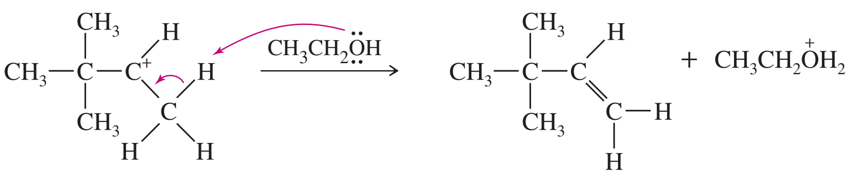
SOLUTION

Silver nitrate forces ionization of the alkyl iodide to give solid silver iodide and a cation.



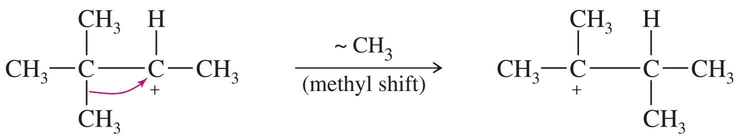
This secondary carbocation can lose a proton to give an unrearranged alkene (**A**), or it can rearrange to a more stable tertiary cation.

Loss of a proton



(A)

Rearrangement

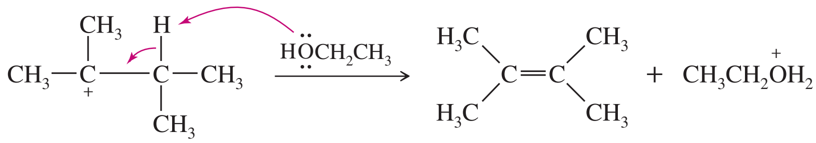


2° carbocation

3° carbocation

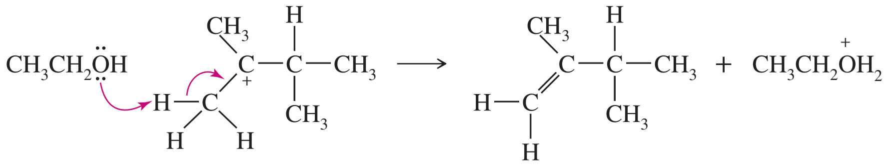
The tertiary cation can lose a proton in either of two positions. One of the products (**B**) is a tetrasubstituted alkene, and the other (**C**) is disubstituted.

Formation of a tetrasubstituted alkene



(B)
(tetrasubstituted)

Formation of a disubstituted alkene



(C)
(disubstituted)

Product **B** predominates over product **C** because the double bond in **B** is more substituted. Whether product **A** is a major product will depend on the specific reaction conditions and whether proton loss or rearrangement occurs faster.

PROBLEM 7-21

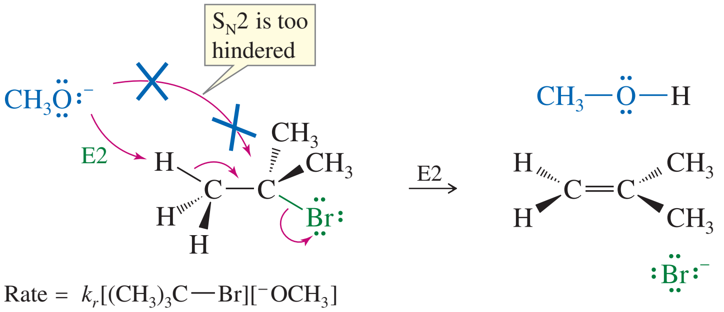
Each of the two carbocations in Solved Problem 7-4 can also react with ethanol to give a substitution product. Give the structures of the two substitution products formed in this reaction.

PROBLEM 7-22

When (1-bromoethyl)cyclohexane is heated in methanol for an extended period of time, five products result: two ethers and three alkenes. Predict the products of this reaction, and propose mechanisms for their formation. Predict which of the three alkenes is the major elimination product.

7-12 Bimolecular Elimination: The E2 Reaction

Eliminations can also take place under second-order conditions with a strong base present. As an example, consider the reaction of *tert*-butyl bromide with methoxide ion in methanol. Methoxide ion is a strong base as well as a strong nucleophile. It attacks the alkyl halide faster than the halide can ionize to give a first-order reaction. No substitution product (methyl *tert*-butyl ether) is observed, however. The S_N2 mechanism is blocked because the tertiary alkyl halide is too hindered. The observed product is 2-methylpropene, resulting from elimination of HBr and formation of a double bond.



Rate = $k_r[(CH_3)_3C—Br][^-OCH_3]$

The rate of this elimination is proportional to the concentrations of both the alkyl halide and the base, giving a second-order rate equation. This is a *bimolecular* process, with both the base and the alkyl halide participating in the transition state, so this mechanism is abbreviated **E2** for *Elimination, bimolecular*.

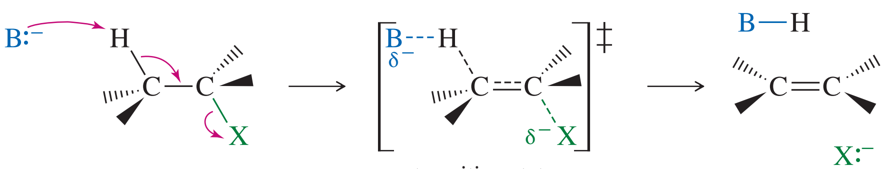
$$\text{E2 rate} = k_r[\text{RX}]\,[\text{B}:^-]$$

In the E2 reaction just shown, methoxide reacts as a *base* (removing a proton) rather than as a *nucleophile* (forming a bond to carbon). Most strong nucleophiles are also strong bases, and elimination commonly results when a strong base/nucleophile is used with a poor S_N2 substrate such as a 3° or hindered 2° alkyl halide. Instead of attacking the back side of the hindered electrophilic carbon, methoxide abstracts a proton from one of the methyl groups. This reaction takes place in one step, with bromide leaving as the base abstracts a proton.

In the general mechanism of the E2 reaction, a strong base abstracts a proton on a carbon atom adjacent to the one with the leaving group. As the base abstracts a proton, a double bond forms and the leaving group leaves. Like the S_N2 reaction, the E2 is a *concerted* reaction in which bonds break and new bonds form at the same time, in a single step.

KEY MECHANISM 7-3 The E2 Reaction

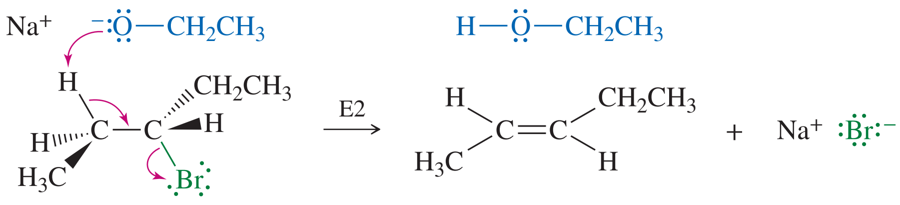
The concerted E2 reaction takes place in a single step. A strong base abstracts a proton on a carbon next to the leaving group, and the leaving group leaves. The product is an alkene.



transition state

(continued)

EXAMPLE: E2 elimination of 3-bromopentane with sodium ethoxide.



3-bromopentane

pent-2-ene (mostly trans)

The order of reactivity for alkyl halides in E2 reactions is 3° > 2° > 1°.

PROBLEM 7-23

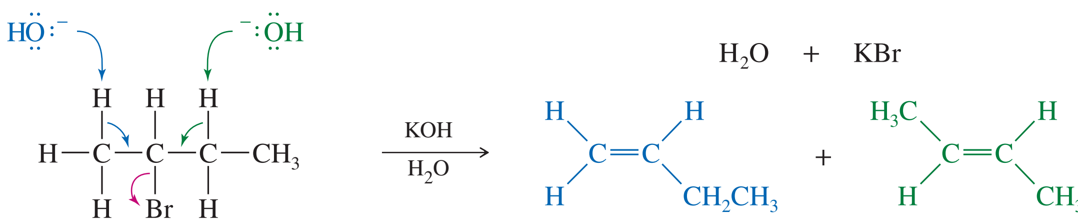
Under second-order conditions (strong base/nucleophile), S_N2 and E2 reactions may occur simultaneously and compete with each other. Show what products might be expected from the reaction of 2-bromo-3-methylbutane (a moderately hindered 2° alkyl halide) with sodium ethoxide.

Reactivity of the Substrate in the E2 The order of reactivity of alkyl halides toward E2 dehydrohalogenation is found to be

$$3° \quad > \quad 2° \quad > \quad 1°$$

This reactivity order reflects the greater stability of highly substituted double bonds. Elimination of a secondary halide gives a more substituted alkene than elimination of a primary halide. Elimination of a tertiary halide gives an even more substituted alkene than a secondary halide. The stabilities of the alkene products are reflected in the transition states, giving lower activation energies and higher rates for elimination of alkyl halides that lead to highly substituted alkenes.

Mixtures of Products in the E2 The E2 reaction requires abstraction of a proton on a carbon atom next to the carbon bearing the halogen. If there are two or more possibilities, mixtures of products may result. In most cases, Zaitsev's rule predicts which of the possible products will be the major product: the most substituted alkene. For example, the E2 reaction of 2-bromobutane with potassium hydroxide gives a mixture of two products, but-1-ene (a monosubstituted alkene) and but-2-ene (a disubstituted alkene). As predicted by Zaitsev's rule, the disubstituted isomer but-2-ene is the major product.

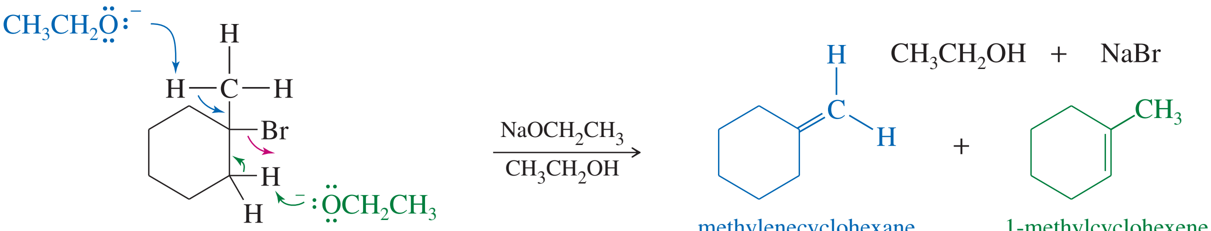


2-bromobutane

19% but-1-ene
monosubstituted

81% but-2-ene
disubstituted

Similarly, the reaction of 1-bromo-1-methylcyclohexane with sodium ethoxide gives a mixture of a disubstituted alkene and a trisubstituted alkene. The trisubstituted alkene is the major product.



1-bromo-1-methylcyclohexane

methylenecyclohexane
disubstituted, minor

1-methylcyclohexene
trisubstituted, major

PROBLEM 7-24

1. Predict the elimination products of the following reactions. When two alkenes are possible, predict which one will be the major product. Explain your answers, showing the degree of substitution of each double bond in the products.
2. Which of these reactions are likely to produce both elimination and substitution products?

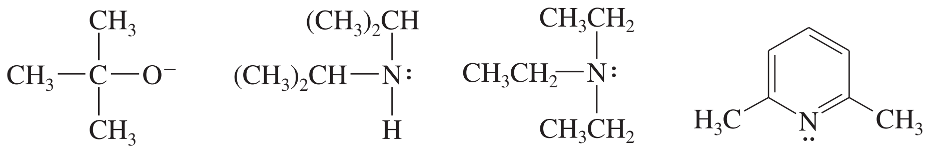
(a) 2-bromopentane + $NaOCH_3$
(b) 3-bromo-3-methylpentane + NaOMe (Me = methyl, CH_3)
(c) 2-bromo-3-ethylpentane + NaOH

PROBLEM-SOLVING HINT

Zaitsev's rule usually applies in E2 reactions unless the base and/or the leaving group are unusually bulky.

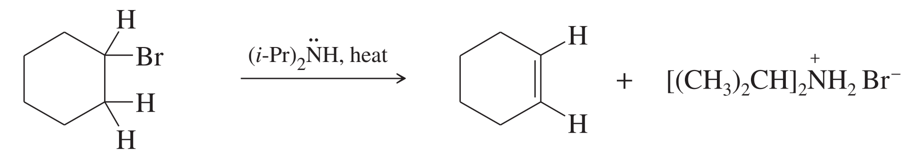
7-13 Bulky Bases in E2 Eliminations; Hofmann Orientation

Use of a Bulky Base If the substrate in an E2 elimination is prone to substitution, we can minimize the amount of substitution by using a bulky base. Large alkyl groups on a bulky base hinder its approach to attack a carbon atom (substitution), yet it can easily abstract a proton (elimination). Some of the bulky strong bases commonly used for elimination are *tert*-butoxide ion, diisopropylamine, triethylamine, and 2,6-dimethylpyridine.



tert-butoxide   diisopropylamine   triethylamine   2,6-dimethylpyridine

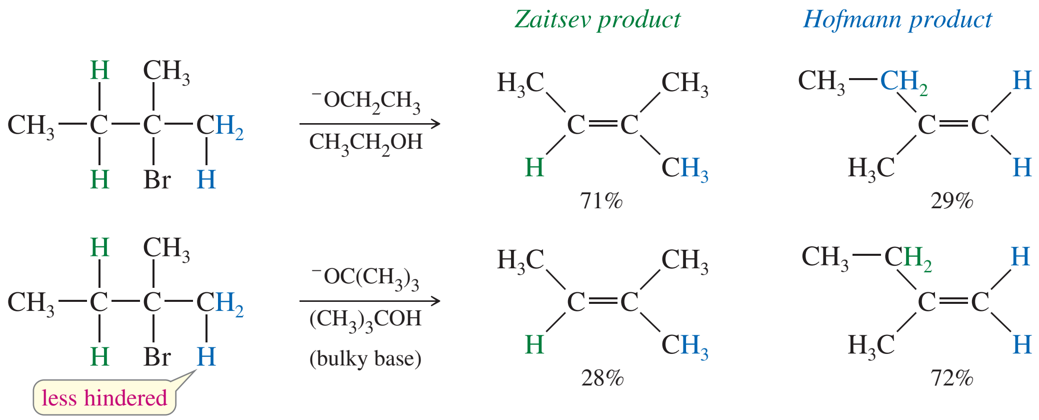
The dehydrohalogenation of bromocyclohexane illustrates the use of a bulky base for elimination. Bromocyclohexane, a secondary alkyl halide, can undergo both substitution and elimination. Elimination (E2) is favored over substitution (S_N2) by using a bulky base such as diisopropylamine. Diisopropylamine is too bulky to be a good nucleophile, but it acts as a strong base to abstract a proton.



bromocyclohexane

cyclohexene
(93%)

Formation of the Hofmann Product Bulky bases can also accomplish dehydrohalogenations that do not follow the Zaitsev rule. Steric hindrance often prevents a bulky base from abstracting the proton that leads to the most highly substituted alkene. In these cases, it abstracts a less hindered proton, often the one that leads to formation of the least highly substituted product, called the **Hofmann product.** The following reaction gives mostly the **Zaitsev product** with the relatively unhindered ethoxide ion, but mostly the Hofmann product with the bulky *tert*-butoxide ion.



PROBLEM 7-25

For each reaction, decide whether substitution or elimination (or both) is possible, and predict the products you expect. Label the major products.

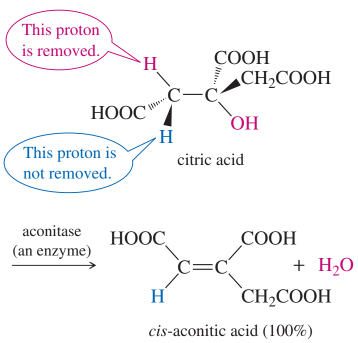
(a) 1-bromo-1-methylcyclohexane + NaOH in acetone
(b) 1-bromo-1-methylcyclohexane + triethylamine (Et_3N:)
(c) chlorocyclohexane + $NaOCH_3$ in CH_3OH
(d) chlorocyclohexane + $NaOC(CH_3)_3$ in $(CH_3)_3COH$

7-14 Stereochemistry of the E2 Reaction

Application: Biochemistry

Enzyme-catalyzed eliminations generally proceed by E2 mechanisms and produce only one stereoisomer. Two catalytic groups are involved: One abstracts the hydrogen, and the other assists in the departure of the leaving group. The groups are positioned appropriately to allow an anti-coplanar elimination.

For example, one of the steps in cellular respiration involves elimination of water from citric acid to give only the *Z* isomer of the product, commonly called *cis*-aconitic acid.



Like the S_N2 reaction, the E2 follows a **concerted mechanism:** Bond breaking and bond formation take place at the same time, and the partial formation of new bonds lowers the energy of the transition state. Concerted mechanisms require specific geometric arrangements so that the orbitals of the bonds being broken can overlap with those being formed and the electrons can flow smoothly from one bond to another. The geometric arrangement required by the S_N2 reaction is a back-side attack; with the E2 reaction, a coplanar arrangement of the orbitals is needed.

E2 elimination requires partial formation of a new pi bond, with its parallel *p* orbitals, in the transition state. The electrons that once formed a C—H bond must begin to overlap with the orbital that the leaving group is vacating. Formation of this new pi bond implies that these two sp^3 orbitals must be parallel so that pi overlap is possible as the hydrogen and halogen leave and the orbitals rehybridize to the *p* orbitals of the new pi bond.

7-14A Anti-Coplanar and Syn-Coplanar E2 Eliminations

Figure 7-10 shows two conformations that provide the necessary coplanar alignment of the leaving group, the departing hydrogen, and the two carbon atoms. When the hydrogen and the halogen are **anti** to each other ($\theta = 180°$), their orbitals are aligned. This is called the **anti-coplanar** conformation. When the hydrogen and the halogen eclipse each other ($\theta = 0°$), their orbitals are once again aligned. This is called the **syn-coplanar** conformation. Make a model corresponding to Figure 7-10, and use it to follow along with this discussion.

Of these possible conformations, the anti-coplanar arrangement is most commonly seen in E2 reactions. The transition state for the anti-coplanar arrangement is a staggered conformation, with the base far away from the leaving group. In most cases, this transition state is lower in energy than that for the syn-coplanar elimination.

The transition state for syn-coplanar elimination is an eclipsed conformation. In addition to the higher energy resulting from eclipsing interactions, the transition state suffers from interference between the attacking base and the leaving group. To abstract the proton, the base must approach quite close to the leaving group. In most cases, the leaving group is bulky and negatively charged, and the repulsion between the base and the leaving group raises the energy of the syn-coplanar transition state.

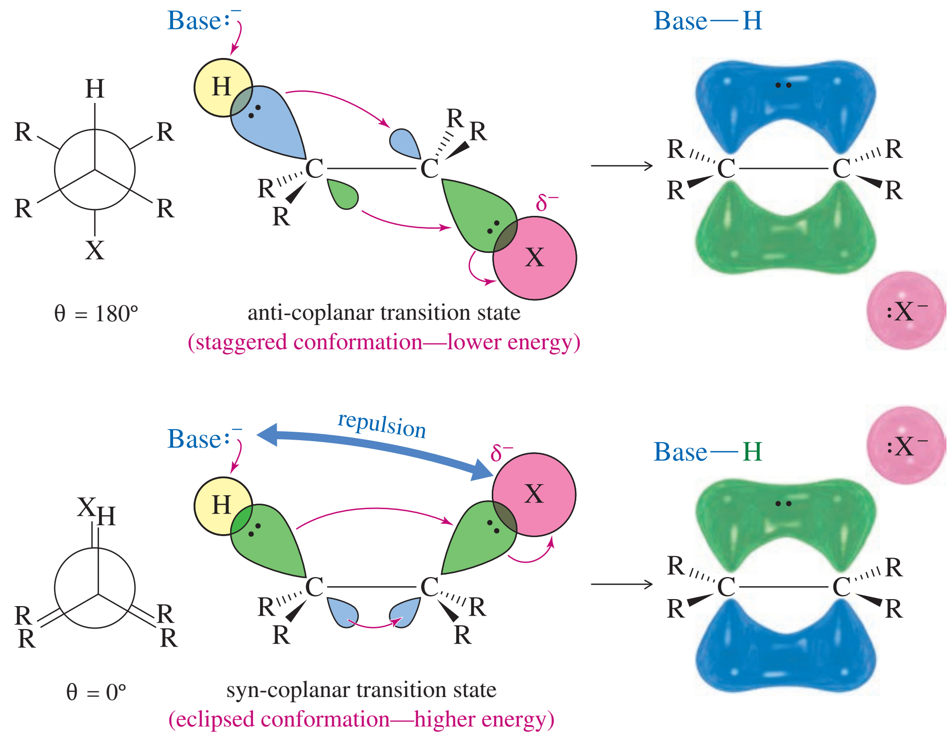
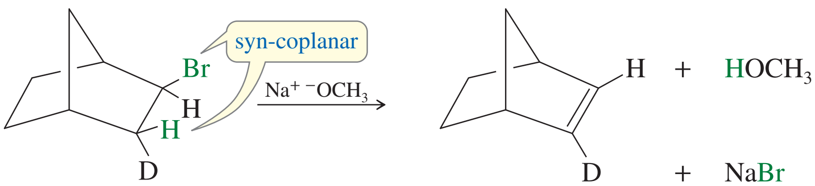


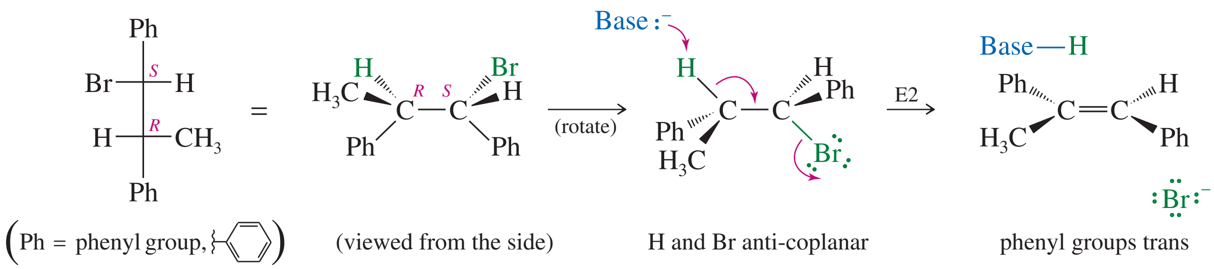
FIGURE 7-10 Concerted transition states of the E2 reaction. The orbitals of the hydrogen atom and the halide must be aligned so that they can begin to form a pi bond in the transition state.

Some molecules are rigidly held in eclipsed (or nearly eclipsed) conformations, with a hydrogen atom and a leaving group in a syn-coplanar arrangement. Such compounds are likely to undergo E2 elimination by a concerted syn-coplanar mechanism. Deuterium labeling (using D, the hydrogen isotope with mass number 2) is used in the following reaction to show which atom is abstracted by the base. Only the hydrogen atom is abstracted, because it is held in a syn-coplanar position with the bromine atom. Remember that syn-coplanar eliminations are unusual, however, and anti-coplanar eliminations are more common.



7-14B Stereospecific E2 Reactions

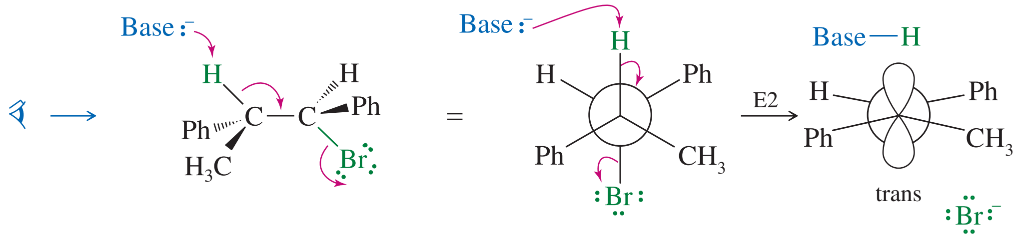
Like the S_N2 reaction (Section 6-12), the E2 is **stereospecific:** Different stereoisomers of the reactant give different stereoisomers of the product. The E2 is stereospecific because it normally goes through an anti-coplanar transition state. The products are alkenes, and different diastereomers of starting materials commonly give different diastereomers of alkenes. The following example shows why the E2 elimination of one diastereomer of 1-bromo-1,2-diphenylpropane gives only the trans isomer of the alkene product.



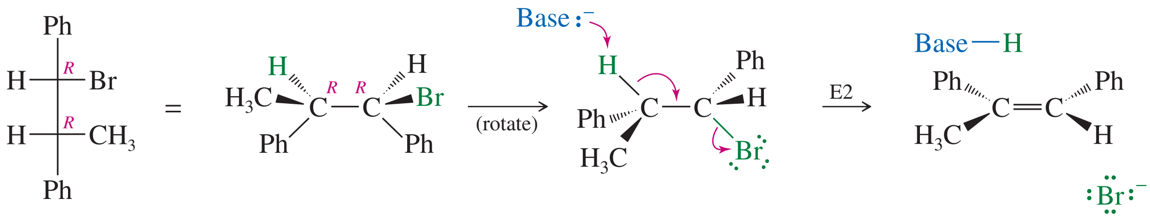
If we make a model and look at this reaction from the left end of the molecule, the anti-coplanar arrangement of the H and Br is apparent.

MECHANISM 7-4 Stereochemistry of the E2 Reaction

Most E2 reactions go through an anti-coplanar transition state. This geometry is most apparent if we view the reaction sighting down the carbon–carbon bond between the hydrogen and leaving group. Viewed from the left:

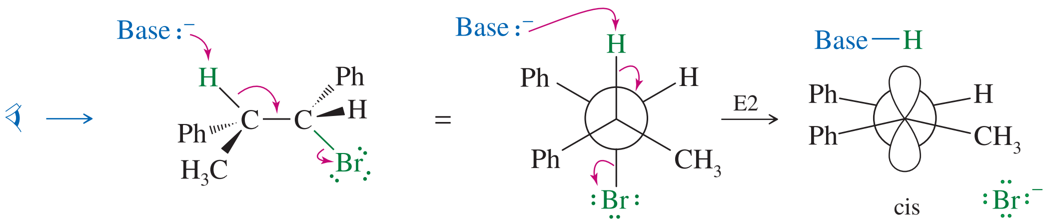


The following reaction shows how the anti-coplanar elimination of the other diastereomer (*R*,*R*) gives only the cis isomer of the product. In effect, the two different diastereomers of the reactant give two different diastereomers of the product: a stereospecific result.



(viewed from the side) H and Br anti-coplanar phenyl groups cis

Viewed from the left end of the molecule:



PROBLEM-SOLVING HINT

Don't try to memorize your way through these reactions. Look at each one, and consider what it might do. Use your models for the ones that involve stereochemistry.

PROBLEM 7-26

Show that the (*S*,*S*) enantiomer of this (*R*,*R*) diastereomer of 1-bromo-1,2-diphenylpropane also undergoes E2 elimination to give the cis diastereomer of the product. (We do not expect these achiral reagents to distinguish between enantiomers.)

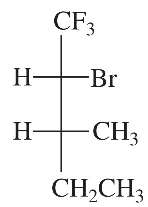
PROBLEM-SOLVING HINT

Anti-coplanar E2 eliminations are common.

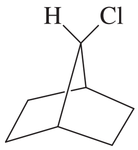
Syn-coplanar E2 eliminations are rare, usually occurring when free rotation is not possible.

PROBLEM 7-27

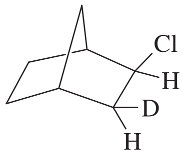
Make models of the following compounds, and predict the products formed when they react with the strong bases shown.

(a)  + KOH ⟶ (substitution and elimination)

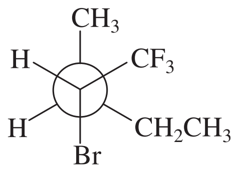
(b) *meso*-1,2-dibromo-1,2-diphenylethane + $(CH_3CH_2)_3N$:
(c) (*d,l*)-1,2-dibromo-1,2-diphenylethane + $(CH_3CH_2)_3N$:

(d) 

+ NaOH in acetone

(e) 

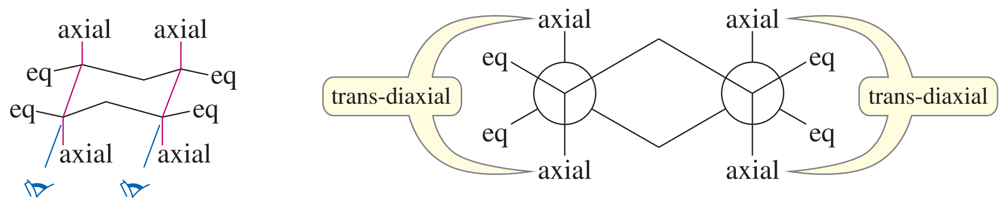
+ $(CH_3)_3CO^-$

(f) 

+ $NaOCH(CH_3)_2$

7-15 E2 Reactions in Cyclohexane Systems

Nearly all cyclohexanes are most stable in chair conformations. In the chair, all the carbon–carbon bonds are staggered, and any two adjacent carbon atoms have axial bonds in an anti-coplanar conformation, ideally oriented for the E2 reaction. (As drawn in the following figure, the axial bonds are vertical.) On any two adjacent carbon atoms, one has its axial bond pointing up and the other has its axial bond pointing down. These two bonds are trans to each other, and we refer to their geometry as **trans-diaxial.**



perspective view Newman projection

An E2 elimination can take place on this chair conformation only if the proton and the leaving group can get into a trans-diaxial arrangement. Figure 7-11 shows the E2 dehydrohalogenation of bromocyclohexane. The molecule must flip into the chair conformation with the bromine atom axial before elimination can occur.

(You should make models of the structures in the following examples and problems so that you can follow along more easily.)

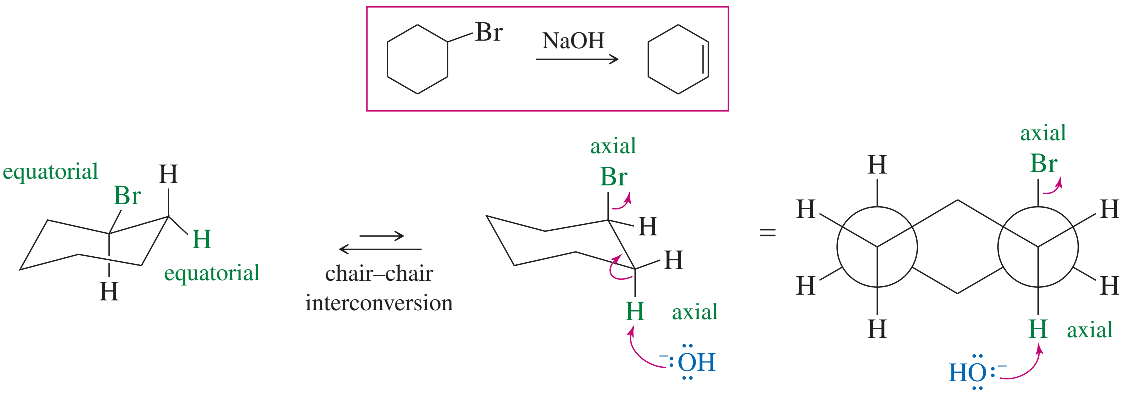
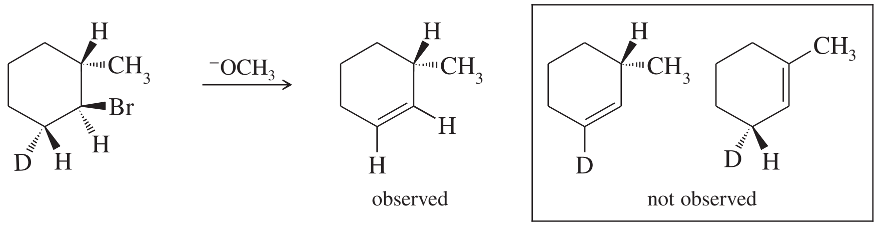


FIGURE 7-11 E2 eliminations on cyclohexane rings. E2 elimination of bromocyclohexane requires that both the proton and the leaving group be trans and both be axial.

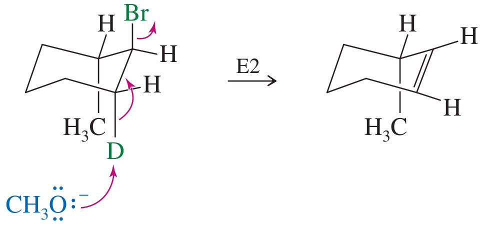
SOLVED PROBLEM 7-5

Explain why the following deuterated 1-bromo-2-methylcyclohexane undergoes dehydrohalogenation by the E2 mechanism to give only the indicated product. Two other alkenes are not observed.



SOLUTION

In an E2 elimination, the hydrogen atom and the leaving group must have a trans-diaxial relationship. In this compound, only one hydrogen atom—the deuterium—is trans to the bromine atom. When the bromine atom is axial, the adjacent deuterium is also axial, providing a trans-diaxial arrangement.



PROBLEM-SOLVING HINT

In a chair conformation of a cyclohexane ring, a trans-diaxial arrangement places the two groups anti and coplanar.

PROBLEM-SOLVING HINT

Look for a hydrogen trans to the leaving group; then see if the hydrogen and the leaving group can become diaxial.

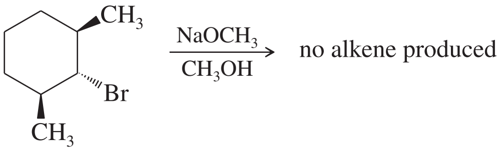
PROBLEM 7-28

Predict the elimination products of the following reactions, and label the major products.

(a) *cis*-1-bromo-2-methylcyclohexane + $NaOCH_3$ in CH_3OH
(b) *trans*-1-bromo-2-methylcyclohexane + $NaOCH_3$ in CH_3OH

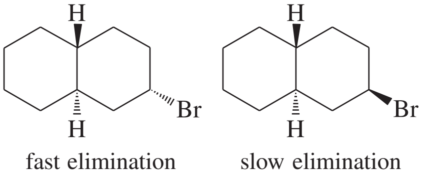
PROBLEM 7-29

When the following stereoisomer of 2-bromo-1,3-dimethylcyclohexane is treated with sodium methoxide, no E2 reaction is observed. Explain why this compound cannot undergo the E2 reaction in the chair conformation.

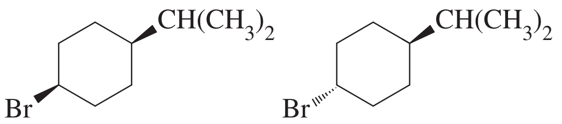


PROBLEM 7-30

(a) Two stereoisomers of a bromodecalin are shown. Although the difference between these stereoisomers may seem trivial, one isomer undergoes elimination with KOH much faster than the other. Predict the products of these eliminations, and explain the large difference in the ease of elimination.



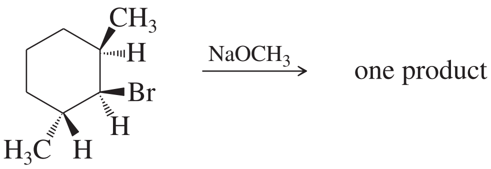
(b) Predict which of the following compounds will undergo elimination with KOH faster, and explain why. Predict the major product that will be formed.



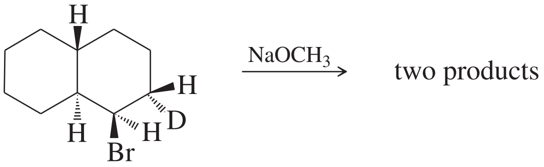
PROBLEM 7-31

Give the expected product(s) of E2 elimination for each reaction. (*Hint*: Use models!)

(a)



(b)



7-16 Comparison of E1 and E2 Elimination Mechanisms

Let's summarize the major points to remember about the E1 and E2 reactions, focusing on the factors that help us predict which of these mechanisms will operate under a given set of experimental conditions.

Effect of the Base The nature of the base is the single most important factor in determining whether an elimination will go by the E1 or E2 mechanism. If a strong base is present, the rate of the bimolecular reaction will be greater than the rate of ionization, and the E2 reaction will predominate (perhaps accompanied by the S_N2).

If no strong base is present, then a good solvent makes a unimolecular ionization likely. Subsequent loss of a proton to a weak base (such as the solvent) leads to elimination. Under these conditions, the E1 reaction usually predominates, usually accompanied by the S_N1.

E1:	Base strength is unimportant (usually weak).
E2:	Requires strong bases.

Effect of the Solvent The slow step of the E1 reaction is the formation of two ions. Like the S_N1, the E1 reaction critically depends on polar ionizing solvents such as water and the alcohols.

In the E2 reaction, the transition state spreads out the negative charge of the base over the entire molecule. There is no more need for solvation in the E2 transition state than in the reactants. The E2 is therefore less sensitive to the solvent; in fact, some reagents are stronger bases in less polar solvents.

E1:	Requires a good ionizing solvent.
E2:	Solvent polarity is not so important.

Effect of the Substrate For both the E1 and the E2 reactions, the order of reactivity is

E1, E2: $3° > 2° > 1°$ (1° usually will not go E1)

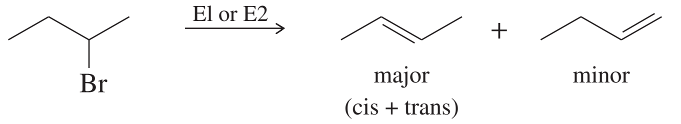
In the E1 reaction, the rate-limiting step is formation of a carbocation, and the reactivity order reflects the stability of carbocations. In the E2 reaction, the more substituted halides generally form more substituted, more stable alkenes.

Kinetics The rate of the E1 reaction is proportional to the concentration of the alkyl halide [RX] but not to the concentration of the weak base. It follows a first-order rate equation.

The rate of the E2 reaction is proportional to the concentrations of both the alkyl halide [RX] and the strong base [B:$^-$]. It follows a second-order rate equation.

$$\text{E1 rate} = k_r[\text{RX}]$$
$$\text{E2 rate} = k_r[\text{RX}][\text{B:}^-]$$

Orientation of Elimination In most E1 and E2 eliminations with two or more possible products, the product with the most substituted double bond (the most stable product) predominates. This principle is called **Zaitsev's rule,** and the most highly substituted product is called the **Zaitsev product.**



E1, E2: Usually Zaitsev orientation.

Stereochemistry The E1 reaction begins with an ionization to give a flat carbocation. No particular geometry is required for ionization.

The E2 reaction takes place through a concerted mechanism that requires a coplanar arrangement of the bonds to the atoms being eliminated. The transition state is usually anti-coplanar, although it may be syn-coplanar in rigid systems.

E1: No particular geometry required for the slow step.
E2: Coplanar arrangement (usually anti) required for the transition state.

Rearrangements The E1 reaction involves a carbocation intermediate. This intermediate can rearrange, usually by the shift of a hydride or an alkyl group, to give a more stable carbocation.

The E2 reaction takes place in one step with no intermediates. No rearrangement is possible in the E2 reaction.

E1: Rearrangements are common.
E2: No rearrangements.

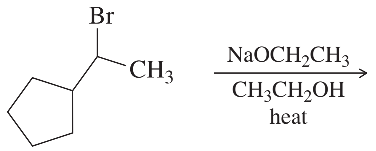
SUMMARY Elimination Reactions

	E1	E2
Promoting factors		
base	weak bases work	strong base required
solvent	good ionizing solvent	wide variety of solvents
substrate	$3° > 2°$	$3° > 2° > 1°$
leaving group	good one required	good one required
Characteristics		
kinetics	first order, $k_r[\text{RX}]$	second order, $k_r[\text{RX}][\text{B:}^-]$
orientation	most substituted alkene	most substituted alkene
stereochemistry	no special geometry	coplanar transition state required
rearrangements	common	impossible

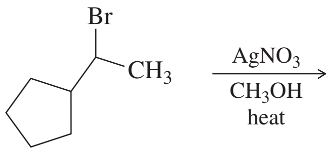
PROBLEM 7-32

Predict the major and minor elimination products of the following proposed reactions (ignoring any possible substitutions for now). In each case, explain whether you expect the mechanism of the elimination to be E1 or E2.

(a)



(b)

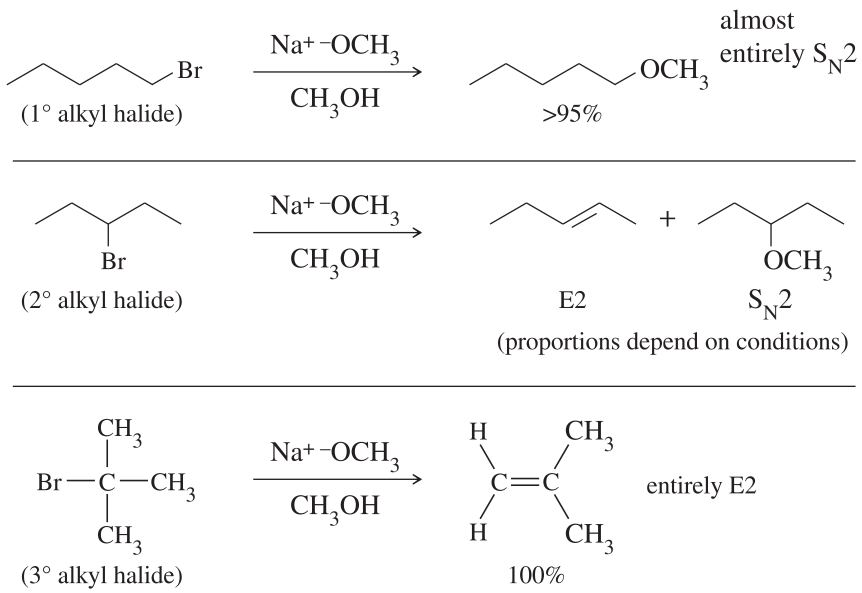


7-17 Competition Between Substitutions and Eliminations

Substitutions require the substrate to react with a nucleophile, and eliminations require it to react with a base. Most bases are also nucleophilic, however, and most nucleophiles are also basic; so both substitution and elimination may be possible in the very same reaction. To predict the products of a reaction, we need to predict both the mechanism and whether substitution or elimination will predominate.

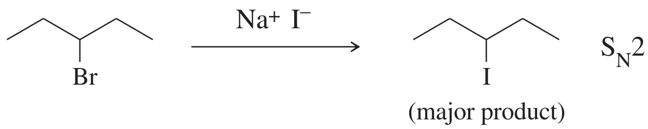
7-17A Bimolecular Substitution Versus Elimination (Strong Base or Nucleophile)

In most cases, the order of a reaction is determined by the strength of the reagent that serves as the base or nucleophile. Strong reagents are likely to attack the substrate faster than the substrate ionizes, so strong reagents favor bimolecular, second-order reactions: S_N2 and E2. For example, sodium methoxide ($NaOCH_3$) is both a strong base and a strong nucleophile. Sodium methoxide narrows the likely reaction mechanisms to S_N2 and E2, but it does not imply which of these will predominate. The following examples show substrates that give almost entirely substitution (1°), entirely elimination (3°), and a mixture of the two reactions (2°).

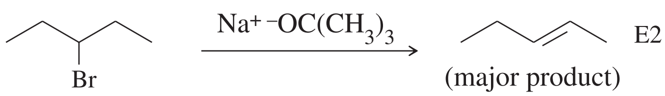


These examples show the principles that apply to bimolecular substitutions and eliminations. Primary halides are most reactive toward S_N2 reactions, and they are least reactive toward E2 reactions. Primary halides usually react with strong nucleophiles to give substitution by the S_N2 mechanism. If the reagent is also a strong base, a small amount of elimination might also occur. Tertiary halides cannot undergo S_N2 reactions, so we can safely predict that a strong base will react with a tertiary halide to give only elimination by the E2 mechanism.

Secondary halides are hardest to predict. They react with strong bases by the E2 mechanism and with strong nucleophiles by the S_N2 mechanism. But most strong nucleophiles are also strong bases, so the proportions of substitution and elimination may be controlled by other factors such as temperature, solvent, and structural effects such as steric hindrance. A few strong nucleophiles are not strong bases, however, and they can promote S_N2 substitution over E2 elimination. Halide ions are the most common examples of strong nucleophiles that are weak bases.



Bulky bases are often used to promote E2 elimination over S_N2 substitution. Bulky bases are hindered from approaching the back side of a carbon atom to do the S_N2 displacement, but they can easily approach a hydrogen atom on a neighboring carbon and abstract a proton.



7-17B Unimolecular Substitution Versus Elimination (Weak Base or Nucleophile)

In the absence of a strong base or strong nucleophile and in the presence of a good ionizing solvent (very polar, often hydroxylic), the reaction rate is first-order, limited by the rate of unimolecular ionization to form a carbocation. Under these conditions, we have little control over the fast second step, when the reagent either attacks the carbocation to give substitution (S_N1) or removes an adjacent proton to give elimination (E1). The carbocation intermediate can also rearrange, leading to even more products. This lack of control explains why unimolecular substitutions and eliminations of alkyl halides are rarely used for synthesis.

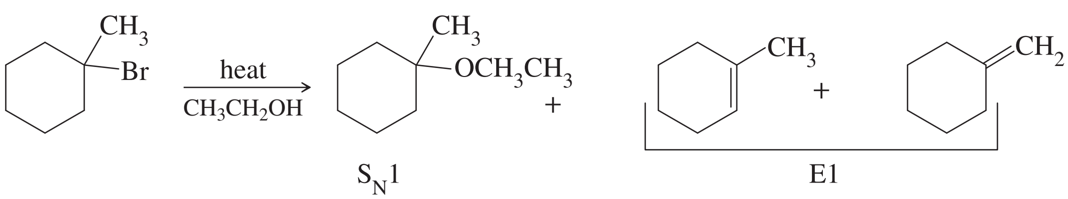
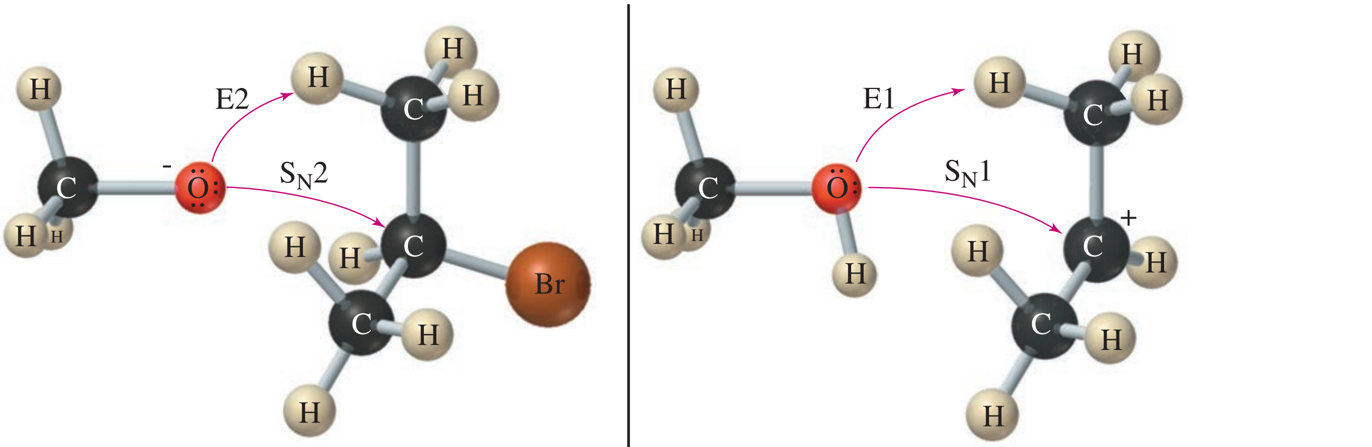


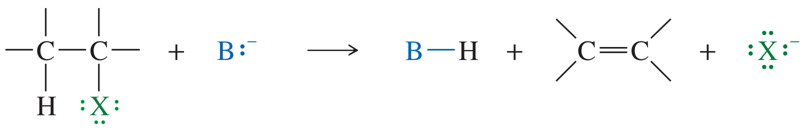
Figure 7-12 compares the substitution and elimination mechanisms under second-order (bimolecular) conditions with those that occur under first-order (unimolecular) conditions. A strong reagent is likely to displace the leaving group from carbon (S_N2) or remove a proton from a neighboring carbon (E2). A weak reagent is unlikely to react until ionization occurs. Then it may form a bond to the carbocation (S_N1) or remove a proton from a neighboring carbon (E1).

Temperature is one variable we can control to influence the competition between substitution and elimination. Unlike substitutions, eliminations produce more products than reactants.



second-order conditions (strong base/nucleophile) | first-order conditions (weak base/nucleophile)

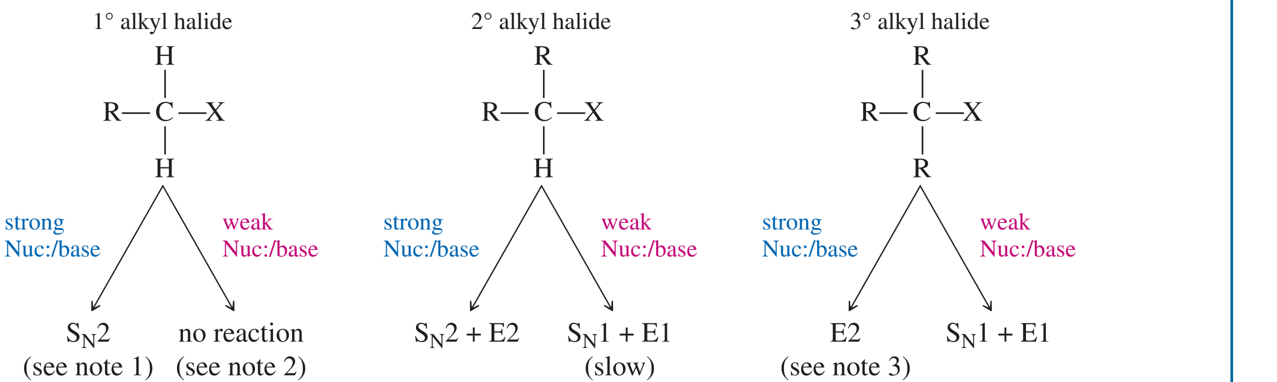
FIGURE 7-12 Under second-order conditions (strong base/nucleophile), a secondary alkyl halide might undergo either substitution (S_N2) or elimination (E2). Under first-order conditions (weak base/nucleophile), S_N1 and E1 are possible.



Reactions that produce more products than reactants generally increase the entropy of the system, meaning $\Delta S > 0$. By raising the temperature, we can increase the $(T\Delta S)$ term in the free-energy equation, $\Delta G = \Delta H - T\Delta S$, contributing to a more favorable (negative) value of ΔG. Therefore, an increase in temperature usually favors elimination, while a decrease in temperature favors substitution. This general principle applies to both unimolecular and bimolecular reactions when other factors are evenly balanced.

SUMMARY TABLE: Substitution and Elimination Reactions of Alkyl Halides

The reaction products and mechanisms depend on the type of alkyl halide and on the strength of the nucleophile/base.

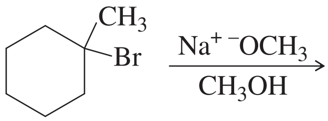


Note 1: In the presence of a strong nucleophile, unhindered 1° alkyl halides usually give good yields of substitution products with little elimination.
Note 2: 1° alkyl halides can be forced to ionize by $AgNO_3$/EtOH and heat, usually with rearrangement, to give S_N1 and E1 products.
Note 3: Any strongly hindered alkyl halide (even if 1° or 2°) will undergo S_N2 reactions very slowly or not at all.

SOLVED PROBLEM 7-6

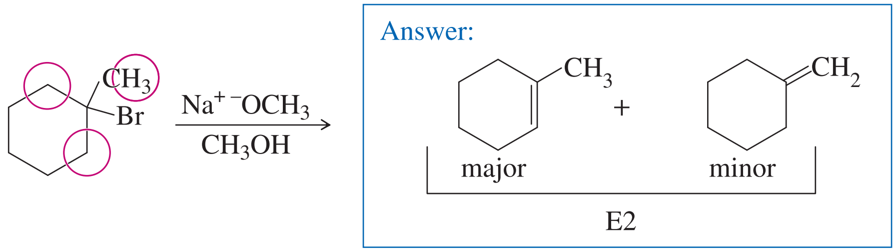
Predict the major products of the following reactions, and indicate the mechanisms (S_N1, S_N2, E1, or E2) that are likely responsible for their formation.

(a) 1-bromo-1-methylcyclohexane reacts with sodium methoxide in methanol.



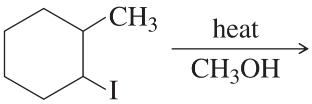
SOLUTION

(a) $NaOCH_3$ is a strong base and a strong nucleophile, suggesting that S_N2 and E2 are the likely reactions. This 3° alkyl halide cannot undergo S_N2 reactions, so the only remaining possibility is elimination by the E2 mechanism. The elimination can take place by abstraction of a methyl proton (circled below) or a ring proton on one of the CH_2 groups (circled) next to the carbon bonded to Br. $NaOCH_3$ is not a bulky base, so Zaitsev's rule probably applies.



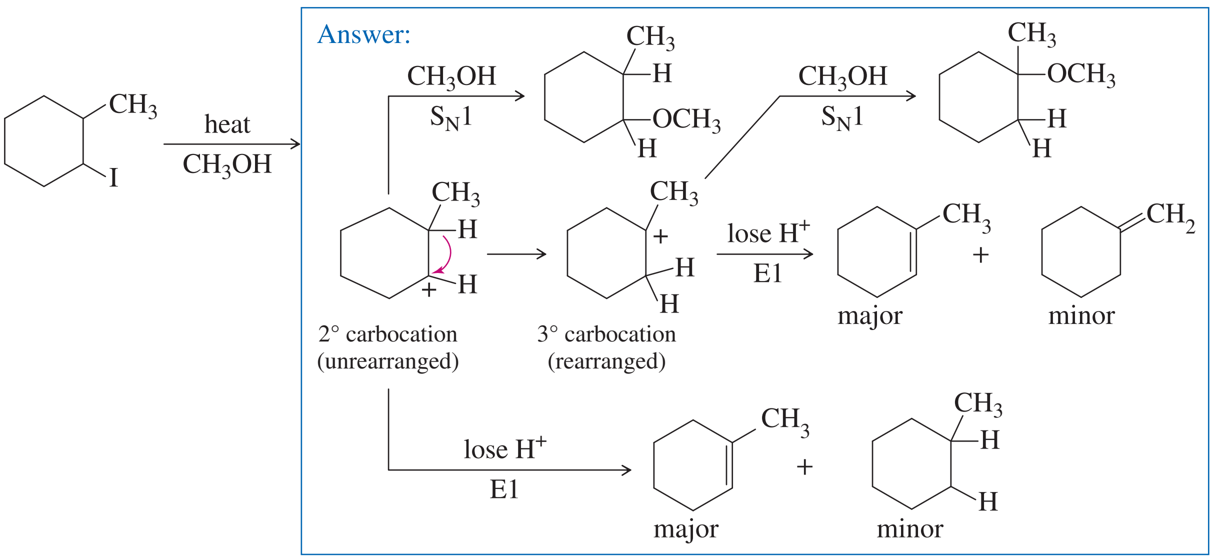
Follow-up question: Draw a chair transition state for this elimination, showing how a trans-diaxial arrangement of the H and Br allows E2 elimination to give the major product.

(b) 1-iodo-2-methylcyclohexane is heated in methanol for an extended period.



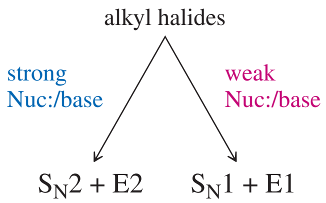
SOLUTION

(b) A slow ionization of this secondary alkyl halide leads to formation of a 2° carbocation, which can lose a proton in either of two directions to give products of E1 elimination. Alternatively, methanol can attack the 2° carbocation, followed by loss of a proton, to give a product of S_N1 substitution. The 2° carbocation can also rearrange by a hydride shift to give a more stable 3° carbocation. That 3° carbocation can lose a proton in either of two directions to give products of E1 elimination, or it can react with methanol, followed by deprotonation, to give a product of S_N1 substitution. We don't know what the major product will be or even if there will be one major product.

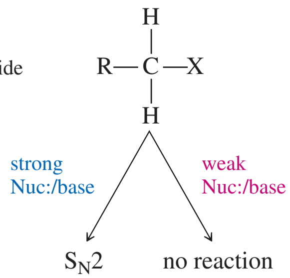


Among the elimination products, Zaitsev's rule tells us that the product with the most substituted double bond will predominate. That is the same product for both the initial carbocation and the rearranged carbocation. However, a substitution product might be the major product. Which one depends on whether the rearrangement is faster than attack by the methanol solvent. If the rearrangement is faster, then we see more rearranged product. This complicated reaction shows why E1 and S_N1 reactions of alkyl halides are rarely used for synthesis.
Follow-up question: Show the mechanistic steps that are omitted in the solution to (b) shown above.

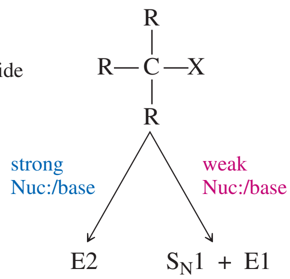
PROBLEM-SOLVING STRATEGY Predicting Substitutions and Eliminations



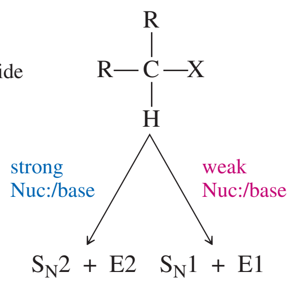
1° alkyl halide



3° alkyl halide



2° alkyl halide



Given a set of reagents and solvents, how can you predict what products will result and which mechanisms will be involved? Should you memorize all this theory about substitutions and eliminations? Students sometimes feel overwhelmed at this point.

Memorizing is not the best way to approach this material because the answers are not absolute and too many factors are involved. Besides, the real world with its real reagents and solvents is not as clean as our equations on paper. Most nucleophiles are also basic, and many solvents can solvate ions or react as nucleophiles or bases.

The first principle you must understand is that *you cannot always predict one unique product or one unique mechanism*. Often, the best you can do is to eliminate some of the possibilities and make some accurate predictions. Remembering this limitation, here are some general guidelines:

1. **The strength of the base or nucleophile determines the order of the reaction.**
   If a strong nucleophile (or base) is present, it will force second-order kinetics, either S_N2 or E2.
   If no strong base or nucleophile is present, you should consider first-order reactions, both S_N1 and E1. Addition of silver salts to the reaction can force some difficult ionizations.
2. **Primary halides usually undergo the S_N2 reaction, occasionally the E2 reaction.**
   Primary halides rarely undergo first-order reactions, unless the carbocation is resonance-stabilized. With good nucleophiles, S_N2 substitution is usually observed. With a strong base, E2 elimination may occasionally be observed.
3. **Tertiary halides usually undergo the E2 reaction (strong base) or a mixture of S_N1 and E1 (weak base).**
   Tertiary halides cannot undergo the S_N2 reaction. A strong base forces second-order kinetics, resulting in elimination by the E2 mechanism. In the absence of a strong base, tertiary halides react by first-order processes, usually a mixture of S_N1 and E1. The specific reaction conditions determine the ratio of substitution to elimination.
4. **The reactions of secondary halides are the most difficult to predict.**
   With a strong base, either the S_N2 or the E2 reaction is possible. With a weak base and a good ionizing solvent, both the S_N1 and E1 reactions are possible, but both are slow. Mixtures of products are common.
5. **Some nucleophiles and bases favor substitution or elimination.**
   To promote elimination, the base should readily abstract a proton but not readily attack a carbon atom. A bulky strong base, such as *tert*-butoxide [$^-OC(CH_3)_3$], enhances elimination. Higher temperatures also favor elimination in most cases. To promote substitution, you need a good nucleophile with limited basicity: a highly polarizable species that is the conjugate base of a strong acid. Bromide (Br^-) and iodide (I^-) are examples of good nucleophiles that are weak bases and favor substitution.

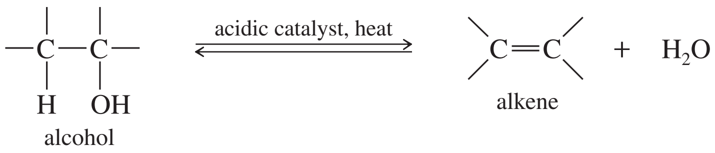
PROBLEM 7-33

Predict the products and mechanisms of the following reactions. When more than one product or mechanism is possible, explain which are most likely.

(a) 1-bromohexane + sodium ethoxide in ethanol
(b) 2-chlorohexane + $NaOCH_3$ in methanol
(c) 2-chloro-2-methylbutane + $NaOCH_2CH_3$ in ethanol
(d) 2-chloro-2-methylbutane heated in ethanol
(e) isobutyl iodide + KOH in ethanol/water
(f) isobutyl chloride + $AgNO_3$ in ethanol/water
(g) 1-bromo-1-methylcyclopentane + NaOEt in ethanol
(h) 1-bromo-1-methylcyclopentane heated in methanol

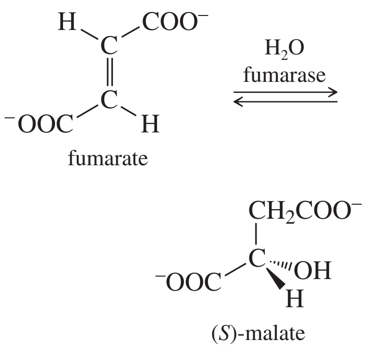
7-18 Alkene Synthesis by Dehydration of Alcohols

Thus far, we have studied the eliminations that convert alkyl halides to alkenes. Alkenes are also made from a variety of other types of compounds. A common laboratory synthesis of alkenes involves the elimination of the elements of water from alcohols under acidic conditions. This reaction is called **dehydration,** which literally means "removal of water."



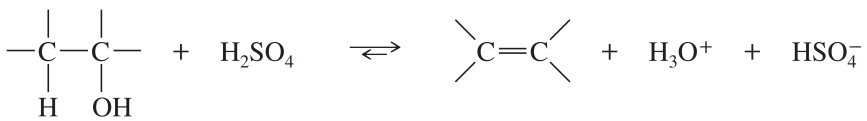
Application: Biochemistry

Hydration and dehydration reactions are common in biological pathways. In one of the steps in cellular respiration, the enzyme fumarase catalyzes the reversible addition of water to the double bond of fumarate to form malate. In contrast to the harsh acidic conditions used in the laboratory reaction, the enzymatic reaction takes place at neutral pH and at 37 °C. The enzymatic reaction forms only one enantiomer of the product.



Dehydration is reversible, and in most cases, the equilibrium constant is not large. In fact, the reverse reaction (hydration) is a method for converting alkenes to alcohols (see Section 8-4). Dehydration can be forced to completion by removing the products from the reaction mixture as they form. The alkene boils at a lower temperature than the alcohol because the alcohol is hydrogen-bonded. A carefully controlled distillation removes the alkene while leaving the alcohol in the reaction mixture.

Concentrated sulfuric acid and/or concentrated phosphoric acid are often used as reagents for dehydration because these acids act both as acidic catalysts and as dehydrating agents. Hydration of these acids is strongly exothermic. The overall reaction (using sulfuric acid) is

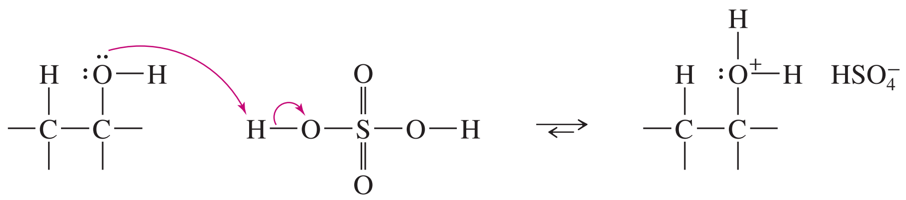


The mechanism of dehydration resembles the E1 mechanism (Key Mechanism 7-1, page 366). The hydroxy group of the alcohol is a poor leaving group (^-OH), but protonation by the acidic catalyst in the first step converts it to a good leaving group (H_2O). In the second step, loss of water from the protonated alcohol gives a carbocation. The carbocation is a very strong acid: Any weak base such as H_2O or HSO_4^- can abstract the proton in the final step to give the alkene.

KEY MECHANISM 7-5 Acid-Catalyzed Dehydration of an Alcohol

Alcohol dehydrations usually involve E1 elimination of the protonated alcohol.

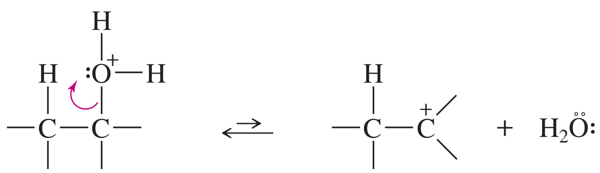
Step 1: Protonation of the hydroxy group (fast equilibrium).



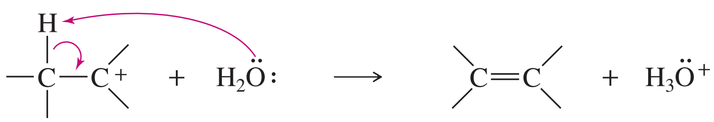
PROBLEM-SOLVING HINT

In acid-catalyzed mechanisms, the first step is often addition of H^+, and the last step is often loss of H^+.

Step 2: Ionization to a carbocation (slow; rate limiting).

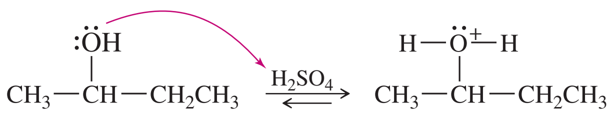


Step 3: Deprotonation to give the alkene (fast).

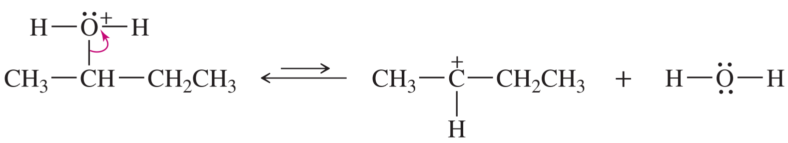


EXAMPLE: Acid-catalyzed dehydration of butan-2-ol

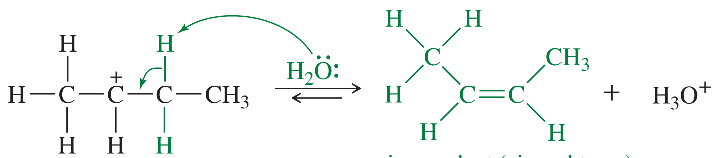
Step 1: Protonation of the hydroxy group (fast equilibrium).



Step 2: Ionization to a carbocation (slow; rate limiting).

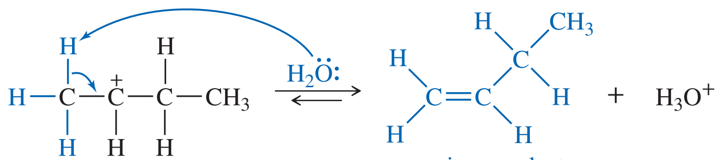


Step 3: Deprotonation to give the alkene (fast).



major product (cis and trans)

or



minor product

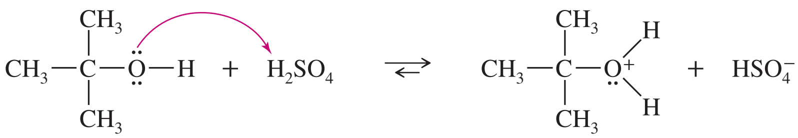
Like other E1 reactions, alcohol dehydration follows an order of reactivity that reflects carbocation stability: 3° alcohols react faster than 2° alcohols, and 1° alcohols are the least reactive. Rearrangements of the carbocation intermediates are common in alcohol dehydrations. In most cases, Zaitsev's rule applies: The major product is usually the one with the most substituted double bond.

SOLVED PROBLEM 7-7

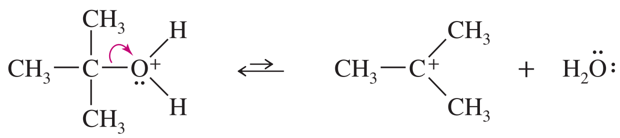
(a) Propose a mechanism for the sulfuric acid-catalyzed dehydration of *tert*-butyl alcohol.

SOLUTION

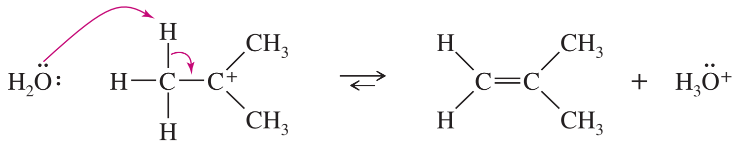
The first step is protonation of the hydroxy group, which converts it to a good leaving group.



The second step is ionization of the protonated alcohol to give a carbocation.



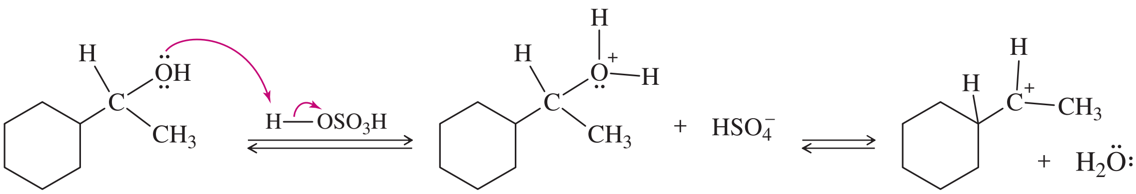
Abstraction of a proton completes the mechanism.



(b) Predict the products and propose a mechanism for the acid-catalyzed dehydration of 1-cyclohexylethanol.

PARTIAL SOLUTION:

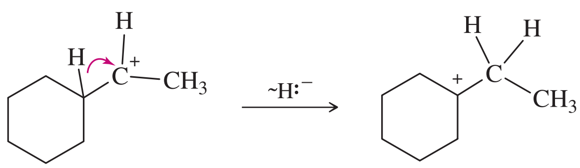
Protonation of the hydroxy group, followed by loss of water, forms a carbocation.



1-cyclohexylethanol

carbocation

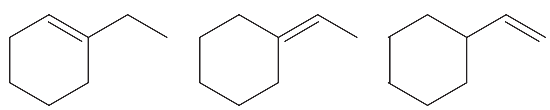
The carbocation can lose a proton, or it can rearrange to a more stable carbocation.



unrearranged 2° carbocation

rearranged 3° carbocation

Complete this problem by showing how the unrearranged 2° carbocation can lose either of two protons to give two of the following products. Also show how the rearranged 3° carbocation can lose either of two protons to give two of the following products.



PROBLEM-SOLVING HINT

Alcohol dehydrations usually go through E1 elimination of the protonated alcohol.

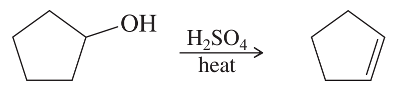
Reactivity is:
3° > 2° >> 1°
Rearrangements are common.

Protonated primary alcohols dehydrate at elevated temperatures with rearrangement (E1), or the adjacent carbon may lose a proton to a weak base at the same time water leaves (E2).

PROBLEM 7-34

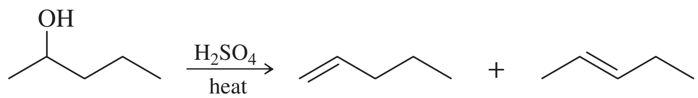
Propose mechanisms for the following reactions.

(a)



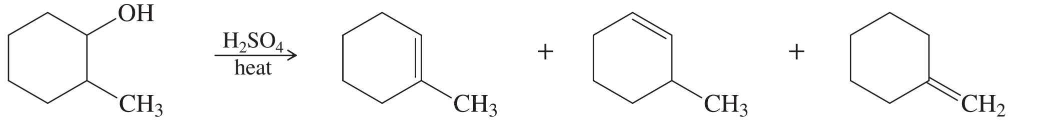
cyclopentanol → cyclopentene

(b)



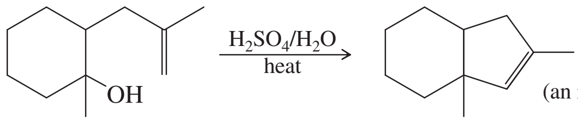
pentan-2-ol → pent-1-ene + pent-2-ene (cis + trans)

(c)



2-methylcyclohexanol → 1-methylcyclohexene + 3-methylcyclohexene + methylenecyclohexane

(d)

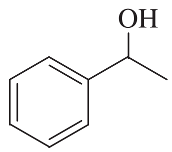


(an interesting minor product)

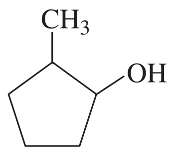
PROBLEM 7-35

Show the product(s) you expect from dehydration of the following alcohols when they are heated in sulfuric or phosphoric acid. In each case, use a mechanism to show how the products are formed.

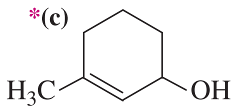
(a)



(b)



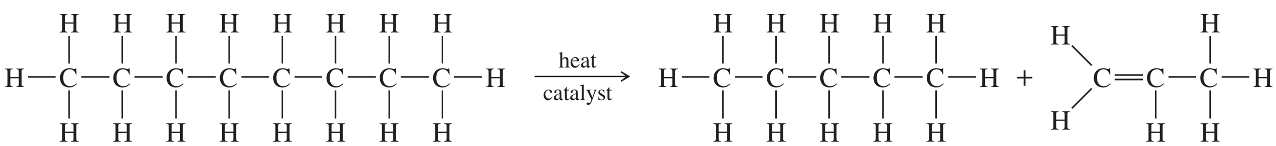
***(c)**



7-19 Alkene Synthesis by High-Temperature Industrial Methods

7-19A Catalytic Cracking of Alkanes

The least expensive way to make alkenes on a large scale is by the **catalytic cracking** of petroleum: heating a mixture of alkanes in the presence of a catalyst (usually aluminosilicates). Alkenes are formed by bond cleavage to give an alkene and a shortened alkane.



long-chain alkane → shorter alkane + alkene

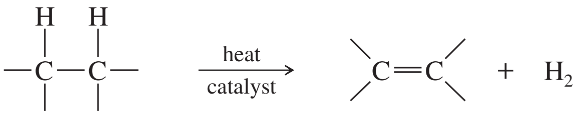
Cracking is used primarily to make small alkenes, up to about six carbon atoms. Its value depends on having a market for all the different alkenes and alkanes produced. The average molecular weight and the relative amounts of alkanes and alkenes can be controlled by varying the temperature, catalyst, and concentration of hydrogen in the cracking process. A careful distillation on a huge column separates the mixture into its pure components, ready to be packaged and sold.

This plant in Tokuyama, Japan, passes ethane rapidly over a hot catalyst. The products are ethylene and hydrogen.

Because the products are always mixtures, catalytic cracking is unsuitable for laboratory synthesis of alkenes. Better methods are available for synthesizing relatively pure alkenes from a variety of other functional groups. Dehydrohalogenation of alkyl halides and dehydration of alcohols are better methods shown in this chapter, and additional methods are included in the Summary on page 385.

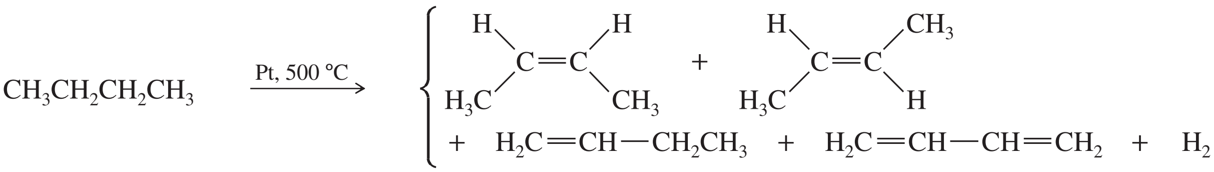
7-19B Dehydrogenation of Alkanes

Dehydrogenation is the removal of H_2 from a molecule, just the reverse of hydrogenation. Dehydrogenation of an alkane gives an alkene. This reaction has an unfavorable enthalpy change but a favorable entropy change.



alkane alkene

$\Delta H° = +80$ to $+120$ kJ/mol ($+20$ to $+30$ kcal/mol) $\quad \Delta S° = +125$ J/kelvin-mol



The hydrogenation of alkenes (Section 7-8) is exothermic, with values of $\Delta H°$ around -80 to -120 kJ/mol (-20 to -30 kcal/mol). Therefore, dehydrogenation is endothermic and has an unfavorable (positive) value of $\Delta H°$. The entropy change for dehydrogenation is strongly favorable ($\Delta S° = +120$ J/kelvin-mol), however, because one alkane molecule is converted into two molecules (the alkene and hydrogen), and two molecules are more disordered than one.

The equilibrium constant for the hydrogenation–dehydrogenation equilibrium depends on the change in free energy, $\Delta G = \Delta H - T\Delta S$. At room temperature, the enthalpy term predominates and hydrogenation is favored. When the temperature is raised, however, the entropy term ($-T\Delta S$) becomes larger and eventually dominates the expression. At a sufficiently high temperature, dehydrogenation is favored.

PROBLEM-SOLVING HINT

When you are doing synthesis problems, avoid using these high-temperature industrial methods. They require specialized equipment, and they produce variable mixtures of products.

PROBLEM 7-36

The dehydrogenation of butane to *trans*-but-2-ene has $\Delta H° = +116$ kJ/mol ($+27.6$ kcal/mol) and $\Delta S° = +117$ J/kelvin-mol ($+28.0$ cal/kelvin-mol).

(a) Compute the value of $\Delta G°$ for dehydrogenation at room temperature (25 °C or 298 °K). Is dehydrogenation favored or disfavored?

(b) Compute the value of ΔG for dehydrogenation at 1000 °C, assuming that ΔS and ΔH are constant. Is dehydrogenation favored or disfavored?

In many ways, dehydrogenation is similar to catalytic cracking. In both cases, a catalyst lowers the activation energy, and both reactions use high temperatures to increase a favorable entropy term ($-T\Delta S$) and overcome an unfavorable enthalpy term (ΔH). Unfortunately, dehydrogenation and catalytic cracking also share a tendency to produce mixtures of products, and neither reaction is well suited for the laboratory synthesis of alkenes.

PROBLEM-SOLVING STRATEGY Proposing Reaction Mechanisms

At this point, we have seen examples of three major classes of reaction mechanisms:

- Those involving strong bases and strong nucleophiles
- Those involving strong acids and strong electrophiles
- Those involving free radicals

Many students have difficulty proposing mechanisms. You can use some general principles to approach this process, however, by breaking it down into a series of logical steps. Using a systematic approach, you can usually come up with a mechanism that is at least possible and that explains the products, without requiring any unusual steps. Appendix 3A contains more complete methods for approaching mechanism problems.

First, Classify the Reaction

Before you begin to propose a mechanism, you must determine what kind of reaction you are dealing with. Examine what you know about the reactants and the reaction conditions:

A free-radical initiator such as chlorine, bromine, or a peroxide (with heat or light) suggests that a free-radical chain reaction is most likely. Free-radical reactions were discussed in Chapter 4.

Strong bases or strong nucleophiles suggest mechanisms such as the S_N2 or E2, involving attack by the strong base or nucleophile on a substrate.

Strong acids or strong electrophiles (or a reactant that can dissociate to give a strong electrophile) suggest mechanisms such as the S_N1, E1, and alcohol dehydration that involve carbocations and other strongly acidic intermediates.

General Principles for Drawing Mechanisms

Once you have decided which type of mechanism is most likely (acidic, basic, or free-radical), some general principles can guide you in proposing the mechanism. Some principles for free-radical reactions were discussed in Chapter 4. Now we consider reactions that involve either strong nucleophiles or strong electrophiles as intermediates. In later chapters, we will apply these principles to more complex mechanisms.

Whenever you start to work out a mechanism, **draw all the bonds** and all the substituents of each carbon atom affected throughout the mechanism. Three-bonded carbon atoms are likely to be the reactive intermediates. If you attempt to draw condensed formulas or line–angle formulas, you will likely misplace a hydrogen atom and show the wrong carbon atom as a radical, a cation, or an anion.

Show only one step at a time; never combine steps, unless two or more bonds really do change position in one step (as in the E2 reaction, for example). Protonation of an alcohol and loss of water to give a carbocation, for example, must be shown as two steps. You must not simply circle the hydroxy and the proton to show water falling off.

Use curved arrows to show the movement of electrons in each step of the reaction. This movement is always *from* the nucleophile (electron donor) to the electrophile (electron acceptor). For example, protonation of an alcohol must show the arrow going from the electrons of the hydroxy oxygen to the proton—never from the proton to the hydroxy group. *Don't* use curved arrows to try to "point out" where the proton (or other reagent) goes.

Reactions Involving Strong Nucleophiles

When a strong base or nucleophile is present, we expect to see intermediates that are also strong bases and strong nucleophiles; anionic intermediates are common. Acids and electrophiles in such a reaction are generally weak. Avoid drawing carbocations, H_3O^+, and other strong acids. They are unlikely to coexist with strong bases and strong nucleophiles.

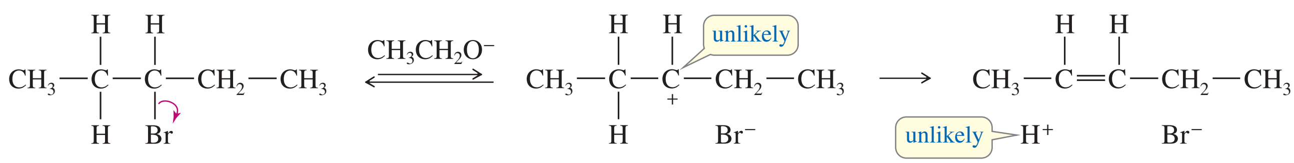
Functional groups are often converted to alkoxides, carbanions, or other strong nucleophiles by deprotonation or reaction with a strong nucleophile. Then the carbanion or other strong nucleophile reacts with a weak electrophile such as a carbonyl group or an alkyl halide.

Consider, for example, the mechanism for the dehydrohalogenation of 3-bromopentane.

$$CH_3{-}CH_2{-}\underset{\displaystyle Br}{\underset{|}{CH}}{-}CH_2{-}CH_3 \xrightarrow{CH_3CH_2O^-} CH_3{-}CH{=}CH{-}CH_2{-}CH_3$$

Someone who has not read Chapter 7 or these guidelines might propose an ionization, followed by loss of a proton:

Incorrect mechanism

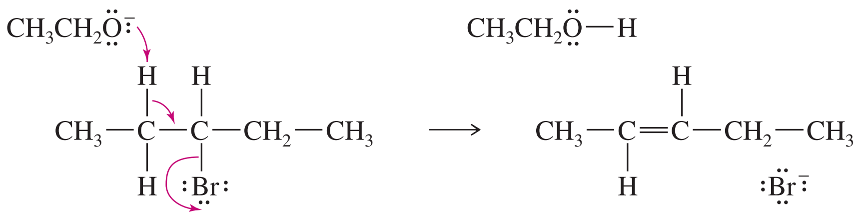


(*continued*)

This mechanism would violate several general principles of proposing mechanisms. First, in the presence of ethoxide ion (a strong base), both the carbocation and the H^+ ion are unlikely. Second, the mechanism fails to explain why the strong base is required; the rate of ionization would be unaffected by the presence of ethoxide ion. Also, H^+ doesn't just fall off (even in an acidic reaction); it must be removed by a base.

The presence of ethoxide ion (a strong base and a strong nucleophile) in the reaction suggests that the mechanism involves only strong bases and nucleophiles and not any strongly acidic intermediates. As shown in Section 7-12, the reaction occurs by the E2 mechanism, an example of a reaction involving a strong base. In this concerted reaction, ethoxide ion removes a proton as the electron pair left behind forms a pi bond and expels bromide ion.

Correct mechanism

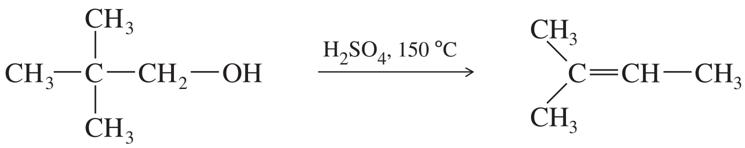


Reactions Involving Strong Electrophiles

When a strong acid or electrophile is present, expect to see intermediates that are also strong acids and strong electrophiles. Cationic intermediates are common, but avoid drawing any species with more than one + charge. Bases and nucleophiles in such a reaction are generally weak. Avoid drawing carbanions, alkoxide ions, and other strong bases. They are unlikely to coexist with strong acids and strong electrophiles.

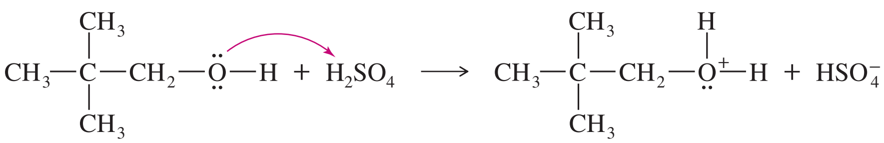
Functional groups are often converted to carbocations or other strong electrophiles by protonation or by reaction with a strong electrophile; then the carbocation or other strong electrophile reacts with a weak nucleophile such as an alkene or the solvent.

For example, consider the dehydration of 2,2-dimethylpropan-1-ol:



The presence of sulfuric acid indicates that the reaction is acidic and should involve strong electrophiles. The carbon skeleton of the product is different from the reactant. Under these acidic conditions, formation and rearrangement of a carbocation would be likely. The hydroxy group is a poor leaving group; it certainly cannot ionize to give a carbocation and ^-OH (and we do not expect to see a strong base such as ^-OH in this acidic reaction). The hydroxy group is weakly basic, however, and it can become protonated in the presence of a strong acid. The protonated OH group becomes a good leaving group.

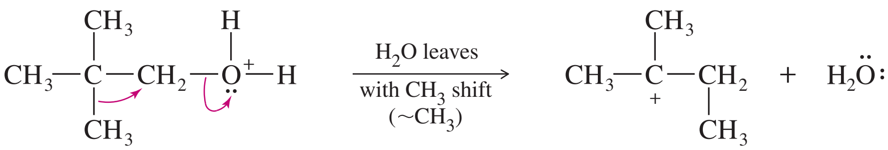
Step 1: Protonation of the hydroxy group



starting alcohol protonated alcohol

The protonated hydroxy group $-\overset{+}{O}H_2$ is a good leaving group. A simple ionization to a carbocation would form a primary carbocation. Primary carbocations, however, are very unstable. Thus, a methyl shift occurs as water leaves, so a primary carbocation is never formed. A tertiary carbocation results. (You can visualize this as two steps if you prefer.)

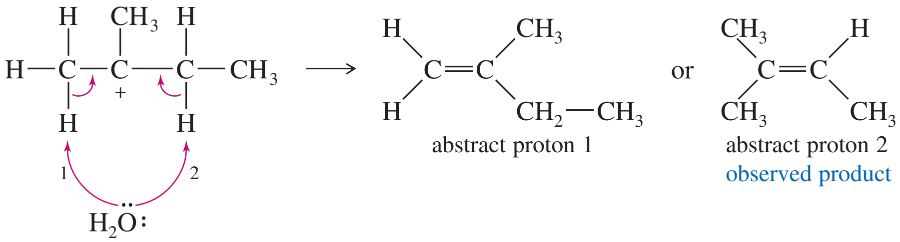
Step 2: Ionization with rearrangement



protonated alcohol tertiary carbocation

The final step is loss of a proton to a weak base, such as H_2O or HSO_4^- (but *not* ^-OH, which is incompatible with the acidic solution). Either of two types of protons, labeled 1 and 2 in the following figure, could be lost to give alkenes. Loss of proton 2 gives the required product.

Step 3: Abstraction of a proton to form the required product



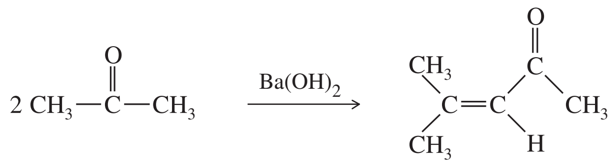
Because abstraction of proton 2 gives the more highly substituted (therefore more stable) product, Zaitsev's rule predicts it will be the major product. Note that in other problems, however, you may be asked to propose mechanisms to explain unusual compounds that are only minor products.

PROBLEM 7-37

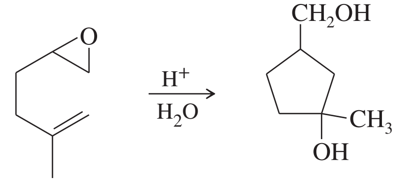
For practice in recognizing mechanisms, classify each reaction according to the type of mechanism you expect:

1. Free-radical chain reaction
2. Reaction involving strong bases and strong nucleophiles
3. Reaction involving strong acids and strong electrophiles

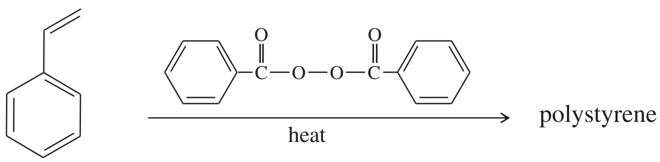
(a)



(b)



(c)



styrene

(d) ethylene $\xrightarrow{BF_3}$ polyethylene

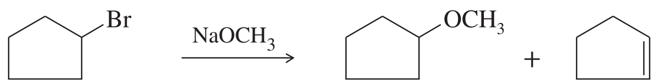
PROBLEM 7-38

Propose mechanisms for the following reactions. Additional products may be formed, but your mechanism only needs to explain the products shown.

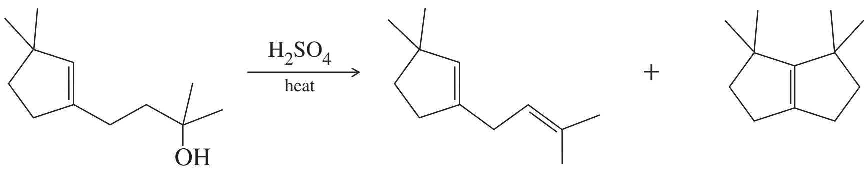
(a) $CH_3{-}CH_2{-}CH_2{-}CH_2{-}OH \xrightarrow{H_2SO_4,\ 140\ °C} CH_3{-}CH{=}CH{-}CH_3 + CH_2{=}CH{-}CH_2CH_3$

(*Hint*: Hydride shift)

(b)



(c)



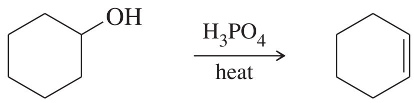
PROBLEM-SOLVING HINT

Alcohol dehydrations usually go through E1 elimination of the protonated alcohol, with a carbocation intermediate. Rearrangements are common.

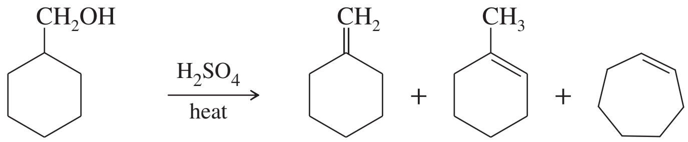
PROBLEM 7-39

Propose mechanisms for the following reactions.

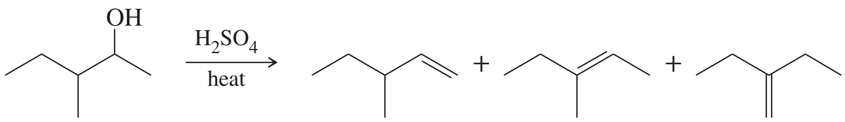
(a)



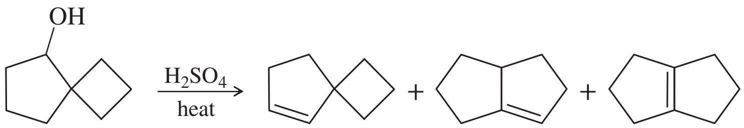
(b)



(c)

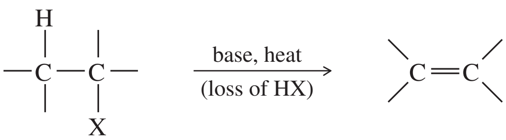


(d)



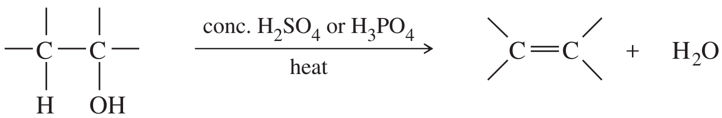
SUMMARY Methods for Synthesis of Alkenes

1. *Dehydrohalogenation of alkyl halides* (Section 7-9)



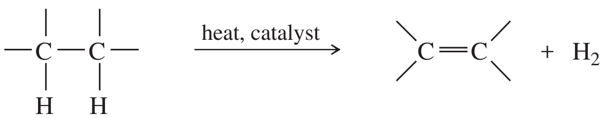
Requires an anti-coplanar conformation for the E2 reaction.

2. *Dehydration of alcohols* (Section 7-18)



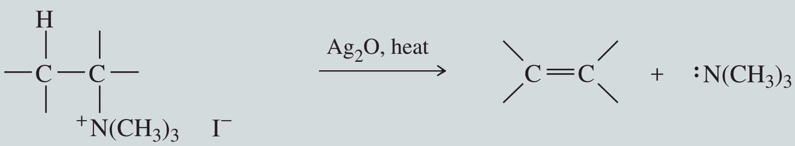
Rearrangements are common.

3. *Dehydrogenation of alkanes* (Section 7-19B)



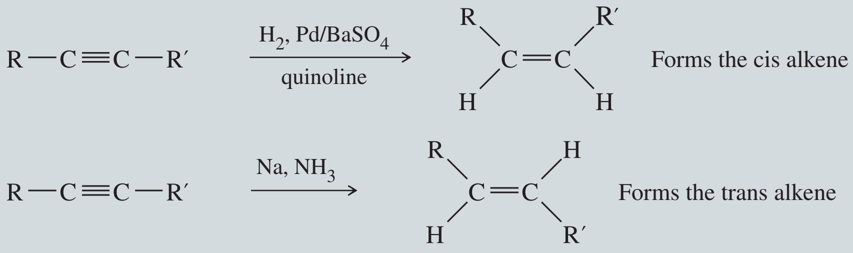
Industrial prep, not used in the lab.
Useful only for small alkenes.
Commonly gives mixtures.

4. *Hofmann and Cope eliminations* (Covered later in Sections 19-14 and 19-15)

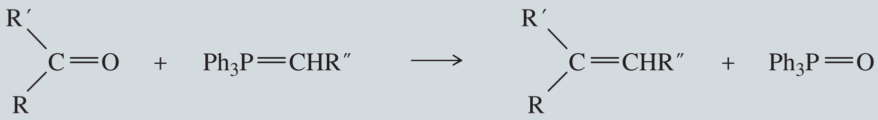


Usually gives the least substituted alkene.

5. *Reduction of alkynes* (Covered later in Section 9-9)



6. *Wittig reaction* (Covered later in Section 18-18)



Essential Terms

alkene — **(olefin)** A hydrocarbon with one or more carbon–carbon double bonds. (p. 342)

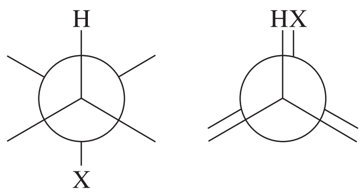
- **diene:** A compound with two carbon–carbon double bonds. (p. 348)
- **triene:** A compound with three carbon–carbon double bonds. (p. 348)
- **tetraene:** A compound with four carbon–carbon double bonds. (p. 348)

alkyl shift — (symbolized **~R**) Movement of an alkyl group with a pair of electrons from one atom (usually carbon) to another. Alkyl shifts are examples of rearrangements that convert carbocations into more stable carbocations. (pp. 330, 370)

allyl group — A vinyl group plus a methylene group: $CH_2{=}CH{-}CH_2{-}$ (p. 349)

anti — Adding to (or eliminating from) opposite faces of a molecule. (p. 376)

- **anti-coplanar** Having a dihedral angle of 180°.
- **syn-coplanar** Having a dihedral angle of 0°.



anti-coplanar   syn-coplanar

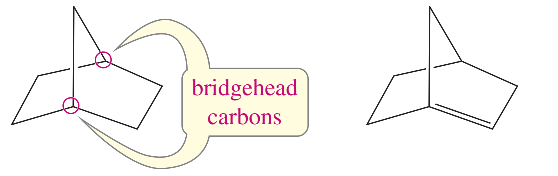
base — An electron-rich species that can abstract a proton. (pp. 107, 311)

- **basicity:** **(base strength)** The thermodynamic reactivity of a base, expressed quantitatively by the base-dissociation constant K_b. (p. 110)
- **Lewis base:** **(nucleophile)** An electron-rich species that can donate a pair of electrons to form a bond. (pp. 126, 306)

Bredt's rule — A stable bridged bicyclic compound cannot have a double bond at a bridgehead position unless one of the rings contains at least eight carbon atoms. (p. 361)

- **bicyclic:** Containing two rings.
- **bridged bicyclic:** Having at least one carbon atom in each of the three links connecting the bridgehead carbons. (p. 362)

bridgehead carbons: Those carbon atoms that are part of both rings, with three bridges of bonds connecting them.



a bridged bicyclic compound a Bredt's rule violation

catalytic cracking The heating of petroleum products in the presence of a catalyst (usually an aluminosilicate mineral), causing bond cleavage to form alkenes and alkanes of lower molecular weight. (p. 391)

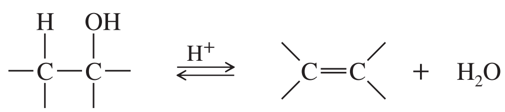
cis-trans isomers **(geometric isomers)** Isomers that differ in their cis-trans arrangement on a ring or double bond. Cis-trans isomers are a subclass of diastereomers. (p. 349)

- **cis:** Having similar groups on the same side of a double bond or a ring.
- **trans:** Having similar groups on opposite sides of a double bond or a ring.
- ***Z*:** Having the higher-priority groups on the same side of a double bond.
- ***E*:** Having the higher-priority groups on opposite sides of a double bond.

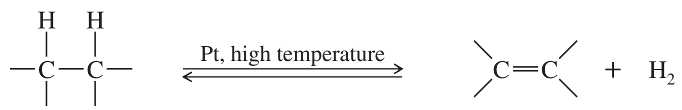
concerted reaction A reaction in which the breaking of bonds and the formation of new bonds occur at the same time (in one step). (p. 373)

conjugated double bonds Double bonds that are separated by only one single bond. The interaction between the pi bonds contributes to extra stability in the conjugated system. (p. 363)

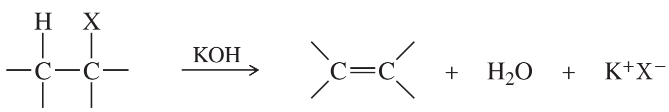
dehydration The elimination of water from a compound; usually acid-catalyzed. (p. 388)



dehydrogenation The elimination of hydrogen (H_2) from a compound; usually done in the presence of a catalyst. (p. 392)



dehydrohalogenation The elimination of a hydrogen halide (HX) from a compound; usually base-promoted. (p. 365)

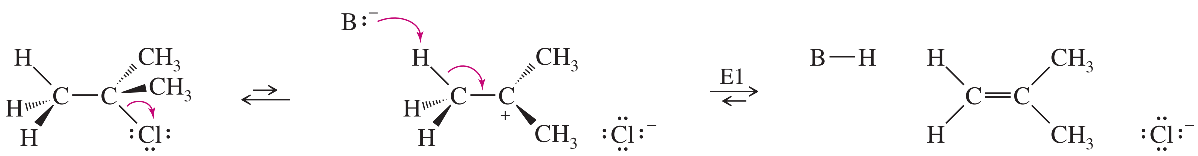


double-bond isomers Constitutional isomers that differ only in the position of a double bond. Double-bond isomers hydrogenate to give the same alkane. (p. 358)

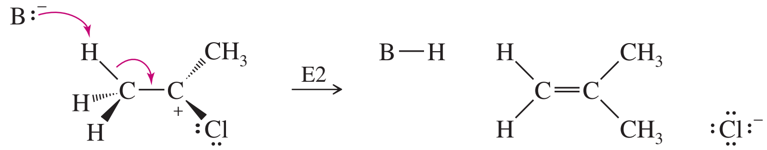
element of unsaturation **(degree of unsaturation, unsaturation number, index of hydrogen deficiency)** A structural feature that results in two fewer hydrogen atoms in the molecular formula. A double bond or a ring is one element of unsaturation; a triple bond is two elements of unsaturation. (p. 345)

elimination A reaction that involves the loss of two atoms or groups from the substrate, usually resulting in the formation of a pi bond. (p. 365)

- **E1 reaction:** **(elimination, unimolecular)** A multistep elimination where the leaving group is lost in a slow ionization step, then a proton is lost in a second step. Zaitsev orientation is generally preferred. (p. 366)



E2 reaction: **(elimination, bimolecular)** A concerted elimination involving a transition state where the base is abstracting a proton at the same time the leaving group is leaving. The anti-coplanar transition state is generally preferred. Zaitsev orientation is usually preferred, unless the base or the leaving group is unusually bulky. (p. 373)



geminal dihalide A compound with two halogen atoms on the same carbon atom.

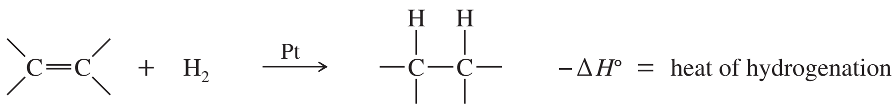
geometric isomers See **cis-trans isomers.** (p. 349)

heteroatom Any atom other than carbon or hydrogen. (p. 346)

Hofmann product The least highly substituted alkene product. (p. 375)

hydride shift (symbolized **~H**) Movement of a hydrogen atom with a pair of electrons from one atom (usually carbon) to another. Hydride shifts are examples of rearrangements that convert carbocations into more stable carbocations. (pp. 328, 370)

hydrogenation Addition of hydrogen to a molecule. The most common hydrogenation is the addition of H_2 across a double bond in the presence of a catalyst (*catalytic hydrogenation*). The value of $(-\Delta H°)$ for this reaction is called the **heat of hydrogenation.** (p. 392)



isolated double bonds Double bonds separated by two or more single bonds. Isolated double bonds react independently, as they do in a simple alkene. (p. 363)

methyl shift (symbolized **~CH_3**) Rearrangement of a methyl group with a pair of electrons from one atom (usually carbon) to another. A methyl shift (or any alkyl shift) in a carbocation generally results in a more stable carbocations. (p. 372)

nucleophile **(Lewis base)** An electron-rich species that can donate a pair of electrons to form a bond. (pp. 126, 367)

nucleophilicity: **(nucleophile strength)** The kinetic reactivity of a nucleophile; a measure of the rate of substitution in a reaction with a standard substrate. (p. 369)

olefin An alkene. (p. 342)

polymer A substance of high molecular weight made by linking many small molecules, called **monomers.** (p. 353)

addition polymer: A polymer formed by simple addition of monomer units.

polyolefin: A type of addition polymer with an olefin (alkene) serving as the monomer.

rearrangement A reaction involving a change in the bonding sequence within a molecule. Rearrangements are common in reactions such as the S_N1 and E1 involving carbocation intermediates. (p. 382)

resonance energy The extra stability of a conjugated system compared with the energy of a compound with an equivalent number of isolated double bonds. (p. 363)

saturated Having only single bonds; incapable of undergoing addition reactions. (p. 345)

Saytzeff Alternate spelling of Zaitsev.

solvolysis A nucleophilic substitution or elimination where the solvent serves as the attacking reagent. Solvolysis literally means "cleavage by the solvent." (p. 367)

$$(CH_3)_3C{-}Br \xrightarrow{CH_3OH.\ heat} (CH_3)_3C{-}OCH_3 \quad + \quad (CH_3)_2C{=}CH_2 \quad + \quad HBr$$

stereogenic Giving rise to stereoisomers. Characteristic of an atom or a group of atoms such that interchanging any two groups creates a stereoisomer. (p. 377)

stereospecific reaction A reaction in which different stereoisomers of the starting material react to give different stereoisomers of the product. (p. 377)

syn Adding to (or eliminating from) the same face of a molecule. (p. 376)

syn-coplanar: Having a dihedral angle of 0°. See **anti-coplanar** for a diagram.

trans-diaxial An anti and coplanar arrangement allowing E2 elimination of two adjacent substituents on a cyclohexane ring. The substituents must be trans to each other, and both must be in axial positions on the ring. (p. 379)

unsaturated Having multiple bonds that can undergo addition reactions. (p. 345)

vicinal dihalide A compound with two halogens on adjacent carbon atoms. (p. 294)

vinyl group An ethenyl group, $CH{=}CH{-}$. (p. 351)

Zaitsev's rule **(Saytzeff's rule)** Alkenes with more highly substituted double bonds are usually more stable. An elimination usually gives the most stable alkene product, commonly the most substituted alkene. Zaitsev's rule does not always apply, especially with a bulky base or a bulky leaving group. (pp. 382, 384, 390)

Zaitsev elimination: An elimination that gives the Zaitsev product.

Zaitsev product: The most substituted alkene product.

Essential Problem-Solving Skills in Chapter 7

Each skill is followed by problem numbers exemplifying that particular skill.

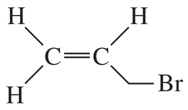
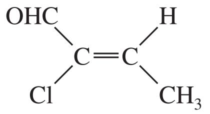
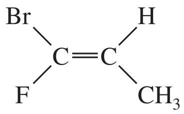
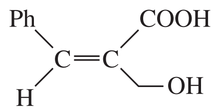
1	Draw and name alkenes, and calculate their elements of unsaturation. Name geometric isomers using both the *E-Z* and *cis-trans* systems.	Problems 7-40, 41, 42, 43, 44, 45, 46, and 47
2	Predict the relative stabilities of alkenes and cycloalkenes based on their structure and stereochemistry.	Problems 7-47, 48, and 70
3	Propose logical mechanisms for dehydrohalogenation of alkyl halides and dehydration of alcohols.	Problems 7-59, 61, 62, 63, 64, 67, 69, 72, 73, 74, 75, and 76
4	Predict the products of dehydrohalogenation and dehydration reactions, and use Zaitsev's rule to predict the major and minor products.	Problems 7-49, 50, 51, 52, 53, 54, 55, 67, 68, and 72
5	Given a set of reaction conditions, predict whether a reaction will be unimolecular or bimolecular, and identify the possible mechanisms and the likely products.	Problems 7-50, 54, 73, 75, and 76
	Problem-Solving Strategy: Predicting Substitutions and Eliminations	Problems 7-61, 62, 63, 67, 68, and 75
6	Draw and compare the mechanisms and reaction-energy diagrams of S_N1, S_N2, E1, and E2 reactions. Predict which substitutions or eliminations will be faster based on differences in substrate, leaving group, solvent, and base or nucleophile.	Problems 7-60, 71, 75, and 76
7	Explain the stereochemistry of E2 eliminations to form alkenes, and predict the products of E1 reactions on stereoisomers and on cyclohexane systems.	Problems 7-55, 65, 66, 67, 68, 69, and 76
	Problem-Solving Strategy: Proposing Reaction Mechanisms	Problems 7-59, 61, 62, 63, 64, 67, 69, 72, 73, 74, 75, and 76
8	Propose effective syntheses of alkenes from alkyl halides and alcohols.	Problems 7-56, 57, 58, 65, and 66

Study Problems

7-40 Draw a structure for each compound (includes old and new names).

(a) 3-methylhex-1-ene **(b)** *trans*-3-methyl-3-hexene **(c)** 3,4-dichlorobut-1-ene
(d) 1,3-cyclopentadiene **(e)** cyclohepta-1,4-diene **(f)** vinylcyclobutane
(g) (*E*)-2-bromo-2-hexene **(h)** (*Z*)-3-methyl-2-nonene **(i)** (*4E,6E*)-1,4,6-octatriene

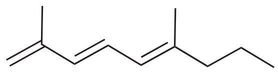
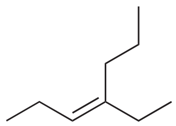
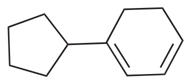
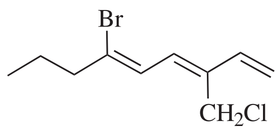
7-41 Label each structure as *Z, E*, or neither.

(a)  **(b)**  **(c)**  **(d)** 

7-42 Determine which compounds show cis-trans isomerism. Draw and label the isomers, using both the cis-trans and *E-Z* nomenclatures where applicable.

(a) pent-1-ene
(b) pent-2-ene
(c) hex-3-ene
(d) 1,1-dibromopropene
(e) 1,2-dibromopropene
(f) 1-bromo-1-chlorohexa-1,3-diene

7-43 Give a correct name for each compound.

(a) $CH_3CH_2C(=CH_2)CH_2CH_2CH_2CH_3$
(b) $(CH_3CH_2)_2C{=}CHCH_2CH_3$
(c) 
(d) 
(e) 
(f) 

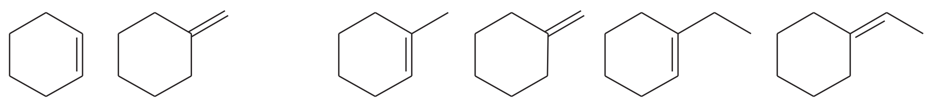
7-44 **(a)** Draw and name all five isomers of formula C_3H_5F.
(b) Draw all 12 acyclic (no rings) isomers of formula C_4H_7Br. Include stereoisomers.
(c) Cholesterol, $C_{27}H_{46}O$, has only one pi bond. With no additional information, what else can you say about its structure?

7-45 Draw and name all stereoisomers of 3-methylhepta-2,4-diene
(a) using the cis-trans nomenclature.
(b) using the *E-Z* nomenclature.

7-46 For each alkene, indicate the direction of the dipole moment. For each pair, determine which compound has the larger dipole moment.
(a) *cis*-1,2-dichloroethene or *trans*-1,2-dichloroethene
(b) *cis*-1,2-difluoroethene or *trans*-2,3-difluorobut-2-ene
(c) *cis*-1,2-dibromo-1,2-difluoroethene or *cis*-1,2-difluoroethene

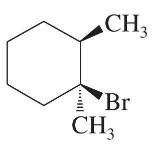
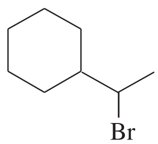
7-47 The energy difference between *cis*- and *trans*-but-2-ene is about 4 kJ/mol; however, the trans isomer of 4,4-dimethylpent-2-ene is nearly 16 kJ/mol more stable than the cis isomer. Explain this large difference.

7-48 A double bond in a six-membered ring is usually more stable in an endocyclic position than in an exocyclic position. Hydrogenation data on two pairs of compounds follow. One pair suggests that the energy difference between endocyclic and exocyclic double bonds is about 9 kJ/mol. The other pair suggests an energy difference of about 5 kJ/mol. Which number do you trust as being more representative of the actual energy difference? Explain your answer.



endocyclic exocyclic 107 116 105 110
heats of hydrogenation (kJ/mol)

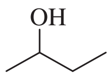
7-49 Predict the products of E1 elimination of the following compounds. Label the major products.

(a) 
(b) 
(c) $(CH_3)_3C{-}CH(Br){-}CH_3$

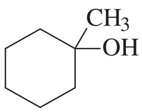
7-50 Predict the products formed by sodium hydroxide-promoted dehydrohalogenation of the following compounds. In each case, predict which will be the major product.
(a) 1-bromobutane
(b) 2-chlorobutane
(c) 3-bromopentane
(d) *cis*-1-bromo-2-methylcyclohexane
(e) *trans*-1-bromo-2-methylcyclohexane

7-51 What halides would undergo E2 dehydrohalogenation to give the following pure alkenes?
(a) hept-1-ene
(b) hex-2-ene
(c) hept-3-ene
(d) methylenecyclopentane
(e) 3-methylcyclobutene

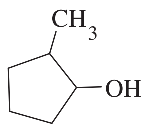
7-52 Predict the major products of acid-catalyzed dehydration of the following alcohols.

(a) 

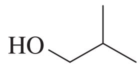
butan-2-ol

(b) 

1–methylcyclohexanol

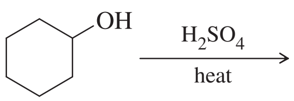
(c) 

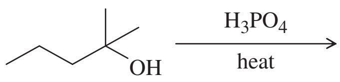
2-methylcyclopentanol

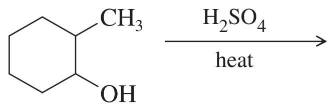
(d) 

2-methylpropan-1-ol

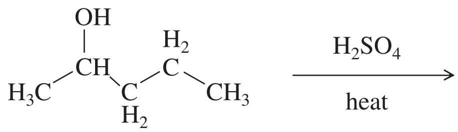
7-53 Predict the products of the following reactions. When more than one product is expected, predict which will be the major product.

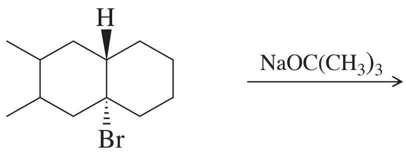
(a) 

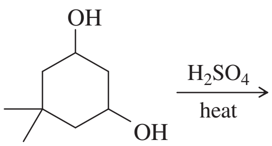
(b) 

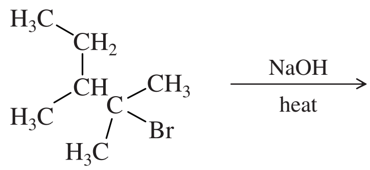
(c) 

7-54 Write a balanced equation for each reaction, showing the major product you expect.

(a) 

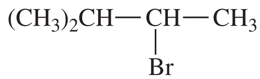
(b) 

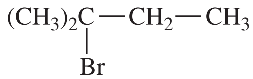
(c) 

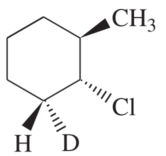
(d) 

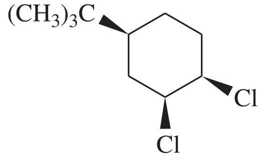
7-55 Predict the dehydrohalogenation product(s) that result when the following alkyl halides are heated in alcoholic KOH. When more than one product is formed, predict the major and minor products.

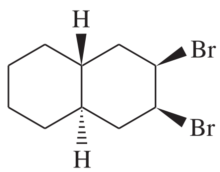
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

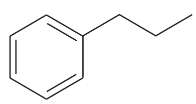
7-56 Using cyclohexane as your starting material, show how you would synthesize each of the following compounds. (Once you have shown how to synthesize a compound, you may use it as the starting material in any later parts of this problem.)

(a) bromocyclohexane
(b) cyclohexene
(c) ethoxycyclohexane
(d) 3-bromocyclohex-1-ene
(e) cyclohexa-1,3-diene
(f) cyclohexanol

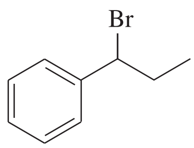
7-57 Show how you would prepare cyclohexene from each compound.

(a) cyclohexanol
(b) cyclohexyl bromide
(c) cyclohexane (not by dehydrogenation)

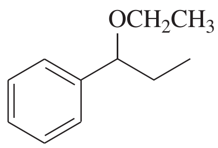
7-58 Show how you would convert (in one or two steps) 1-phenylpropane to the three products shown below. In each case, explain what unwanted reactions might produce undesirable impurities in the product.



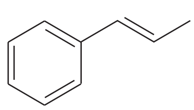
1-phenylpropane

(a) 

1-bromo-1-phenylpropane

(b) 

1-ethoxy-1-phenylpropane

(c) 

1-phenylprop-1-ene

7-59 E1 eliminations of alkyl halides are rarely useful for synthetic purposes because they give mixtures of substitution and elimination products. Explain why the sulfuric acid-catalyzed dehydration of cyclohexanol gives a good yield of cyclohexene even though the reaction goes by an E1 mechanism. (*Hint*: What are the nucleophiles in the reaction mixture? What products are formed if these nucleophiles attack the carbocation? What further reactions can these substitution products undergo?)

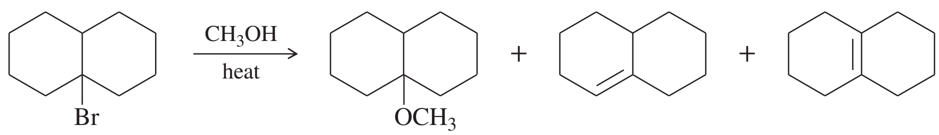
7-60 Propose mechanisms and draw reaction-energy diagrams for the following reactions. Pay particular attention to the structures of any transition states and intermediates. Compare the reaction-energy diagrams for the two reactions, and explain the differences.

(a) 2-Bromo-2-methylbutane reacts with sodium methoxide in methanol to give 2-methylbut-2-ene (among other products).

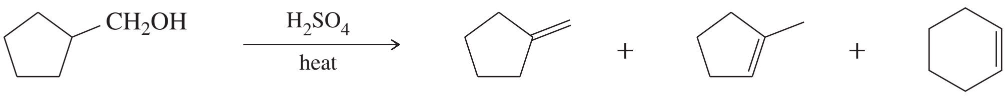
(b) 2-Bromo-2-methylbutane reacts in boiling methanol to give 2-methylbut-2-ene (among other products).

7-61 Propose mechanisms for the following reactions. Additional products may be formed, but your mechanism only needs to explain the products shown.

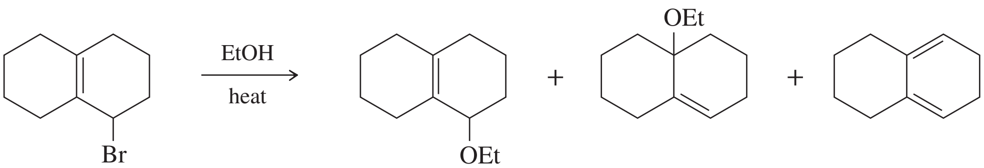
(a)



(b)

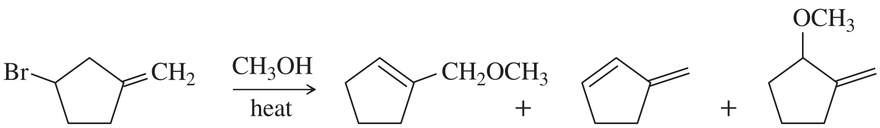


(c)

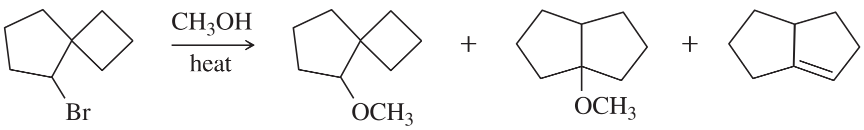


7-62 Propose mechanisms to account for the observed products in the following reactions. In some cases, more products are formed, but you only need to account for the ones shown here.

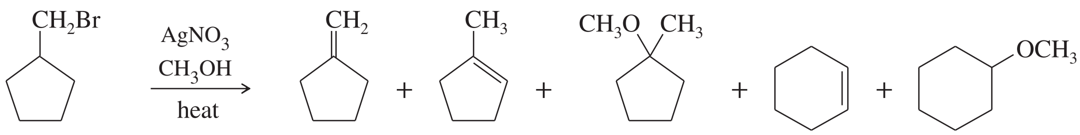
(a)



(b)

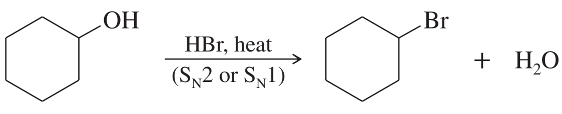


7-63 Silver-assisted solvolysis of bromomethylcyclopentane in methanol gives a complex product mixture of the following five compounds. Propose mechanisms to account for these products.

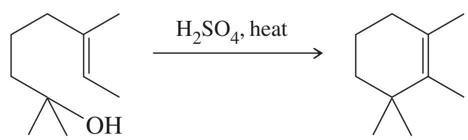


7-64 Protonation converts the hydroxy group of an alcohol to a good leaving group. Suggest a mechanism for each reaction.

(a)

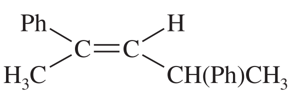


(b)



7-65 **(a)** Design an alkyl halide that will give *only* 2,4-diphenylpent-2-ene upon treatment with potassium *tert*-butoxide (a bulky base that promotes E2 elimination).

(b) What stereochemistry is required in your alkyl halide so that *only* the following stereoisomer of the product is formed?



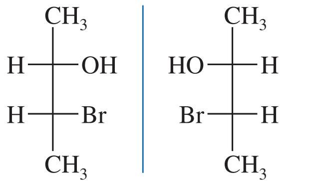
7-66 A chemist allows some pure (2*S*,3*R*)-3-bromo-2,3-diphenylpentane to react with a solution of sodium ethoxide ($NaOCH_2CH_3$) in ethanol. The products are two alkenes: **A** (cis-trans mixture) and **B,** a single pure isomer. Under the same conditions, the reaction of (2*S*,3*S*)-3-bromo-2,3-diphenylpentane gives two alkenes, **A** (cis-trans mixture) and **C.** Upon catalytic hydrogenation, all three of these alkenes (**A, B,** and **C**) give 2,3-diphenylpentane. Determine the structures of **A, B,** and **C;** give equations for their formation; and explain the stereospecificity of these reactions.

***7-67** Pure (*S*)-2-bromo-2-fluorobutane reacts with methoxide ion in methanol to give a mixture of (*S*)-2-fluoro-2-methoxybutane and three fluoroalkenes.

(a) Use mechanisms to show which three fluoroalkenes are formed.

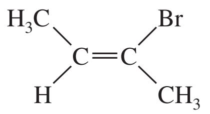
(b) Propose a mechanism to show how (*S*)-2-bromo-2-fluorobutane reacts to give (*S*)-2-fluoro-2-methoxybutane. Has this reaction gone with retention or inversion of configuration?

***7-68** When (±)-2,3-dibromobutane reacts with potassium hydroxide, some of the products are (2*S*,3*R*)-3-bromobutan-2-ol and its enantiomer and *trans*-2-bromobut-2-ene. Give mechanisms to account for these products. Why is no *cis*-2-bromobut-2-ene formed?



(2*S*,3*R*) (2*R*,3*S*)

3-bromobutan-2-ol



trans-2-bromobut-2-ene

***7-69** When 2-bromo-3-phenylbutane is treated with sodium methoxide, two alkenes result (by E2 elimination). The Zaitsev product predominates.

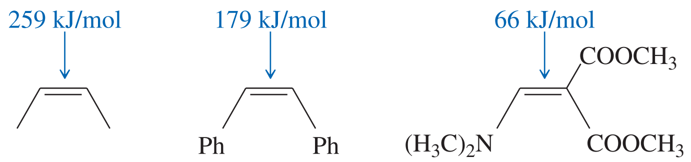
(a) Draw the reaction, showing the major and minor products.

(b) When one pure stereoisomer of 2-bromo-3-phenylbutane reacts, one pure stereoisomer of the major product results. For example, when (*2R,3R*)-2-bromo-3-phenylbutane reacts, the product is the stereoisomer with the methyl groups cis. Use your models to draw a Newman projection of the transition state to show why this stereospecificity is observed.

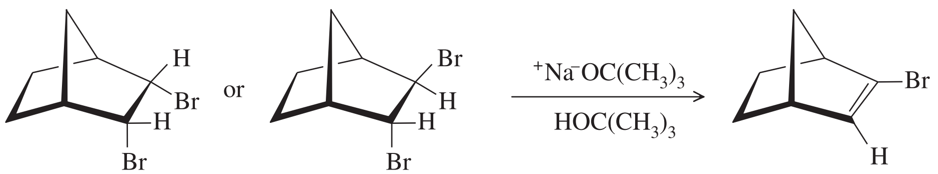
(c) Use a Newman projection of the transition state to predict the major product of elimination of (*2S,3R*)-2-bromo-3-phenylbutane.

(d) Predict the major product from elimination of (*2S,3S*)-2-bromo-3-phenylbutane. This prediction can be made without drawing any structures, by considering the results in part (b).

***7-70** Explain the dramatic difference in rotational energy barriers of the following three alkenes. (*Hint:* Consider what the transition states must look like.)

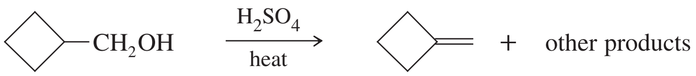


***7-71** One of the following dibromonorbornanes undergoes elimination much faster than the other. Determine which one reacts faster, and explain the large difference in rates.



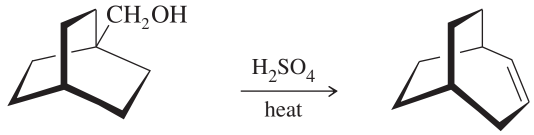
cis trans

***7-72** A graduate student wanted to make methylenecyclobutane, and he tried the following reaction. Propose structures for the other products, and give mechanisms to account for their formation.

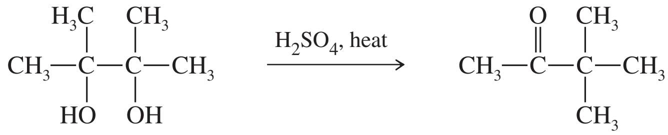


methylenecyclobutane
(minor)

***7-73** Write a mechanism that explains the formation of the following product. In your mechanism, explain the cause of the rearrangement, and explain the failure to form the Zaitsev product.



***7-74** The following reaction is called the *pinacol rearrangement*. The reaction begins with an acid-promoted ionization to give a carbocation. This carbocation undergoes a methyl shift to give a more stable, resonance-stabilized cation. Loss of a proton gives the observed product. Propose a mechanism for the pinacol rearrangement.



pinacol pinacolone

***7-75** Deuterium (D) is the isotope of hydrogen of mass number 2, with a proton and a neutron in its nucleus. The chemistry of deuterium is nearly identical to the chemistry of hydrogen, except that the C—D bond is slightly (5.0 kJ/mol, or 1.2 kcal/mol) stronger than the C—H bond. Reaction rates tend to be slower if a C—D bond (as opposed to a C—H bond) is broken in a rate-limiting step. This effect on the rate is called a *kinetic isotope effect*. (Review Problem 4-57.)

(a) Propose a mechanism to explain each product in the following reaction.

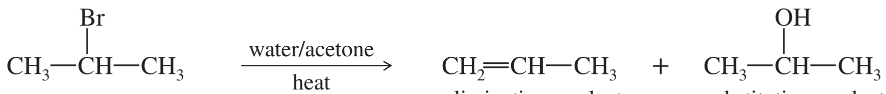
elimination product substitution product

(b) When the following deuterated compound reacts under the same conditions, the rate of formation of the substitution product is unchanged, while the rate of formation of the elimination product is slowed by a factor of 7.

seven times slower rate unchanged

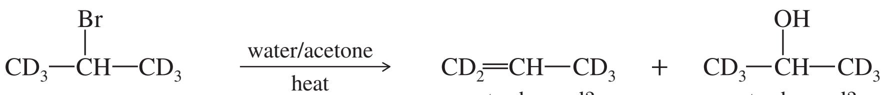
Explain why the elimination rate is slower, but the substitution rate is unchanged.

(c) A similar reaction takes place on heating the alkyl halide in an acetone/water mixture.



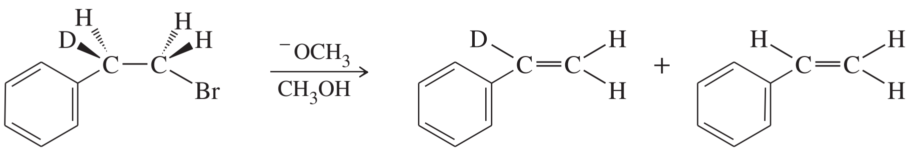
elimination product substitution product

Give a mechanism for the formation of each product under these conditions, and predict how the rate of formation of each product will change when the deuterated halide reacts. Explain your prediction.



rate changed? rate changed?

***7-76** When the following compound is treated with sodium methoxide in methanol, two elimination products are possible. Explain why the deuterated product predominates by about a 7:1 ratio (refer to Problem 7-75).



87% 13%